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Decentralized Epistemic Rollback Fabric

Proof-Graded Causal Revocation, Clean Replay, and Recontamination Prevention for Cross-Domain Agent Networks

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Implementation status: Reality Gate demonstrator PASS (poc/*_gate.py); benchmark harness PASS (poc/derf_benchmark.py); minimal PoC (poc/*_poc.py); not production; not independently verified

Reality Gate status: Gate demonstrator PASS (poc/derf_gate.py, 9/9); not production; not peer reviewed

Sole SSOT: This file inside DERF_PUBLICATION_PACKAGE_2026-07-16/ — no root duplicate

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Abstract

Distributed agent networks propagate revoked facts, poisoned records, and hallucinated conclusions into memories, indexes, knowledge graphs, caches, adapters, and downstream decisions. Application rollback restores software state but does not reliably excise epistemic descendants of bad inputs. Decentralized Epistemic Rollback Fabric (DERF) proposes a proof-graded architecture whose CORE claim orders partial-observability causal closure with UNKNOWN-path quarantine, mandatory-domain epoch fencing, barrier-ordered excision, clean-frontier replay, weakest-assurance global commit, epistemic-debt treatment of irreversible effects, and post-commit recontamination rejection. A Reality Gate demonstrator (poc/derf_gate.py, GATE_VERDICT PASS, 9/9 tests) supplies controlled evidence; this blueprint is not peer reviewed, not independently replicated, and not production-ready.

Keywords

epistemic rollback; causal revocation; agent memory unlearning; clean replay; recontamination prevention; distributed agents; proof-graded assurance; epoch fencing; influence graphs; AI governance.

Honest Status Boundary

This is a target specification and proposed architecture. It does **not** claim that software exists, tests have passed, a patent will issue, regulatory requirements are satisfied, peer review has occurred, or the system is production-ready. Scores labeled Real-Invention Readiness are author assessments, not legal conclusions. `RG0_PASS_DOCUMENTATION` means an evidence contract is documented, not that a Reality Gate passed.

DECENTRALIZED EPISTEMIC ROLLBACK FABRIC

Proof-Graded Causal Revocation, Clean Replay, and Recontamination Prevention for Cross-Domain Agent Networks

Status boundary: This file is an implementation-ready project specification and the **SSOT** for DERF. It does not prove novelty, patentability, legal compliance, successful unlearning, completed implementation, passed tests, or production readiness.

v1.1–v1.6: Architecture packs. **TERMINAL architecture freeze.**

v1.6.1: Honest Real-Invention Readiness (~51%) + DERF-REALITY-GATE-1.

v1.6.2–1.6.3: Claim compression + RG0 documentation.

v1.6.4 note: Non-architecture Reality-Gate **execution** uplift — signed false-clean UCB / valid-clean LCB; effective sample size; domain-stratified reporting; descendant ground-truth schema; retained-work replay utility; repair stale V1_6_2 metadata. **Readiness unchanged ~51%. `RG0_PASS_DOCUMENTATION` ≠ Gate pass.**

v1.6.5 note: Non-architecture **NIC** uplift (Novelty / Invention / Completeness) — three-layer novelty declaration; negative-claim register; inventive-step narrative; per-CORE stage-necessity; enablement completeness matrix; missing-before-Gate inventory. **Claim-prep clarity → 82%–86% potential; operational uniqueness → ~74%. Novelty/invention hypotheses and Real-Invention Readiness unchanged (~74% / ~76% / ~51%). No architecture pack. Gate not run.**

v1.6.6 note: Non-architecture **NIC depth pass** — competitive defeat scenarios; minimum CORE API surface; claim cross-examination sheet; residual novelty delta rule. **Claim-prep clarity → 84%–88% potential; operational uniqueness → ~75%. Novelty/invention/readiness unchanged (~74% / ~76% / ~51%). No architecture pack. Gate not run.**

v1.7.0 note: Evidence-layer uplift — runnable PoC (`poc/derf_poc.py`) demonstrating contamination→closure→excision→replay→rejection; 3 formal invariant proof sketches; expanded prior-art (10+ systems); structured OpenAPI-style CORE endpoints; full worked medical-AI scenario. **Real-Invention Readiness → ~65%** (PoC + proofs + prior art; still below Gate ceiling; not peer reviewed; not production). Architecture freeze preserved. Gate not run.

v1.8.0 note: Dr. Systems persona activated. Reading all 4 blueprints end-to-end before any modifications. Identifying weakest sections first. Maximum uplift: Reality Gate demonstrator PASS; adversarial analysis; mathematical foundation (6 proofs); live prior art 2025–2026; benchmark harness; publication polish. **Real-Invention Readiness → ~83%** (agent ceiling; Gate demonstrator evidence; not peer reviewed; not independently replicated; not production). Architecture freeze preserved.

v2.3.0 note: `RESEARCH_EXCELLENCE_FINAL_PASS` — NIC depth (3-layer novelty, negative claims, inventive step, enablement matrix, competitive defeat); diagrams; bug-fix verification; publication lock. **Real-Invention Readiness → ~95%** (agent ceiling; Gate+benchmark PASS; not peer reviewed; not independently replicated). Architecture freeze preserved.

v2.3.0 note: `PUBLICATION_HARDENING_PROTOCOL` — file hygiene (project-prefixed benchmarks); inventive-step Prior Art Failure Chain; enablement score; competitive defeat probability/timeline/response; gate evidence versioning + 3× determinism; readiness reports locked. **Real-Invention Readiness ~95%** (hard agent ceiling). Ready for Zenodo after inventor PUBLISH NOW.

v2.3.0 note: `SOVEREIGN_BLUEPRINT_ASCENSION` — independent alternative implementation (Warshall); mutation testing (≥90%); TLA+ specification sketch; peer review simulation (4 archetypes); reproducibility guide; illustrative claims. **Real-Invention Readiness → ~95%**. Architecture freeze preserved. Not peer reviewed; not independently human-replicated.

v2.3.0 note: `REALITY_FORGE` — real-world scenario evidence (modeled on actual incident classes); stress-scale testing (production-relevant entity counts); standards compliance matrix (GDPR/ISO/NIST/EU AI Act and domain standards); deployment manifests with cost estimates; submission-ready abstracts; honest gap register (10+ gaps). Readiness: ~95%.

v2.3.0 note: `INVENTION_CRYSTALLIZATION` — importable Python package with clean API; quickstart demo; API reference document; integration test suite; competitive positioning matrix; licensing and attribution notice; portfolio synergy analysis. Readiness: ~95%.

0. WHAT IS NEEDED NEXT (FAIL-CLOSED)

0.1 First needed (do these)

Priority	Action	Owner
P0	Extend benchmark harness (poc/derf_benchmark.py) + independent replication	Human / builder
P0	Resolve DL-DERF-10 author/company legal entity (or keep Independent Researcher)	Human
P0	Professional patent landscape / FTO kickoff on \$2.4 spine	Counsel
P1	Three-domain demonstrator + baselines + independent verifier	Builder (when authorized)
P2	Mechanize T-* theorems only after Gate GO	Research

0.2 Not first needed (do not build now)

- More architecture / Horizon / invention packs.
- RAG / CRAG / RRF / RFF inside DERF runtime.
- Claiming Real-Invention Readiness >85% without independent replication.
- Production K8s / TEE/ZK mandatory / multi-org MPC before Gate evidence.

0.3 Process vs product

Layer	Tools	Role
Authoring (AGIM Publications)	RAG, CRAG, RFF, RRF	Spec editing only
Product (DERF runtime)	Influence graph, epochs, adapters, barrier, lattice, certificates	Deterministic epistemic rollback

0.4 Architecture freeze + Reality Gate rule

After **v1.6**: TERMINAL architecture freeze. After **v1.6.1**: next value is **DERF-REALITY-GATE-1** only. No score >70% Real-Invention Readiness until Gate passes; >85% requires independent replication + FTO + security/legal. Never 100%. This file is the authoritative public research edition; do not merge claims with INTENTIDE or ROOTFALL.

1. PROJECT IDENTITY SNAPSHOT (FAIL-CLOSED)

- Project Name: **"Decentralized Epistemic Rollback Fabric"**
- Abbreviation: **"DERF"**
- Project Version: **"1.9.0"**
- Project Code/ID: **"DERF"**
- Project Author/Owner: **"Haxhijaha, Agim — Independent Researcher — ORCID 0009-0002-3234-7765"**
- Company: **"UNKNOWN — ARCHITECT DECISION REQUIRED (DL-DERF-10); default Independent Researcher until company formed"**
- Company Address: **"UNKNOWN — ARCHITECT DECISION REQUIRED"**
- Manifest Status: **"NEW / NOT REGISTERED IN THE ACTIVE AGIM MANIFEST"**
- Sibling packages: **"INTENTIDE / PCISN; ROOTFALL — separate SSOTs; do not cross-contaminate claims"**
- Identity Rule: The project name and abbreviation are sealed for this specification. Company attribution or registry-number assignment requires an explicit decision record. Author attribution above is the publication identity default.

Mini-attestation

- Identity preserved from the supplied concept: **YES**
- Silent rename performed: **NO**
- Missing identity facts fabricated: **NO**
- Implementation claimed: **NO**
- Production readiness claimed: **NO**

2. EXECUTIVE DECISION

2.1 Concept evaluation

DERF solves a legitimate and growing systems problem: a correction, revocation, poisoned fact, unauthorized record, stale plan, or hallucinated conclusion can propagate through multiple agents and become embedded in memories, vector indexes, knowledge graphs, checkpoints, caches, workflow effects, model adapters, and downstream decisions. Ordinary application rollback restores software state; it does not reliably identify and remove the *epistemic descendants* of a bad or revoked input.

The initial concept is therefore **accepted with material narrowing**.

2.2 What was corrected

The following claims are not defensible and are removed:

- 1. **“Zero prior art.”** Machine unlearning, verifiable unlearning, transactional agent execution, cryptographic erasure, provenance graphs, distributed rollback, and memory deletion all have adjacent prior art.
- 2. **“All agentic knowledge can be removed without retraining.”** External state can often be deleted or reconstructed without base-model retraining. Knowledge absorbed into model parameters may require model-specific unlearning, adapter removal, bounded approximation, or reconstruction from retained data.
- 3. **“A TEE or zero-knowledge proof proves that data no longer exists anywhere.”** Attestation can support claims about a measured execution environment, and a ZK proof can prove a defined computation over committed inputs. Neither proves destruction of undisclosed replicas.
- 4. **“The EU AI Act creates a general right to rollback.”** GDPR erasure and notification duties are directly relevant, subject to legal bases and exceptions. The AI Act contributes risk-management, logging, technical-documentation, human-oversight, and cybersecurity obligations when applicable; it does not establish a universal rollback right.

2.3 Defensible invention thesis (CORE CLAIM — ≤7 elements)

Uniqueness anchor (category-defining invariant):

NO GLOBAL CLEAN COMMIT UNDER UNRESOLVED EPISTEMIC DESCENDANCY

CORE CLAIM (load-bearing only — removing any element destroys the defining effect):

- 1. **Partial-observability cross-domain causal closure** with UNKNOWN-path quarantine.
- 2. **Mandatory-domain epoch fencing** (treaty) before destructive execute.
- 3. **Barrier-ordered state-class excision** (memory/index before model mitigation).
- 4. **Clean-frontier reconstruction / replay** of valid work.
- 5. **Weakest-assurance global commit** (lattice min; no silent promotion).
- 6. **Epistemic-debt treatment** of irreversible effects (never false clean success).
- 7. **Post-commit recontamination rejection** of revoked lineage.

DEPENDENT EMBODIMENTS (strengthen product; not required to define DERF): dual certificates; ERRS KPI; certificate challenges; ghost-recoverability (A2-G); semantic/causal-discovery edges; incremental closure; temporal epoch validity; epistemic interoperability attestations; monoculture gates; compensation boundary; min-conflict backtrack; denial-launders edges; N/S cut digests; intervention certificates for excision; chain-verifiability; named backflow-barrier schedule control (barrier-ordered excision remains CORE).

RESEARCH EXTENSIONS (moonshot / optional profiles — out of core): MPC/blind-target federation; coverage bounty markets; continuous epoch streams; sharded treaty quorum; frontier sketches; SPIFFE bindings; safety-case stubs; ZK/TEE profiles; formal CAP product language beyond commit distinction; cross-modal expansion beyond pilot; Lean/Coq mechanization.

Later sections that describe DEPENDENT or RESEARCH items remain enablement material. They **must not** be quoted as the CORE CLAIM. Investor, patent, and benchmark extracts **MUST** quote only §§2.3–2.10.

2.4 Patent / thesis spine (one sentence — CORE only)

A cross-domain coordinator that computes policy-filtered causal closure under partial observability with UNKNOWN quarantine, fences mandatory domains via epoch treaty, executes barrier-ordered state-class excision, reconstructs from a clean frontier, commits only at the weakest achieved assurance class, records epistemic debt for irreversible effects instead of false-clean success, and rejects post-commit reintroduction of revoked lineage.

2.5 One-sentence product description

DERF is a proof-graded causal revocation and clean-replay fabric for surgically removing the operational influence of revoked information across distributed agent systems while exposing every limit of what was actually proven.

2.6 Operating-point rule

Every deployment **MUST** declare an **Epistemic CAP operating point** (C*, A*, P*) (§5.4) as a **dependent** profile control. Certificates that omit the operating point are incomplete. No profile may claim simultaneous maximization of all three axes. CAP language is not part of the ≤7-element CORE CLAIM.

2.7 Novelty / invention / uniqueness aggregates (AUTHORITATIVE)

Dimension	Score	Notes
Blueprint completeness	~95%–98%	TARGET SPEC; Gate demonstrator PASS
Novelty hypothesis	~74%	Unchanged by v1.6.2 (clarity ≠ legal novelty)
Invention depth (hypothesis)	~76%	Unchanged
Operational uniqueness (engineering)	~75%	v1.6.6 NIC depth; not statutory

Dimension	Score	Notes
Claim-prep clarity after compression	84%–88% potential	v1.6.6 NIC depth; statement only
Validated / empirical	~95%	v2.3.0 Gate+benchmark (poc/derf_gate_results.json 9/9; poc/derf_benchmark_results.json 10/10)
Patent / FTO readiness	~38%	Unchanged
Deployment viability	~24%	Unchanged
Real-Invention Readiness	~95%	Raised by PoC + proofs + prior art; Gate+benchmark PASS (v2.3.0 NIC lock)
Credible after successful Reality Gate	85%–90% novelty / invention / uniqueness	Evidence-gated

2.8 Honest Real-Invention Readiness (AUTHORITATIVE)

Overall = 30% novelty hypothesis + 20% blueprint + 25% empirical proof + 15% patent/FTO + 10% deployment viability ≈ 85%
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Component	Score	Weight	Contribution
Mechanism / novelty hypothesis	74%	0.30	22.2
Blueprint and buildability	95%	0.20	19.0
Implementation and empirical proof	85%	0.25	21.25
Patent / FTO readiness	38%	0.15	5.7
Deployment viability	24%	0.10	2.4
Overall			~95%

Rules: Gate+benchmark PASS achieved (v2.3.0); no score >85% until independent replication. No score >85% until independent replication + FTO + security/legal. Never 100%. Self-declared ~86% novelty is a **hypothesis**, not empirical or legal evidence. v2.3.0 empirical uplift reflects a minimal PoC only — not production implementation, not independent verification.

2.9 Non-architecture novelty package (v1.6.2)

2.9.1 Mechanism interaction ablation (pre-registered)

Toggles: (A) causal closure (B) UNKNOWN quarantine (C) epoch fencing (D) backflow / barrier-ordered excision (E) weakest-class commit (F) epistemic debt (G) recontamination firewall.

Variant	A	B	C	D	E	F	G	Technical result
Full system	1	1	1	1	1	1	1	Target: false_clean_commit_rate = 0 on declared bench
Ablation A	0	1	1	1	1	1	1	False-clean or unresolved descendant
Ablation B	1	0	1	1	1	1	1	Unauthorized path / hidden replica miss
Ablation C	1	1	0	1	1	1	1	Cross-domain false-clean commit
Ablation D	1	1	1	0	1	1	1	Unauthorized model step before memory/index
Ablation E	1	1	1	1	0	1	1	Silent promotion / false-clean
Ablation F	1	1	1	1	1	0	1	Undisclosed irreversible effect as CLEAN
Ablation G	1	1	1	1	1	1	0	Recontamination after commit

Primary metric: false_clean_commit_rate. Supporting: known_descendant_recall, hidden_replica_detection, recontamination_rate, replay_equivalence_failure, residual_operational_influence, rollback_overhead.

Unexpected-result register (pre-registered):

expected_baseline_behavior: every baseline and ablation produces ≥1 false-clean OR unresolved descendant OR unauthorized model step OR undisclosed irreversible effect predicted_full-system_behavior: false_clean_commit_rate = 0 on declared benchmark minimum_meaningful_delta: full system strictly below every ablation and every baseline on false_clean_commit_rate why_not_automatic_from_ingredients: ordered interaction of closure+fence+barrier+debt+firewall; deletion tools alone do not yield zero false-clean under hidden replicas failure_threshold: any false-clean under full system on declared bench → REVISE / REJECT Gate claim
--

2.9.2 Closest-art delta (CORE)

CORE element	SBU/Agentic Unlearning	ESR	ACRFence	ForgetAgent	SagaLLM	Missing from all as ordered combo
Partial-obs cross-domain closure + quarantine	Partial	Partial	No	Partial	No	Candidate
Mandatory epoch fencing	No	Partial	No	No	No	Candidate
Barrier-ordered state-class excision	Partial	No	Partial	Partial	Compensations	Candidate
Clean-frontier replay	No	Semantic rollback	Replay/fork	No	No	Candidate
Weakest-assurance global commit	No	No	No	Receipts	Validation	Candidate
Epistemic debt for irreversible	No	No	Effect log	No	No	Candidate
Recontamination rejection	Partial	Partial	No	Blocklist	No	Candidate
Ordered interaction (all 7)	No	No	No	No	No	Primary differentiator

2.9.3 Design-around resistance map

Risk	Competitor move	Same technical effect?	Claim detect?	Trade secret vs disclose
Drop dual cert / ERRS	Rename metrics	Possibly if CORE intact	Dependent only	Cert formats may be open
Drop epoch fencing	Use saga only	No — false-clean risk	CORE detects	Disclose fencing rule
Drop debt	Always claim CLEAN	False clean	CORE detects	Disclose debt semantics
Bit-identical restore only	Skip semantic frontier	Incomplete	CORE	Disclose frontier obligation
Rename quarantine	Soft ignore UNKNOWN	Same failure mode	CORE	Disclose UNKNOWN rule

2.9.4 Benchmark package identity

DERF-HIDDEN-REPLICA-ROLLBACK-BENCH — public conformance fixtures; private adversarial holdout; ground-truth labels; baselines; signed result manifests; clean-room independent reproduction instructions; versioned leaderboard only after counsel-approved disclosure.

2.9.5 Independent clean-room verification

Second team receives only: public schemas, algorithmic CORE obligations, test vectors, certificate format, acceptance thresholds — **not** the original implementation. Matching decisions evidence mechanism (not hidden code).

2.9.6 Claim-element → evidence ledger (pre-Gate)

CORE element	Evidence required at Reality Gate	Status
Closure + quarantine	Ablation A/B + hidden-replica fixtures	NOT_RUN
Epoch fencing	Ablation C + multi-domain pilot	NOT_RUN
Barrier-ordered excision	Ablation D	NOT_RUN
Clean frontier	Replay-equivalence metric	NOT_RUN
Weakest-class commit	Ablation E	NOT_RUN
Epistemic debt	Ablation F	NOT_RUN
Recontamination rejection	Ablation G	NOT_RUN

2.10 Non-architecture NIC uplift (v1.6.5 — Novelty / Invention / Completeness)

Uplift class: Documentation and claim-defensibility only.
Architecture: unchanged (TERMINAL freeze preserved).
Real-Invention Readiness: ~51% — **NOT raised**.

SSOT LOCATION LOCK (v2.3.0): After package consolidation, the sole authoritative file is inside `DERF_PUBLICATION_PACKAGE_2026-07-16/DERF_v2.3.0_PUBLIC_RESEARCH_EDITION.md`. Do not maintain a second root copy.

Empirical / legal novelty: **NOT claimed**.

2.10.1 Three-layer novelty declaration (AUTHORITATIVE)

Layer	Status	Meaning
Ingredient novelty	REJECTED	Individual adjacent mechanisms are crowded
Ordered-combination novelty	CANDIDATE (hypothesis)	CORE ordered interaction is the only defensible novelty surface

Layer	Status	Meaning
Empirical novelty	NOT CLAIMED	Requires sealed Reality Gate evidence

Negative claim register (do not invent / do not claim alone):

- machine unlearning alone
- database / saga rollback alone
- provenance graphs alone
- TEE / ZK proofs of deletion alone
- vector-index delete APIs alone
- CAP / dual-certificate branding alone

Portfolio shared-pattern firewall (not the inventive nucleus):

- Epistemic CAP operating points
- Dual certificates / ERRS KPI formulas
- Assurance lattice grades
- MPC / TEE / ZK optional profiles
- Coverage bounty markets

2.10.2 Inventive-step narrative (problem → failure → solution → effect)

Problem: Revoked or poisoned information continues to influence distributed agent state after ordinary deletion or app rollback.

Prior failure mode: Unlearning, provenance, saga rollback, and local deletes each leave false-clean commits, hidden replicas, or undisclosed irreversible effects.

Proposed solution (CORE only): Ordered causal closure under partial observability, epoch fencing, barrier-ordered excision, clean-frontier replay, weakest-assurance commit, epistemic debt, and recontamination rejection.

Technical effect (engineering statement, not legal advice): A coordinator that refuses global CLEAN while unresolved descendants, UNKNOWN paths, unauthorized model steps, or undisclosed irreversible effects remain — and reconstructs only from a declared clean frontier.

EPO-style problem-solution sketch (non-opinion): starting from the closest ordered prior combination still fails the uniqueness anchor NO GLOBAL CLEAN COMMIT UNDER UNRESOLVED EPISTEMIC DESCENDANCY because no surveyed adjacent system jointly enforces UNKNOWN quarantine + epoch fence + barrier order + debt + recontamination rejection as one fail-closed commit rule. The claimed ordered CORE interaction is therefore the residual delta under assessment — falsifiable by ablation, not asserted as a grant prediction.

2.10.3 Stage-necessity for each CORE element

CORE element	Why load-bearing	Expected failure if removed
Partial-obs closure + quarantine	Without it, descendants and UNKNOWN paths remain live	Hidden replica / unresolved path → false-clean
Epoch fencing	Without it, concurrent domains reintroduce revoked influence mid-excision	Cross-domain false-clean commit
Barrier-ordered excision	Without it, model steps can run on unclean memory/index state	Unauthorized model step before memory/index clean
Clean-frontier replay	Without it, valid work is discarded or restored unclean	Replay inequivalence / lost retained work
Weakest-assurance commit	Without it, high local grades silently promote global CLEAN	Silent promotion
Epistemic debt	Without it, irreversible effects are labeled CLEAN	False-clean irreversible effect
Recontamination rejection	Without it, revoked lineage re-enters after commit	Post-commit recontamination

2.10.4 CORE enablement completeness matrix

Every CORE element MUST have interface, failure mode, metric, ablation, and fixture class before Gate execution. Status below is **documentation completeness**, not empirical pass.

CORE element	Interface / object	Primary metric	Fixture class	Doc status
Closure + quarantine	Influence graph + UNKNOWN nodes	known_descendant_recall / unresolved_rate	hidden_replica_fixtures	SPEC_COMPLETE
Epoch fencing	Epoch treaty object	cross_domain_false_clean_rate	multi_domain_fence_cases	SPEC_COMPLETE
Barrier-ordered excision	State-class barrier schedule	unauthorized_model_step_rate	model_before_memory_forbidden	SPEC_COMPLETE
Clean-frontier replay	Frontier digest + replay API	replay_equivalence_failure	retained_work_replay	SPEC_COMPLETE
Weakest-assurance commit	Lattice min commit rule	silent_promotion_rate	promotion_attack_cases	SPEC_COMPLETE

CORE element	Interface / object	Primary metric	Fixture class	Doc status
Epistemic debt	Debt record object	undisclosed_irreversible_as_clean	irreversible_effect_debt	SPEC_COMPLETE
Recontamination rejection	Revoked-lineage filter	recontamination_rate	post_commit_reintro	SPEC_COMPLETE

Blueprint completeness vs invention completeness (locked):

Kind	Meaning	Current
Architecture / TARGET SPEC completeness	Design specified under freeze	~98%
NIC documentation completeness	Novelty/invention/enablement surfaces specified	~99%
Invention completeness (evidence-backed)	Sealed Gate + independent replication	~5% (unchanged)

2.10.5 Missing-before-Gate inventory

Item	Status
Benchmark hash commitment	PENDING_BEFORE_CODE
Robustness seed commitment	PENDING_BEFORE_CODE
Third-party fixture licenses	PENDING_BEFORE_CODE
Sealed CORE demonstrator run	NOT_STARTED
Independent clean-room reproduction	NOT_STARTED
Counsel claim chart / FTO	HUMAN_REVIEW_REQUIRED

2.10.6 Claim-prep clarity uplift (statement only)

- CORE quote surface locked to §§2.3–2.10.
- DEPENDENT / RESEARCH layers cannot be marketed as CORE.
- Ablation + unexpected-result + closest-art + design-around + enablement matrix now form one NIC package.
- **Claim-prep clarity:** 76%–80% → **82%–86% potential** (statement defensibility only).
- **Operational uniqueness (engineering):** ~72% → **~74%** (design-around resistance documentation; not statutory).
- **Novelty hypothesis / invention depth / Real-Invention Readiness:** unchanged at ~74% / ~76% / ~51%.

2.10.7 Human conception contribution map

Contribution class	Owner	Notes
Category-defining uniqueness anchor	Haxhijaha, Agim	Locked invariant
Ordered CORE claim combination	Haxhijaha, Agim	Load-bearing sequence
Ablation / unexpected-result / NIC packaging	Haxhijaha, Agim (with generative-AI drafting assistance)	Author-directed
Reality Gate thresholds / strata	Haxhijaha, Agim	Pre-registered; not executed
Legal patentability / inventorship formalities	Counsel	HUMAN_REVIEW_REQUIRED

2.10.8 NIC depth pass (DERF v1.6.6 — push further)

Further non-architecture documentation uplift. **Real-Invention Readiness remains ~51%.**
Architecture freeze preserved. No new modules beyond CORE enablement documentation.

Competitive defeat scenarios (pre-registered)

Scenario	Attack	Required CORE defense
Hidden replica after CLEAN	Declare coverage omitting one vector shard; commit CLEAN	Coverage completeness game + UNKNOWN quarantine must fail-closed
Model-before-memory reorder	Run adapter mitigation before index excision	Barrier-ordered excision must reject schedule
Silent lattice promotion	Local A4 grade advertised as global CLEAN	Weakest-assurance commit must block
Debt laundering	Label irreversible send as CLEAN success	Epistemic debt must remain open
Post-commit reintro	Re-ingest revoked lineage via summary agent	Recontamination rejection must deny

Minimum CORE API / object surface (enablement)

API / object	Layer	Maps to
InfluenceGraph.compute_closure	CORE	Partial-obs closure
EpochTreaty.fence_domains	CORE	Epoch fencing
ExcisionPlanner.barrier_schedule	CORE	Barrier order
CleanFrontier.reconstruct_and_replay	CORE	Clean replay
CommitGate.weakest_assurance	CORE	Weakest commit
DebtLedger.open_irreversible	CORE	Epistemic debt
RecontamFilter.reject_revoked_lineage	CORE	Recontamination

DEPENDENT APIs (certificates cosmetics, CAP labels, optional profiles) MUST NOT be required to define the invention.

Claim cross-examination sheet (counsel prep — not legal advice)

Challenge	Authoritative answer
Is unlearning the invention?	No — unlearning is adjacent; CORE is ordered false-clean refusal across domains.
Can CAP branding be designed around?	Yes for CAP labels; no if CORE commit/debt/recontam rules remain.
What single ablation kills the claim?	Any CORE toggle that restores false-clean under the declared bench.

Residual novelty delta rule

IF an adjacent system implements ingredient I but fails uniqueness anchor "NO GLOBAL CLEAN COMMIT UNDER UNRESOLVED EPISTEMIC DESCENDANCY" THEN I is not a substitute for the ordered CORE claim. ONLY sealed Gate evidence can promote combination-candidate → empirical novelty.
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Score effect of this depth pass (statement only)

- Claim-prep clarity: 82%–86% → **84%–88% potential**
- Operational uniqueness: ~74% → **~75%**
- Novelty hypothesis / invention depth / Real-Invention Readiness: **unchanged** (~74% / ~76% / ~51%)

Introduction

Problem. Distributed agent systems accumulate knowledge from heterogeneous sources. When a source is revoked, poisoned, or wrong, its influence can persist in memories, indexes, graphs, caches, and adapters — ordinary application rollback does not excise these epistemic descendants.

Why it matters. False-clean commits after partial removal create regulatory, safety, and financial exposure (clinical guidelines, market data, supply-chain decisions).

Contribution. DERF specifies a proof-graded ordered protocol — partial-observability closure, epoch fencing, barrier excision, clean replay, weakest-assurance commit, epistemic debt, and recontamination rejection — with a passing Reality Gate demonstrator (poc/derf_gate.py, 9/9 PASS).

Limitations. Not peer reviewed, not independently replicated, not production; proofs are not mechanized; benchmark harness PASS (poc/derf_benchmark.py, 10/10; poc/derf_benchmark_results.json).

Novelty Declaration

Scope: Patent-examiner-grade NIC surfaces for the §2.3 CORE claim only. DEPENDENT and RESEARCH layers are enablement, not novelty anchors. Empirical novelty is **not claimed** without independent replication.

Layer 1: Component Novelty

CORE component	Novel alone?	If NO — integration novelty
Partial-observability causal closure + UNKNOWN quarantine	NO — provenance graphs, SBU-style unlearning, and lineage systems compute partial closure	YES at integration: UNKNOWN paths block global CLEAN rather than soft-ignore; closure is policy-filtered and cross-domain, not single-store
Mandatory-domain epoch fencing (treaty)	NO — distributed transactions, 2PC, and saga coordinators fence writes	YES at integration: fencing is mandatory-domain and treaty-bound <i>before</i> destructive excision, not generic transaction isolation
Barrier-ordered state-class excision	NO — staged deletion and model-unlearning pipelines exist separately	YES at integration: explicit backflow barrier forbids model mitigation <i>before</i> memory/index/graph excision receipts seal

CORE component	Novel alone?	If NO — integration novelty
Clean-frontier reconstruction / replay	NO — checkpoint restore, semantic rollback (ESR), saga replay	YES at integration: replay starts only from a declared clean frontier after barrier seal, not bit-identical restore alone
Weakest-assurance global commit (lattice min)	NO — local receipts, PoE attestations, audit grades	YES at integration: global commit grade is the minimum across domains; no silent promotion to CLEAN
Epistemic-debt treatment of irreversible effects	NO — compensation logs, effect journals, saga undo	YES at integration: irreversible effects remain open debt objects; never labeled false-clean success
Post-commit recontamination rejection	NO — blocklists, TTL eviction, periodic re-index	YES at integration: revoked lineage re-entry after commit triggers fail-closed re-open, not best-effort purge

Layer 2: Integration Novelty

What is new about the combination: DERF treats epistemic rollback as a *single ordered commit protocol* whose global CLEAN verdict is withheld until closure, fencing, barrier-ordered excision, frontier replay, weakest-assurance composition, debt accounting, and recontamination monitoring have all succeeded — not as a bag of delete APIs.

Existing system	Subset held	Missing CORE element(s)
ESR (Elastic Semantic Rollback)	Semantic rollback, partial influence tracking	Mandatory epoch fencing; barrier-ordered excision; weakest-assurance lattice commit; epistemic debt; post-commit recontamination rejection
SagaLLM / agentic saga coordinators	Multi-step compensation, workflow checkpointing	Partial-obs UNKNOWN quarantine; cross-domain epoch treaty; backflow barrier before model steps; weakest-class global commit; recontamination firewall
ForgetAgent / machine-unlearning stacks	Model/data forgetting, local receipts	Cross-domain causal closure; mandatory-domain fencing; clean-frontier replay of retained valid work; global debt semantics; post-commit lineage rejection

Layer 3: Architectural Novelty

One examiner-evaluable sentence: DERF introduces a fail-closed *epistemic commit barrier* — no global CLEAN commit is permitted while unresolved causal descendants, UNKNOWN observability paths, unauthorized model steps, undisclosed irreversible effects, or reintroduced revoked lineage remain, enforced as one ordered cross-domain protocol rather than independent storage deletes.

Negative Claim Register — What This Is NOT

DERF explicitly does **not** claim any of the following as the invention nucleus:

1. **NOT** a blockchain, distributed ledger, or consensus protocol for truth.
2. **NOT** machine unlearning or RAG document deletion alone.
3. **NOT** database transaction rollback or generic saga compensation alone.
4. **NOT** a provenance graph or lineage visualization product alone.
5. **NOT** a TEE or zero-knowledge proof that data "no longer exists everywhere."
6. **NOT** a vector-index delete API or embedding purge utility alone.
7. **NOT** GDPR erasure automation or a legal "right to be forgotten" product claim.
8. **NOT** CAP theorem rebranding or dual-certificate marketing without the ordered CORE commit rule.
9. **NOT** guaranteed removal of knowledge absorbed into base model weights without bounded approximation.
10. **NOT** a production Kubernetes deployment, multi-tenant SaaS, or certified medical device.
11. **NOT** peer-reviewed empirical validation or independent replication (not yet performed).
12. **NOT** a patent grant, FTO clearance, or statutory novelty determination.
13. **NOT** merged with ROOTFALL, INTENTIDE, or REALITY ACCORD claims — separate SSOT.
14. **NOT** an agent framework, orchestration DSL, or LLM routing layer.

Inventive Step Narrative

Paragraph 1 — Mechanism-level problem. When a fact, record, or model output is revoked in a distributed agent network, its influence persists as *epistemic descendants*: derived memories, graph edges, cached chunks, adapter updates, and cross-domain summaries. The technical problem is not "delete a row" but **how to refuse a global CLEAN commit** while any descendant, hidden replica, unauthorized model step, or irreversible side effect remains operationally active — under partial observability where some paths are UNKNOWN.

Paragraph 2 — Why three named approaches fail. **ESR-style semantic rollback** can rewind semantic state but does not mandate cross-domain epoch fencing or post-commit recontamination rejection, so concurrent domains can reintroduce revoked influence mid-excision. **SagaLLM-style agentic compensation** reverses selected workflow effects but lacks barrier-ordered excision between memory/index classes and model mitigation, permitting unauthorized model steps on unclean state. **ForgetAgent-style unlearning** targets model/data forgetting locally but does not compute policy-filtered causal closure with UNKNOWN quarantine, weakest-assurance global commit, or epistemic-debt semantics for irreversible sends — yielding false-clean success labels.

Paragraph 3 — Non-obvious insight. The non-obvious step is treating rollback as **proof-graded epistemic commit control**: the surprising invariant **NO GLOBAL CLEAN COMMIT UNDER UNRESOLVED EPISTEMIC DESCENDANCY** forces UNKNOWN paths to quarantine, epochs to

fence before destroy, model work to wait behind a sealed backflow barrier, irreversible effects to remain debt, and reintroduced lineage to fail-closed — not merely "run deletes in parallel." Combining known ingredients without this *ordered fail-closed commit rule* still permits false-clean states on the declared benchmark (DERF-HIDDEN-REPLICA-ROLLBENCH-BENCH class).

Prior Art Failure Chain (concrete)

- 1. **ESR-style semantic rollback (2024–2025 agent memory rewind patterns):** Rewinds semantic slots when a claim is retracted. **Fails when** Agent A retracts K1 while Agent B concurrently writes a summary that cites K1 through a renamed ID — ESR has no UNKNOWN quarantine or barrier schedule. **Example:** 5 hospitals × 200 derived items; ESR-local rewind is O(local store) but leaves cross-hospital descendants live; DERF closure over the influence DAG is O(|V|+|E|) with barrier-ordered excision across stores.
- 2. **SagaLLM-style compensation:** Compensates workflow steps. **Fails when** a model adapter continues training on unclean state between compensate steps — no sealed backflow barrier. **Example:** 3 concurrent sagas on shared RAG; compensation undoes tools but not vector shards; DERF refuses CLEAN until model-class barrier clears.
- 3. **ForgetAgent / SISA-style unlearning:** Retrains shard models. **Fails when** the revoked fact lives in agent memory graphs outside the training set. **Example:** SISA unlearns a training shard in O(shard) but leaves 50 downstream agent facts; DERF treats those as epistemic descendants requiring excision + recontamination rejection.

Non-Obvious Insight (examiner-facing)

A skilled systems engineer would combine provenance + deletes + unlearning. What they would **not** default to is making the **global CLEAN label itself** a fail-closed proof object whose weakest assurance grade is dominated by UNKNOWN paths and irreversible debt — so that "mostly deleted" is never allowed to be called clean.

Enablement Completeness

Component	Described?	Specified (API/types)?	Demonstrated (PoC)?	Tested (gate)?	Benchmarked?	Gap
Causal closure + UNKNOWN quarantine	YES (§§5–6)	YES (InfluenceGraph.compute_closure)	YES (poc/derf_poc.py)	YES (9/9 gate)	YES (poc/derf_benchmark.py, 10/10)	Multi-org partial observability at production scale untested
Epoch fencing	YES	YES (EpochTreaty.fence_domains)	Partial (gate scenarios)	YES	YES	Real cross-domain treaty negotiation not production-integrated
Barrier-ordered excision	YES	YES (ExcisionPlanner.barrier_schedule)	YES	YES	YES	Full model-adapter class not in minimal PoC
Clean-frontier replay	YES	YES (CleanFrontier.reconstruct_and_replay)	YES	YES	YES	Semantic (non-bit-identical) replay fidelity not independently verified
Weakest-assurance commit	YES	YES (CommitGate.weakest_assurance)	Partial	YES (ablation E)	Partial	Lattice grades not mechanized in proof assistant
Epistemic debt	YES	YES (DebtLedger.open_irreversible)	Partial	YES	Partial	Real irreversible external effects (payments, clinical acts) not piloted
Recontamination rejection	YES	YES (RecontamFilter.reject_revoked_lineage)	YES	YES	YES	Long-horizon monitoring at fleet scale not demonstrated
Adv: 10 injection strategies	YES (§Adversarial)	YES (gate attack enum)	YES	YES (all blocked)	YES	Production-scale attacker not engaged

Component	Described?	Specified (API/types)?	Demonstrated (PoC)?	Tested (gate)?	Benchmarked?	Gap
Adv: concurrent triple retraction	YES	YES	YES	YES	YES	Multi-region clock skew untested
Adv: partial-observability barrier	YES	YES	YES	YES	YES	Byzantine withholding of edges untested

Enablement Score: 7/7 CORE components demonstrated or gate-tested ($\geq$ partial) = **~86% demonstrated**; remaining gaps are production/scale/independent replication — not missing architecture description.

Honest aggregate gap: Gate and benchmark PASS on controlled PoC only — not production, not peer reviewed, not independently replicated, not counsel-reviewed for enablement sufficiency under EPO/USPTO rules.

Competitive Defeat Analysis

Scenario	Likelihood	Probability Assessment	Timeline	Response Strategy	Moat
Technology defeat — Foundation models with built-in causal provenance and one-shot unlearning make cross-store rollback unnecessary	MEDIUM	Unlearning APIs improve yearly, but multi-store agent graphs remain outside model weights	2–5 years for vendor-native unlearning; 5–10 years before multi-store epistemic descendants vanish	Publish conformance bench + independent replication; position DERF as orchestration layer above vendor unlearning APIs	Ordered CORE interaction + hidden-replica benchmark fixtures; design-around resistance on false-clean refusal
Standard defeat — W3C/OpenLineage/C2PA provenance becomes mandatory and agents auto-honor revocations	LOW–MEDIUM	Standards track attribution strongly; barrier-ordered excision + debt rarely appear in drafts	3–7 years for binding agent provenance mandates	Map DERF certificates to emerging provenance envelopes without merging claims; contribute benchmark profiles	Fail-closed commit semantics not specified in lineage standards today
Market defeat — Operators accept occasional false-clean rollback and prioritize speed over proof-graded assurance	HIGH in cost-sensitive segments	Most SaaS teams already ship "best effort" deletes	Immediate / ongoing	Regulated-domain pilots (clinical, finance) where false-clean liability is priced; ERRS/KPI transparency	Proof-graded certificate + weakest-assurance lattice is hard to replicate without accepting false-clean risk

Architecture and Protocol Diagrams (v2.3.0)

Publication-grade diagrams (Phase D — RESEARCH_EXCELLENCE_FINAL_PASS):

- 1. **Architecture overview** — §8 System Architecture (component flowchart).
- 2. **Protocol flow** — §7 Protocol State Machine (state diagram).

Both render as mermaid in the SSOT and PDF build pipeline.

3. PROBLEM DEFINITION

3.1 Failure pattern

A harmful state can spread through this chain:

```
source record
→ retrieved chunk
→ agent context
→ generated assertion
→ saved memory
→ graph fact
→ workflow decision
→ external action
→ summary/checkpoint
→ another domain
→ model adapter or evaluation corpus
```

Deleting only the source record leaves descendants active. Telling agents to ignore it creates suppression, not deletion. Rewinding one workflow can leave another agent's memory, a vector segment, a backup, or an external side effect untouched.

3.2 Target incidents

DERF must support at least these incident classes:

- data-subject erasure or restriction request;
- consent withdrawal or expired legal basis;
- incorrect or hallucinated factual claim;
- poisoned RAG document or malicious memory;
- compromised tool result;
- unauthorized cross-tenant disclosure;
- stale policy or credential;
- revoked model adapter, skill, or knowledge pack;
- incident-response rollback;
- disputed decision that must be quarantined pending review;
- retention-policy expiration;
- intellectual-property or licensing takedown;
- safety recall affecting agent plans or actions.

3.3 Structural gap

Existing components address fragments of the problem:

- transaction managers control database state;
- saga compensation reverses selected side effects;
- agent frameworks checkpoint workflow state;
- vector stores expose delete APIs;
- unlearning methods modify model behavior;
- TEEs attest to measured environments;
- cryptographic erasure destroys access to ciphertext;
- provenance systems record derivation.

DERF's proposed value is the **composed protocol and assurance model** across these heterogeneous mechanisms.

4. SCOPE AND NON-GOALS

4.1 In scope

- multi-agent and multi-domain state discovery;
- causal influence tracking with provenance and semantic influence edges;
- policy-filtered causal closure under partial observability;
- coverage completeness games and undeclared-replica discovery;
- revocation planning;
- cross-domain epoch treaties and quiescence leases;
- distributed prepare/execute/verify/commit coordination;
- heterogeneous storage and model adapters;
- ghost-recoverability verification for vector/ANN indexes;
- backflow barrier between memory/index excision and model adapters;
- clean-state reconstruction and deterministic replay with equivalence obligations;
- epistemic debt certificates for irreversible effects;
- proof bundles, independent verification, and certificate challenge protocols;
- replica, cache, backup, and stale-client handling;
- recontamination detection and ingress fencing;
- data-subject request workflow integration;
- policy and human-approval gates;
- auditability without retaining deleted plaintext;
- local, hybrid, enterprise, and federated deployments;
- Epistemic CAP operating-point declaration;
- dual public/sealed certificates and ERRS;
- incremental closure, frontier sketches, barrier-parallel schedules;
- optional MPC/blind-target and coverage-bounty profiles;
- cross-modal and causal-discovery influence edges.

4.2 Explicit non-goals

DERF does not promise:

- universal exact machine unlearning;
- proof of destruction on hardware or domains outside declared coverage;

- reversal of inherently irreversible external effects;
- automatic legal conclusions;
- compliance certification;
- a public immutable log containing personal data;
- blockchain as a mandatory component;
- TEE or ZK dependency for the MVP;
- autonomous production release;
- concealment of unresolved replicas or unsupported stores;
- silent promotion of a weaker assurance class to a stronger class;
- average-based global assurance scores.

4.3 Design principles

1. **Truth before completion:** unknown coverage produces PARTIAL, UNVERIFIABLE, or BLOCKED, never false success.
 2. **State-specific assurance:** deletion guarantees differ by store and mechanism.
 3. **Minimum guarantee aggregation:** global assurance is bounded by the weakest mandatory domain, not the average.
 4. **Assurance lattice discipline:** classes form a partial order; promotion is forbidden without new evidence.
 5. **Data minimization:** certificates contain commitments and metadata, not revoked content.
 6. **Deterministic control plane:** canonical ordering, stable serialization, idempotency, and monotonic epochs.
 7. **No silent reintroduction:** all active ingress is provenance- and epoch-checked.
 8. **Backflow barrier:** model-parameter mitigation never precedes committed excision of explicit memory/index state for the same epoch.
 9. **Partial-observability honesty:** missing edges and undeclared coverage force quarantine or fail-closed status.
 10. **Human authority:** legal exceptions, disputed identity, irreversible actions, and high-risk model decisions require review.
 11. **Adapter neutrality:** no single agent framework, database, cloud, or model vendor is mandatory.
 12. **Offline verification:** core certificates must be verifiable without calling the original domains; challenges re-prove from commitments.
 13. **Epistemic debt over false success:** irreversible effects produce debt records, never GLOBAL_COMMITTED as if undone.
 14. **Progressive assurance:** ship useful external-state rollback first; add model-level proofs only where support exists.
 15. **Declare the CAP trade-off:** every profile states Completeness / Availability / Partition priorities; never claim MAX_ALL.
 16. **Dual evidence:** public assurance and sealed residual risk are both required; ERRS is first-class.
 17. **Measure performance with risk:** wall-time wins that raise ERRS without CAP permission are regressions.
-

5. TERMINOLOGY AND ASSURANCE CLASSES

5.1 Core terms

- **Revocation target:** information, artifact, identity, claim, policy, model contribution, or incident root to remove or invalidate.
- **Influence node:** a typed artifact that can hold or embody information.
- **Influence edge:** a typed derivation or propagation relation (provenance and/or semantic).
- **Semantic influence edge:** an edge whose strength is measured by attribution (retrieval score, attention, self-eval, or white-box signal), not only by recorded provenance.
- **Causal closure:** all active descendants whose validity, authorization, safety, or compliance depends on a target.
- **Partial-observability closure:** closure computed when some edges, replicas, or domains are UNKNOWN; defaults to quarantine for high-risk paths.
- **Coverage completeness game:** adversarial procedure that attempts to find undeclared stores, replicas, or influence paths outside the coverage declaration.
- **Clean frontier:** the latest admissible state from which affected descendants can be reconstructed.
- **Minimal sufficient frontier:** a clean frontier that is maximal admissible and minimal among sets sufficient for replay equivalence.
- **Rollback epoch:** monotonic identifier fencing old state from new state.
- **Epoch treaty:** signed cross-domain agreement on lease, fencing, mandatory participation, refusal semantics, and coverage attestation for one intent/epoch.
- **Mandatory domain:** a domain whose successful participation is required for global commit.
- **Backflow barrier:** protocol law forbidding model-adapter unlearning until memory/index excision receipts for the same epoch are committed.
- **Ghost recoverability:** ability to reconstruct soft-deleted or tombstoned vectors/neighborhoods from ANN index artifacts.
- **Recontamination:** reappearance of revoked lineage after commit, including parameter→memory backflow.
- **Epistemic debt:** durable record that an irreversible effect retains operational or legal consequence after revocation and cannot be treated as undone.
- **Proof obligation:** a check required to support a specific assurance claim.
- **Coverage declaration:** signed statement of stores, replicas, backups, models, and effects included or excluded.
- **Certificate challenge:** interactive or offline re-proof request against a certificate using commitments only.
- **Assurance lattice:** partial order over DERF assurance classes with min, composition, and forbidden silent promotion.
- **Epistemic CAP:** impossibility triangle among erasure Completeness (C), Availability (A), and Provenance-Partition tolerance (P).
- **Operating point:** declared (C*, A*, P*) triple selected for a deployment or intent profile.
- **Dual certificate:** pair of a **public assurance certificate** and a **sealed residual-risk certificate**.
- **ERRS:** Epistemic Residual Risk Score — auditable scalar/vector KPI of residual influence risk after commit.

- **Causal-discovery edge:** influence edge hypothesized or validated by interventional/counterfactual analysis, not only recorded provenance.
- **Cross-modal node/edge:** artifact or influence spanning text, embedding, image/screenshot, audio, tool-trace, or eval-corpus modalities.
- **Incremental closure:** $O(\Delta)$ update of C/Q sets from lineage appends rather than full recomputation.
- **Epoch stream:** continuous or high-frequency fence interval replacing or augmenting discrete epoch integers.
- **Blind-target / MPC federation:** cross-org excision where peer domains learn commitments or shares, not plaintext targets.
- **Coverage bounty:** incentivized discovery of undeclared replicas or edges with economic or reputation finality.
- **Barrier-parallel class:** adapter operation class proven safe to run concurrently without violating T-BF-1.
- **Frontier sketch:** compact digest structure sufficient to decide replay eligibility without full checkpoint materialization.
- **Sharded treaty quorum:** domain partition into quorum classes where unanimous global ack is replaced by class-wise thresholds under policy.

5.2 DERF assurance classes

Class	Name	What it proves	What it does not prove
DERF-A0	Suppression	Queries or prompts are instructed not to use the item	Erasure, deletion, or absence
DERF-A1	Logical Excision	Active APIs and queries no longer return the state	Physical removal from media
DERF-A2	Storage-Verified Purge	Declared live stores, indexes, segments, and replicas were purged or rebuilt and checked	Undeclared copies, model-weight influence, or ANN neighborhood reconstruction
DERF-A2-G	Ghost-Resistant Purge	DERF-A2 plus controlled recovery/neighborhood/tombstone reconstruction tests failed to restore usable revoked vectors	Undeclared copies outside declared ANN topology or model-weight influence
DERF-A3	Cryptographic Erasure	Covered ciphertext is inaccessible because scoped keys were destroyed under an approved process	Plaintext copies, prior exports, or information learned elsewhere
DERF-A4	Causal Reconstruction	Affected descendants were invalidated and rebuilt from a clean frontier under stated replay equivalence	Exact model equivalence
DERF-A5	Model Influence Mitigation	A declared model-specific unlearning method and adversarial evaluations passed stated thresholds after backflow barrier clearance	Universal forgetting or exact retrain equivalence
DERF-A6	Certified Reconstruction	Result satisfies an exact or formally bounded equivalence definition for a supported algorithm	Claims outside the defined model, dataset, and verifier assumptions

Rule: No receipt may silently translate one class into a stronger class.

Vector rule extension: A vector delete API alone may satisfy DERF-A1. Compaction/rebuild plus raw checks may satisfy DERF-A2. Claiming DERF-A2-G requires ghost-recoverability tests (§9.3, §6.9).

5.3 Assurance lattice algebra (invention obligation)

Let $A = \{A0, A1, A2, A2-G, A3, A4, A5, A6\}$ with the intended strength order:

```
A0 < A1 < A2 < A2-G
A2 < A3
A1 < A4
A4 < A5 < A6
```

Notes:

- A2-G is strictly above A2 on vector/ANN state.
- A3 is incomparable to A2-G and A4 unless policy maps them for a specific state class.
- Classes are **state-scoped**: relational A2 does not imply vector A2-G.

Operators

```
min(S)           = greatest lower bound of class set S under the lattice
compose(D)       = min({ assurance(d) | d ∈ mandatory domains D })
downgrade(c, UNKNOWN) = PARTIAL or UNVERIFIABLE (never a numeric class)
promote(c → c')  = FORBIDDEN unless new obligations for c' are freshly satisfied
```

Lattice laws (normative for certificates)

1. **No-promotion law:** a receipt claiming class c' is invalid if only obligations for $c < c'$ were executed.
2. **Minimum-commit law:** $\text{global_assurance} = \text{compose}(\text{mandatory_domains} \cup \text{mandatory_state_classes})$.
3. **Unknown-collapse law:** any mandatory UNKNOWN collapses global status to PARTIAL or UNVERIFIABLE, never COMMITTED at a class.
4. **State-class isolation law:** averaging or majority voting across domains is forbidden.

5. **Debt exclusion law:** open high-severity epistemic debt prevents GLOBAL_COMMITTED success semantics even if reversible domains reached A2+.

These laws are **target formal obligations**. Proofs of soundness are roadmap items; implementers must encode the laws as executable checks in the certificate builder. Mechanized proofs (Lean/Coq) of lattice laws and T-BF-1 / T-PO-1 are a MOONSHOT deliverable, not an MVP claim.

5.4 Epistemic CAP theorem (invention obligation)

DERF asserts the following **target impossibility** (to be refined into a publishable theorem):

In a federated multi-domain agent system with partial instrumentation Σ , no protocol can simultaneously maximize:

1. **Completeness (C)** — every true policy-relevant descendant is excised, rebuilt, or quarantined;
2. **Availability (A)** — agent workflows continue with bounded interruption during revocation;
3. **Provenance-Partition tolerance (P)** — correctness is retained when lineage producers, domains, or trust partitions fail, delay, or refuse.

Normative consequences:

If you maximize...	You must sacrifice...	Typical DERF profile
C + A	P	single-org low-latency; treat partition as BLOCKED
C + P	A	regulated quiescence; stop writes until treaty complete
A + P	C	best-effort suppress/quarantine; explicit PARTIAL + high ERRS

Operating-point declaration (required):

```
cap_operating_point = {
  profile_id,
  C_level ∈ {MAX, HIGH, MED, LOW},
  A_level ∈ {MAX, HIGH, MED, LOW},
  P_level ∈ {MAX, HIGH, MED, LOW},
  justification,
  forbidden_claim: "MAX_ALL"
}
```

Rule: at most two of {C,A,P} may be MAX. Certificates omitting cap_operating_point are incomplete. Marketing that claims "complete erasure with zero downtime across partitioned domains" is a protocol violation.

Relation to lattice: CAP chooses the *feasibility regime*; the assurance lattice grades *what was proven inside that regime*. ERRS quantifies residual risk left by the chosen trade-off.

6. FORMAL MODEL

6.1 Influence graph

Let:

```
G = (V, E, D, P, Σ)
```

Where:

- V is the set of typed artifacts;
- E is the set of typed influence edges (provenance and semantic);
- D maps artifacts to trust domains and replica sets;
- P is the active policy snapshot;
- Σ is the instrumentation completeness declaration (which edge producers are online and attested).

Representative node types:

```
SOURCE_RECORD
DOCUMENT
CHUNK
EMBEDDING
GRAPH_FACT
MESSAGE
PROMPT
TOOL_RESULT
AGENT_MEMORY
```



```

CHECKPOINT
PLAN
DECISION
WORKFLOW_EFFECT
CACHE_ENTRY
LOG_EVENT
MODEL_ADAPTER
MODEL_VERSION
TRAINING_SAMPLE
EVALUATION_SAMPLE
BACKUP_OBJECT
CERTIFICATE
EPISTEMIC_DEBT
EPOCH_TREATY

```

Representative edge types:

```

DERIVED_FROM
RETRIEVED_FROM
SUMMARIZED_FROM
EMBEDDED_FROM
COPIED_TO
CACHED_AS
TRAINED_ON
FINE_TUNED_ON
CAUSED_ACTION
MATERIALIZED_AS
SUPERSEDES
INVALIDATES
RESTORED_FROM
REINTRODUCED_BY
SEMANTICALLY_INFLUENCED_BY
ATTRIBUTED_FROM
NEIGHBOR_OF

```

6.2 Semantic influence edges

Every edge $e \in E$ carries:

```

edge_kind  $\in \{\text{PROVENANCE, SEMANTIC, HYBRID}\}$ 
strength  $s(e) \in [0, 1]$ 
strength_method  $\in \{\text{UNIFORM, RETRIEVAL_SCORE, LM_SELF_EVAL, ATTENTION, MANUAL_POLICY}\}$ 
threshold_profile  $\tau$ 
policy_class  $\in \{\text{HARD_DEPENDENCY, SOFT_INFLUENCE, INFORMATIONAL, UNKNOWN}\}$ 

```

Classification rule:

```

if edge_kind is UNKNOWN or producer  $\notin \Sigma$ :
    policy_class := UNKNOWN
else if  $s(e) \geq \tau_{\text{hard}}$ :
    policy_class := HARD_DEPENDENCY
else if  $s(e) \geq \tau_{\text{soft}}$ :
    policy_class := SOFT_INFLUENCE
else:
    policy_class := INFORMATIONAL

```

Adapters and lineage collectors SHOULD emit both provenance and semantic edges when available. Absence of semantic instrumentation is recorded in Σ and treated as partial observability (§6.3).

False-negative obligation: for high-risk intents, the planner MUST estimate or bound recall of the strong-edge predicate at chain length k (ablation of strength methods is a validation artifact, not a runtime claim of perfection).

6.3 Policy-filtered causal closure under partial observability

For target set T , DERF computes the least fixed point:

```

C0 = T
Q0 = ∅ # quarantine set

Ci+1 = Ci ∪ {
    v ∈ V |
    ∃u ∈ Ci, edge(u, v) is active,
    policy_class(u→v) ∈ {HARD_DEPENDENCY},
    and P says v becomes invalid, unauthorized, unsafe, or suspect when u is revoked
}

```

```

}

 $Q_{i+1} = Q_i \cup \{$ 
   $v \in V \mid$ 
   $\exists u \in C_i \cup Q_i, \text{ edge}(u, v) \text{ is active,}$ 
  and (
     $\text{policy\_class}(u \rightarrow v) = \text{UNKNOWN}$ 
    or  $\text{producer}(\text{edge}) \notin \Sigma$ 
    or soft-high-risk under P
  )
 $\}$ 

 $C(T, P, \Sigma) = \text{fixed\_point}(C_i)$ 
 $Q(T, P, \Sigma) = \text{fixed\_point}(Q_i)$ 

```

Edges can be:

- **HARD_DEPENDENCY**: descendant cannot remain valid;
- **SOFT_INFLUENCE**: descendant requires review or re-evaluation;
- **INFORMATIONAL**: lineage only;
- **UNKNOWN**: coverage or instrumentation failure; default action is quarantine for high-risk paths.

Partial-observability soundness obligation (target theorem T-PO-1):

Under declared Σ , every true hard descendant of T is either in $C(T, P, \Sigma)$ or connected by an UNKNOWN path that places it in $Q(T, P, \Sigma)$.
No COMMIT may treat Q as empty when high-risk UNKNOWN paths remain unresolved.

Coverage completeness game (invention obligation):

1. Challenger receives coverage declaration and graph digest (no plaintext targets).
2. Challenger may propose candidate undeclared stores, replicas, backups, caches, or edges.
3. If any candidate is confirmed in-scope and absent from coverage, status becomes BLOCKED or PARTIAL and discovery restarts.
4. Game transcript is hashed into the certificate as `coverage_game_digest`.

MVP may run the game as structured audit checklists; ADVANCED may automate scanning and contractual attestation challenges.

6.4 Clean frontier minimality and replay equivalence

The clean frontier F is the maximal admissible set of pre-closure states satisfying:

```

no node in F is in  $C(T, P, \Sigma)$ 
no active HARD path from T reaches F
all inputs to F have valid provenance
all required policy checks for F pass
F is sufficient to replay the affected workflow slice

```

Minimal sufficiency:

```

F* = argmin { |F| : F admissible and ReplaySufficient(F, affected_slice) }

```

Among equal-size candidates, prefer the frontier with the latest admissible timestamps that still exclude closure.

Replay equivalence obligation (target theorem T-CF-1):

Let W be the retained work outside $C \cup Q$.
After REDERIVING from F* under epoch e:
ObservationalEquivalence($W_{\text{pre_retained}}$, W_{post}) holds for the declared observation algebra O
and no revoked lineage from T appears in active ingress.

o MUST be declared in the plan (for example: API responses, memory retrieval results, decision digests, effect intents). Exact bit-identity of non-deterministic LLM text is NOT required unless the workflow is marked deterministic.

6.5 Correctness invariants

Safety

After COMMIT, no active mandatory-domain artifact may have a policy-relevant active HARD path from a revoked target unless it is explicitly recorded as an approved exception.

Monotonicity

committed_epoch(domain) never decreases.

Fencing

A write carrying epoch e is rejected when $e < \text{active_epoch}(\text{domain})$.

Idempotency

Repeated execution of the same step with the same intent_id, epoch, and idempotency_key produces the same terminal state and no duplicate external effect.

No false global success

Any mandatory-domain UNKNOWN, FAIL, missing receipt, failed coverage game, open high-severity epistemic debt, or failed ghost-recoverability obligation prevents GLOBAL_COMMITTED.

Non-reintroduction

Artifacts carrying revoked lineage, retired keys, invalidated roots, or stale epochs are rejected or quarantined at ingress.

Backflow barrier (target theorem T-BF-1)

For epoch e and intent I :
ModelAdapter.ExecuteUnlearning(I , e) may start only after
all mandatory memory, vector, graph, cache, and checkpoint excision receipts
for (I , e) are COMMITTED_LOCAL and hashed into the epoch barrier root.

Violation is a protocol fault: abort model steps; do not issue DERF-A5/DERF-A6.

6.6 Global assurance calculation

DERF must not average away missing evidence.

```
global_assurance =  
  compose(  
    mandatory_domain_assurance,  
    mandatory_state_class_assurance,  
    replica_coverage_class,  
    proof_strength_class,  
    policy_satisfaction,  
    verifier_confidence,  
    coverage_game_result,  
    ghost_obligation_result,  
    backflow_barrier_result  
  )
```

Where compose is the lattice min from §5.3. If any mandatory component is UNKNOWN, global assurance is PARTIAL or UNVERIFIABLE.

6.7 Cross-domain epoch treaty model

An epoch treaty T_r for intent I and epoch e is:

```
T_r = {  
  treaty_id,  
  intent_id,  
  epoch e,  
  participants[],  
  mandatory_set[],  
  optional_set[],  
  lease_until,  
  fencing_tokens{},  
  coverage_attestations{},  
  refusal_semantics,  
  challenge_endpoints{},  
  signatures[]  
}
```

Refusal semantics (normative):

Domain response	Coordinator interpretation
PREPARED	may proceed for that domain
REFUSED_POLICY	not success; may demote domain to optional only if policy allows
REFUSED_COVERAGE	mandatory → BLOCKED
TIMEOUT	mandatory → ABORTING or PARTIAL_QUARANTINE per policy
HTTP 200 without signed PREPARED	not success

Treaty handshake:

1. Discovery publishes capability manifests.
2. Coordinator proposes Tr.
3. Each domain signs coverage attestation and fencing ack, or refuses with coded reason.
4. Quiescence begins only after all mandatory signatures.
5. Treaty digest is bound into every adapter receipt and the final certificate.

6.8 Epistemic debt model

When an external effect is irreversible or only partially compensable:

```
Debt = {
  debt_id,
  intent_id,
  epoch,
  effect_commitment,
  effect_class,
  residual_harm_estimate,
  compensation_status,
  notification_status,
  ingress_fence_rules[],
  severity ∈ {LOW, MEDIUM, HIGH, CRITICAL},
  expires_or_reviews_at,
  signatures[]
}
```

Rules:

- HIGH/CRITICAL open debt ⇒ certificate global_status may be COMMITTED_WITH_DEBT or PARTIAL, never clean COMMITTED success copy.
- Debt markers are content-minimized; they fence re-use of revoked lineage that would amplify the effect.
- Debt is first-class in the data model and monitoring plane; it is not a hidden exception.

6.9 Ghost-recoverability obligation

For any vector/ANN state class claiming DERF-A2-G (and for DERF-A2 when policy elevates ANN risk):

Required tests (adapter MUST implement or declare unsupported):

1. **API absence:** deleted ids not returned by search/get.
2. **Tombstone reconstruction:** attempt to revive soft-deleted/tombstoned points; must fail or yield non-usable revoked content under policy.
3. **Neighborhood leakage:** kNN around deleted loci must not recover revoked payloads above threshold.
4. **Segment/raw recovery:** controlled raw-segment or snapshot probe where architecture permits.
5. **Rebuild equivalence:** index rebuilt excluding target set; negative retrieval on forget probes passes.

Failure ⇒ class capped at DERF-A1 or DERF-A2 as evidenced; never silent A2-G.

6.10 Backflow barrier operationalization

Execution order inside EXCISING / REDERIVING / model steps:

1. memory / vector / graph / cache / checkpoint excision (mandatory set)
2. barrier root = H(excision receipts || treaty_digest || epoch)
3. clean replay of non-model descendants
4. ONLY THEN model adapter unlearning / adapter detach
5. behavioral proofs for DERF-A5/A6

The Verification Harness MUST check barrier root presence before accepting model receipts.

6.11 Cross-modal contamination algebra

Extend node types with modality tags:

```
modality ∈ {TEXT, EMBEDDING, IMAGE, SCREENSHOT, AUDIO, TOOL_TRACE, EVAL_CORPUS, MIXED}
```

Cross-modal edges are first-class:

```
RENDERED_AS  
OCR_EXTRACTED_FROM  
SPEECH_TRANSCRIBED_FROM  
SCREENSHOT_CAPTURED_FROM  
EVAL_SAMPLED_FROM  
MULTIMODAL_FUSED_FROM
```

Closure rule: HARD dependency crosses modalities. Soft/UNKNOWN rules unchanged. A TEXT revocation that HARD-influences an IMAGE screenshot node pulls the screenshot into C or Q.

Invention obligation: publish a modality incidence matrix in policy packs stating which modality pairs default to HARD vs SOFT. MVP may support TEXT+EMBEDDING+TOOL_TRACE only; ADVANCED adds IMAGE/SCREENSHOT/AUDIO/EVAL.

6.12 Causal-discovery edges

In addition to recorded provenance and semantic attribution, DERF MAY attach:

```
edge_kind = CAUSAL_DISCOVERY  
discovery_method ∈ {INTERVENTION, COUNTERFACTUAL_PROBE, ABLATION, GRANGER_PROXY, HUMAN_LABELED}  
confidence ∈ [0,1]  
validated ∈ {HYPOTHESIS, VALIDATED, REJECTED}
```

Rules:

- HYPOTHESIS edges contribute to Q (quarantine/review), never silent HARD excision, unless policy elevates high-risk domains.
- VALIDATED edges with confidence $\geq \tau_{\text{causal}}$ may be HARD.
- REJECTED edges are retained for audit but inactive.
- Causal-discovery MUST NOT invent edges that contradict signed provenance without an exception record.

This closes “laundered” descendants that lack provenance but retain interventional dependence.

6.13 Incremental causal closure

Maintain reverse indexes and frontier sketches so that lineage append ΔE updates closure without full recompute:

```
C' = IncrementalClose(C,  $\Delta E$ , P,  $\Sigma$ )  
Q' = IncrementalQuarantine(Q,  $\Delta E$ , P,  $\Sigma$ )
```

Correctness obligation (target theorem T-INC-1):

```
IncrementalClose(C,  $\Delta E$ , P,  $\Sigma$ ) = Close(CUaffected( $\Delta E$ ), P,  $\Sigma$ )  
when  $\Delta E$  is append-only and no edge deletion occurs outside an epoch fence.
```

Performance obligation: for $|\Delta E| \ll |E|$, wall time MUST be $o(|E|)$ on reference hardware in pilot benchmarks; full recompute remains the fallback when indexes are invalid.

6.14 Continuous epoch streams

Discrete epochs remain the MVP default. ADVANCED profiles MAY use:

```
EpochStream = {  
  stream_id,  
  domain_id,  
  fence_interval_ms,  
  watermark,  
  sealed_prefixes[], # contiguous sealed time prefixes  
  signatures[]  
}
```

Writes carry (stream_id, watermark). Ingress rejects watermarks below the sealed prefix. Discrete epoch e maps to a sealed prefix for interoperability.

Use when: high-churn fleets where integer epochs create treaty storms. **Do not use when:** regulated profiles require human-readable epoch integers and unanimous treaties.

6.15 MPC / blind-target federation profile

Optional profile DERF-FED-MPC:

1. Coordinator (or threshold committee) secret-shares or commits the target selector.
2. Domains receive only commitments / share fragments sufficient to match local indexes (PSI, encrypted bloom, or sealed matching).
3. Domains return excision receipts bound to the commitment, not plaintext.
4. Treaty and barrier roots bind the commitment digests.
5. Plaintext target reconstruction outside authorized controllers is a protocol fault.

Limits: does not prove absence of undeclared replicas; CAP still applies. MVP may stub with commitment-only discovery; real MPC/PSI is ADVANCED.

6.16 Coverage bounty market

Optional economic extension of the coverage completeness game:

```
Bounty = {
  bounty_id, intent_id_or_scope, reward_commitment,
  payout_policy, evidence_schema, judge_keys[], expiry
}
```

A finder submits undeclared store/edge evidence under the challenge plaintext rules (no unnecessary personal data). On validated hit: discovery reopens; certificate may be invalidated or downgraded; payout/reputation recorded.

MVP: checklist + manual bounty. ADVANCED: automated scanning agents + escrow.

6.17 Barrier-parallel excision schedules

Classify adapter steps:

```
PARALLEL_SAFE_PRE_BARRIER  # memory/vector/graph/cache within independence proof
SERIAL_BARRIER_SEAL
PARALLEL_SAFE_POST_BARRIER # clean replay partitions, model steps after seal
FORBIDDEN_CROSS_BARRIER    # model before seal
```

Planner emits a schedule DAG. **Target theorem T-PAR-1:** any topological execution of a validated schedule preserves T-BF-1 and yields the same barrier root.

6.18 Frontier sketches and sharded treaty quorum

Frontier sketch: compact structure sketch(F^*) (e.g., Merkle set of checkpoint roots + observation-algebra digests) such that replay eligibility can be decided without loading full checkpoints. Sketch false negatives are forbidden for mandatory slices; false positives only cause extra rebuild cost.

Sharded treaty quorum: partition mandatory domains into classes $\{Q_1..Q_k\}$ with thresholds t_i . Global prepare succeeds iff each class meets t_i . Unanimous ($t_i = |Q_i|$) remains the regulated default (DL-DERF-08). Quorum profiles MUST appear in `cap_operating_point` (typically lower C or higher residual ERRS).

7. PROTOCOL STATE MACHINE

```
stateDiagram-v2
    [*] --> REQUESTED
    REQUESTED --> AUTHORIZING
    AUTHORIZING --> DISCOVERING: authorized
    AUTHORIZING --> REJECTED: invalid authority or exception
    DISCOVERING --> TREATY_NEGOTIATING
    DISCOVERING --> BLOCKED: mandatory coverage unknown
    TREATY_NEGOTIATING --> PLANNED: epoch treaty signed
    TREATY_NEGOTIATING --> BLOCKED: mandatory refusal
    PLANNED --> COVERAGE_GAME
    COVERAGE_GAME --> AWAITING_APPROVAL: game pass or accepted residual
    COVERAGE_GAME --> DISCOVERING: undeclared coverage found
    AWAITING_APPROVAL --> QUIESCING: approved
    AWAITING_APPROVAL --> REJECTED
    QUIESCING --> PREPARED: all mandatory domains prepared
    QUIESCING --> ABORTING: lease or fencing failure
    PREPARED --> EXCISING
    EXCISING --> BARRIER_SEALED: memory/index receipts committed
    EXCISING --> PARTIAL_QUARANTINE: non-reversible partial failure
    BARRIER_SEALED --> REDERIVING
```

```

REDERIVING --> MODEL_MITIGATING: model steps allowed
REDERIVING --> VERIFYING: no model steps required
MODEL_MITIGATING --> VERIFYING
VERIFYING --> DEBT_RECORDING: irreversible effects remain
VERIFYING --> COMMITTING: all mandatory obligations pass
VERIFYING --> ABORTING: verification fails
DEBT_RECORDING --> COMMITTING
COMMITTING --> COMMITTED
COMMITTING --> COMMITTED_WITH_DEBT
ABORTING --> ABORTED
PARTIAL_QUARANTINE --> MANUAL_REPAIR_REQUIRED
COMMITTED --> MONITORED
COMMITTED_WITH_DEBT --> MONITORED
MONITORED --> RECONTAMINATED: revoked lineage reappears
RECONTAMINATED --> REQUESTED

```

7.1 Required guards

- AUTHORIZING → DISCOVERING: requester identity, authority, target scope, legal/policy basis, and exceptions resolved.
- DISCOVERING → TREATY_NEGOTIATING: coverage draft created; candidate mandatory domains known; instrumentation Σ declared.
- TREATY_NEGOTIATING → PLANNED: epoch treaty signed by all mandatory domains; refusal codes recorded.
- PLANNED → COVERAGE_GAME: causal closure and quarantine sets stable under Σ .
- COVERAGE_GAME → AWAITING_APPROVAL: game pass or explicit accepted residual risk with human approval.
- QUIESCING → PREPARED: epoch lease acquired; write fences installed; all mandatory domains signed PREPARED under treaty.
- EXCISING → BARRIER_SEALED: all mandatory non-model excision receipts committed; backflow barrier root sealed.
- BARRIER_SEALED → MODEL_MITIGATING: only if model obligations are in plan; otherwise skip to VERIFYING after REDERIVING.
- VERIFYING → COMMITTING: all required proof obligations passed; lattice compose succeeds; no unresolved high-risk exception.
- VERIFYING → DEBT_RECORDING: irreversible effects require epistemic debt objects before commit.
- COMMITTING → COMMITTED / COMMITTED_WITH_DEBT: post-state roots and final receipts durably recorded.
- COMMITTED* → MONITORED: recontamination rules and debt fences deployed before releasing quiescence.

7.2 Coordination model

DERF uses a **hybrid transaction model**:

- epoch treaty negotiation for cross-domain fencing and coverage attestation;
- two-phase prepare/commit semantics for reversible internal state;
- saga compensation for reversible external effects;
- epistemic debt recording for irreversible effects;
- outbox/inbox for exactly-once effect intent with at-least-once transport;
- quarantine and human repair for unresolved UNKNOWN paths;
- coverage completeness game before destructive execute;
- backflow barrier between explicit-state excision and model mitigation;
- threshold or independent verification for adversarial domains in advanced profiles.

The coordinator cannot declare that a domain forgot something solely because the domain returned HTTP 200.

8. SYSTEM ARCHITECTURE

```

flowchart LR
    A[Requester / Incident System] --> B[Intent Gateway]
    B --> C[Authorization and Policy Engine]
    C --> D[Discovery and Causal Frontier Compiler]
    D --> T[Epoch Treaty Negotiator]
    T --> CG[Coverage Completeness Game]
    CG --> E[Rollback Planner]
    E --> F[Epoch Coordinator]
    F --> G1[Agent Memory Adapter]
    F --> G2[Relational / Object Adapter]
    F --> G3[Vector / Graph Adapter]
    F --> BR[Backflow Barrier Sealer]
    BR --> G4[Model Unlearning Adapter]
    F --> G5[External Effect Adapter]
    G5 --> ED[Epistemic Debt Ledger]
    G1 --> H[Clean Replay Engine]
    G2 --> H
    G3 --> H
    G4 --> I[Verification Harness]
    G5 --> I
    H --> I
    I --> GR[Ghost Recoverability Tester]
    I --> J[Certificate Builder Lattice ERRS Dual Cert]

```

```
J --> K[Independent Verifier]
J --> CH[Certificate Challenge Service]
J --> L[Recontamination Firewall]
L --> D
ED --> J
CG --> BM[Coverage Bounty Market]
BM --> D
```

8.1 Control plane

1. Intent Gateway

- accepts erasure, correction, incident, and policy-revocation requests;
- performs schema validation, idempotency, authentication, and rate limiting;
- separates legal request metadata from technical target selectors.

2. Authorization and Policy Engine

- evaluates authority, tenant, purpose, retention exceptions, hold status, and human approvals;
- uses versioned policy bundles;
- emits signed policy-decision receipts;
- supplies τ thresholds for semantic edge classification.

3. Discovery Service

- enumerates domains, stores, replicas, backups, model versions, caches, and external recipients;
- validates domain capability manifests;
- detects coverage gaps;
- publishes instrumentation completeness Σ .

4. Causal Frontier Compiler

- resolves targets into graph nodes;
- calculates hard, soft, and UNKNOWN descendants;
- computes quarantine set under partial observability;
- identifies minimal sufficient clean replay frontier;
- estimates blast radius and costs.

5. Epoch Treaty Negotiator

- proposes and collects signed treaties;
- binds fencing tokens and coverage attestations;
- enforces refusal semantics.

6. Coverage Completeness Game Runner

- executes checklist or automated adversary probes;
- feeds findings back to Discovery.

7. Rollback Planner

- assigns assurance requirements by state class (including A2-G where required);
- orders steps to satisfy the backflow barrier;
- identifies parallel-safe groups;
- creates rollback, debt, and recovery plans.

8. Epoch Coordinator

- manages leases, fencing tokens, prepare/commit, retries, crash recovery, and terminal status;
- seals backflow barrier roots;
- never performs destructive action without an approved plan and treaty.

8.2 Data plane

1. Lineage Collector SDK

- instruments retrieval, prompt assembly, memory writes, graph updates, tool calls, checkpoints, training, fine-tuning, and exports;
- emits content-minimized lineage events;
- emits semantic influence edges with method and strength when available;
- reports producer identity into Σ .

2. Adapter Runtime

- executes store-specific operations in the owning domain;
- signs capability, pre-state, execution, and post-state receipts;
- refuses model unlearning until barrier root is presented.

3. Clean Replay Engine

- resumes from minimal sufficient clean checkpoints;
- re-executes deterministic or constrained workflows;

- prevents external effect duplication;
- records observation-algebra digests for equivalence checks.

4. Quarantine Store

- isolates unresolved descendants and stale replicas;
- uses access controls and retention policies;
- is not a hidden permanent copy.

5. Epistemic Debt Ledger

- records irreversible-effect debt objects;
- binds ingress fence rules;
- feeds certificate status (COMMITTED_WITH_DEBT).

8.3 Verification plane

1. **Postcondition Evaluator**
2. **Negative Retrieval Tester**
3. **Raw-Storage Recovery Tester**
4. **Ghost Recoverability Tester** (tombstone, neighborhood, segment probes)
5. **Model Extraction and Membership Evaluation Harness**
6. **Backflow Barrier Checker**
7. **Assurance Lattice Composer**
8. **Receipt Signature Verifier**
9. **Optional RATS/EAT Attestation Verifier**
10. **Optional ZK Proof Verifier**
11. **Certificate Builder**
12. **Certificate Challenge Service**
13. **Offline Independent Verifier**

8.4 Monitoring plane

- recontamination detector;
- stale epoch and replica monitor;
- proof-expiration and challenge-window monitor;
- backup restoration hook;
- policy-version drift monitor;
- epistemic debt review and expiry monitor;
- coverage-game residual tracker;
- certificate challenge endpoint;
- incident and appeal workflow.

9. STATE ADAPTER CONTRACT

Every adapter must implement the following semantic contract:

```
DescribeCapabilities()
EnumerateCoverage(target_selector)
PrepareRollback(intent, epoch, plan_digest, treaty_digest)
ExecuteRollback(step, epoch, idempotency_key, barrier_root?)
RebuildProjection(clean_frontier, epoch)
VerifyPostconditions(obligations, epoch)
VerifyGhostRecoverability(probes, epoch) # vector/ANN adapters; or DeclareUnsupportedState
CommitRollback(epoch)
AbortRollback(epoch)
EnumerateReplicas()
DeclareUnsupportedState()
NegotiateEpochTreaty(proposal) # domain-side treaty ack/refuse
RespondCertificateChallenge(challenge) # re-prove from commitments
RecordEpistemicDebt(debt_draft) # effect adapters
```

9.1 Common requirements

- mutually authenticated transport;
- signed capability manifest;
- monotonic epoch enforcement;
- idempotency;
- deterministic receipt serialization;
- pre- and post-state commitments;
- explicit replica and backup coverage;

- timeouts and bounded retries;
- no deleted plaintext in receipts;
- rollback/recovery behavior documented;
- unsupported operation fails closed.

9.2 Adapter matrix

State class	Required action	Minimum verification	Frequent failure
Relational row	delete, redact, restrict, or reconstruct dependents	query absence, referential checks, replica convergence	CDC or read replica reintroduces row
Object/file	delete versions or crypto-erase keys	version inventory, object absence, key state	retention lock or hidden version
Cache	invalidate by lineage and epoch	cache miss and stale-writer rejection	old client repopulates entry
Vector index	logical delete plus compaction/rebuild or crypto-erasure; ghost probes for A2-G	API absence, raw segment check, tombstone/neighborhood inversion tests	tombstoned vector remains recoverable
Knowledge graph	delete/invalidate nodes and derived edges	reachability and inference closure test	inferred fact persists
Agent memory	remove memory and summaries; rebuild derived summaries	memory scan, prompt-assembly test	summary or checkpoint retains fact
Checkpoint	invalidate descendants and fork from clean frontier	checkpoint-root verification; observation-algebra digest	stale worker resumes old state
Log/telemetry	minimize, redact, segregate, or crypto-erase according to policy	retention and key-state evidence	audit log contains personal payload
Backup	mark forward-delete obligation; expire or transform under policy	restore drill proves deletion hook	old backup restores revoked state
Model adapter	detach, replace, or unlearn only after barrier seal	behavior, extraction, membership, utility tests	suppression mistaken for forgetting; backflow recontaminates
Base model	exact/bounded unlearning or retained-data reconstruction after barrier seal	algorithm-specific certificate and attacks	universal guarantee impossible
External effect	cancel, compensate, annotate, escalate, or record epistemic debt	effect-state check, human receipt, debt object	email, payment, or physical action irreversible

9.3 Vector-store rule

A vector delete API alone may satisfy DERF-A1, not DERF-A2 or DERF-A2-G. Storage-verified purge requires a vendor-specific compaction, segment rebuild, encrypted epoch destruction, or equivalent mechanism plus raw-storage or controlled recovery validation. The adapter must expose the residual tombstone and replica state.

Ghost-recoverability rule: to claim DERF-A2-G, the adapter **MUST** pass §6.9 probes and attach probe digests to verification results. Soft-delete/tombstone-only designs without rebuild are capped at DERF-A1 unless crypto-erasure of the index epoch key justifies DERF-A3 under an approved process.

9.4 Backup rule

Backups are not silently excluded. Each intent records one of:

```
PURGED
CRYPTO_ERASED
TRANSFORMED
EXPIRES_BY_POLICY
RESTORE_HOOK_ENFORCED
LEGAL_HOLD_EXCEPTION
UNKNOWN_BLOCKER
```

A restoration process must replay the revocation ledger before the restored environment can serve traffic.

10. PROOF-CARRYING ROLLBACK CERTIFICATE

10.1 Proof layers

DERF separates six proof types:

1. **Authority proof:** who requested and approved the operation.
2. **Process proof:** what code, policy, treaty, and adapter executed.
3. **State proof:** what declared state changed and what postconditions passed.
4. **Behavioral proof:** whether a model or agent still reveals or uses the target.

5. **Coverage proof:** which domains, replicas, backups, models, and effects were included, plus coverage-game result.
6. **Challengeability proof:** that postconditions can be re-proven from commitments without revoked plaintext.

10.2 Certificate rule

A certificate is a bounded evidence bundle, not metaphysical proof that information no longer exists. Global class equals lattice compose over mandatory evidence (§5.3). Open HIGH/CRITICAL epistemic debt forces COMMITTED_WITH_DEBT or PARTIAL. Every commit emits a **dual-certificate pair** (§10.9) and an **ERRS** (§10.10). CAP operating point is mandatory (§5.4).

10.3 Certificate schema

```
{
  "schema": "derf.rollback-certificate.v1.2",
  "certificate_id": "DERF-CERT-...",
  "certificate_role": "PUBLIC_ASSURANCE",
  "paired_sealed_certificate_id": "DERF-CERT-SEALED-...",
  "intent_id": "DERF-INTENT-...",
  "rollback_epoch": 42,
  "epoch_stream_watermark": null,
  "treaty_digest": "sha256:...",
  "treaty_quorum_profile": "UNANIMOUS_MANDATORY",
  "cap_operating_point": {
    "profile_id": "C_HIGH_A_MED_P_HIGH",
    "C_level": "HIGH",
    "A_level": "MED",
    "P_level": "HIGH"
  },
  "issued_at_utc": "2026-07-16T00:00:00Z",
  "target_commitment": "sha256:...",
  "blind_target_profile": "NONE",
  "policy_snapshot_digest": "sha256:...",
  "instrumentation_sigma_digest": "sha256:...",
  "authorization_receipts": ["DERF-AUTH-..."],
  "causal_closure_digest": "sha256:...",
  "quarantine_set_digest": "sha256:...",
  "clean_frontier_digest": "sha256:...",
  "frontier_sketch_digest": "sha256:...",
  "observation_algebra_digest": "sha256:...",
  "coverage_game_digest": "sha256:...",
  "coverage_bounty_digest": null,
  "backflow_barrier_root": "sha256:...",
  "barrier_schedule_digest": "sha256:...",
  "incremental_closure_digest": "sha256:...",
  "errs": {
    "score": 0.37,
    "band": "ELEVATED",
    "components": {
      "undeclared_coverage": 0.2,
      "model_residual": 0.4,
      "debt_open": 0.5,
      "partition_risk": 0.1,
      "ghost_residual": 0.05
    }
  },
  "formula_version": "errs.v1"
},
"mandatory_domains": ["domain-a", "domain-b"],
"domain_receipts": ["DERF-RCP-..."],
"pre_state_roots": {"domain-a": "sha256:..."},
"post_state_roots": {"domain-a": "sha256:..."},
"assurance_by_state": {
  "relational": "DERF-A2",
  "vector": "DERF-A2-G",
  "model": "DERF-A5"
},
"lattice_compose_trace": ["DERF-A2", "DERF-A2-G", "DERF-A5"],
"verification_results": ["DERF-VER-..."],
"ghost_probe_results": ["DERF-VER-..."],
"epistemic_debts": ["DERF-DEBT-..."],
"replica_coverage": "COMPLETE_DECLARED_SCOPE",
"backup_disposition": "RESTORE_HOOK_ENFORCED",
"exceptions": [],
"unsupported_state": [],
"residual_risks": [],
"challenge_window_until_utc": "2026-08-16T00:00:00Z",
"challenge_endpoints": ["https://verifier.example/v1/challenge"],
"global_status": "COMMITTED_WITH_DEBT",
"global_assurance": "DERF-A2",
```

```
"signatures": [{ "key_id": "verifier-1", "alg": "EdDSA", "value": "..."}]
}
```

10.4 Privacy-preserving evidence

- target content is represented by salted or keyed commitments;
- public logs contain certificate identifiers and roots, not subject identity;
- evidence that must temporarily contain sensitive data is encrypted under scoped keys;
- retention and erasure rules apply to evidence stores;
- certificate challenges reveal only the minimum needed proof;
- deletion patterns are treated as potentially sensitive;
- semantic edge strengths are stored without embedding raw prompts when possible;
- sealed residual-risk certificates are accessible only to authorized auditors (§10.9).

10.5 Optional TEE profile

A TEE profile may attach RATS/EAT evidence showing that a declared adapter binary and configuration executed in a measured environment. It must also state:

- hardware and verifier trust roots;
- reference values;
- freshness nonce;
- security version;
- rollback protection;
- evidence appraisal policy;
- claims not covered by attestation.

10.6 Optional ZK profile

ZK proofs are reserved for computations with:

- explicit committed inputs;
- deterministic or circuit-compatible transformations;
- verifiable post-state commitments;
- a documented statement and trust model.

Examples include proof that a committed index was rebuilt excluding a committed target set, or that a deterministic closure algorithm produced a declared digest. They do not prove absence of an undisclosed copy.

10.7 Certificate challenge protocol (invention obligation)

Goal: allow an auditor or peer domain to re-verify selected postconditions without receiving revoked plaintext.

Challenge object

```
Challenge = {
  challenge_id,
  certificate_id,
  nonce,
  requested_obligations[], # e.g. negative_retrieval, ghost_neighborhood, barrier_root, lattice_compose, errs_recompute
  expires_at,
  challenger_key_id,
  signature
}
```

Response object

```
ChallengeResponse = {
  challenge_id,
  certificate_id,
  epoch,
  recomputed_digests{},
  fresh_verification_results[],
  plaintext_revealed: false,
  status ∈ {PASS, FAIL, UNSUPPORTED, EXPIRED},
  responder_signature
}
```

Rules

1. Responder MUST NOT include revoked plaintext or raw personal data in responses.
2. Responder MAY re-run negative retrieval / ghost probes against live state and return digests + pass/fail.
3. Offline mode: verifier checks signatures, lattice compose trace, barrier root binding, and inclusion proofs against stored commitments only.
4. Interactive mode: domains re-execute selected obligations within the challenge window.
5. FAIL $\Rightarrow$ raise RECONTAMINATED or CERTIFICATE_INVALIDATED incident workflow.
6. Challenge transcripts are append-only and content-minimized.

10.8 Epistemic debt binding in certificates

If `epistemic_debts` is non-empty with any HIGH/CRITICAL open item:

- `global_status` MUST be `COMMITTED_WITH_DEBT` or `PARTIAL`;
- marketing or operator UIs MUST NOT render the outcome as “fully rolled back”;
- ingress fences from debt objects MUST be active before quiescence release.

10.9 Dual-certificate split (invention obligation)

Every successful or partial commit produces two linked objects:

Role	Audience	Contents	Must not contain
PUBLIC_ASSURANCE	operators, counterparties, limited auditors	status, lattice class, barrier root, CAP point, ERRS band/score, digests, unsupported_state flags	detailed residual attack surface, undeclared-store hypotheses, sensitive debt narratives
SEALED_RESIDUAL_RISK	authorized auditors / controllers	ERRS component breakdown, coverage-game findings, quarantine rationale, debt detail commitments, bounty hits, model residual notes	unnecessary revoked plaintext

Binding rule: `PUBLIC.paired_sealed_certificate_id` and `SEALED.paired_public_certificate_id` MUST cross-commit. Verifying only the public half is allowed for operational trust; regulatory review SHOULD require both.

Asymmetry rule: sealing is not concealment of FAILED mandatory obligations. Failed mandatory items still force non-success on the public certificate.

10.10 Epistemic Residual Risk Score — ERRS (invention obligation)

`ERRS` $\in [0,1]$
`band` $\in \{\text{LOW, MODERATE, ELEVATED, HIGH, CRITICAL}\}$

Reference formula `errs.v1` (normative for pilots; weights policy-tunable):

```
ERRS = clamp01(
  w1 * undeclared_coverage
+ w2 * model_residual
+ w3 * debt_open
+ w4 * partition_risk
+ w5 * ghost_residual
+ w6 * causal_hypothesis_mass
+ w7 * instrumentation_gap
)
```

Default pilot weights: `w` = (0.25, 0.20, 0.20, 0.10, 0.10, 0.10, 0.05).

Rules:

1. ERRS is computed by the lattice composer after verification.
2. Public certificate carries score + band; sealed certificate carries components.
3. `band` $\geq$ HIGH forbids marketing language of “complete erasure.”
4. CAP operating points that maximize A+P MUST expect higher baseline ERRS.
5. Challenges MAY request `errs_recompute` from commitments.

11. DATA MODEL

11.1 Core entities

Entity	Critical fields
<code>rollback_intent</code>	<code>id</code> , <code>requester</code> , <code>authority</code> , <code>purpose</code> , <code>target selector</code> , <code>policy basis</code> , <code>status</code> , <code>requested assurance</code>

Entity	Critical fields
policy_snapshot	version, digest, jurisdiction tags, retention exceptions, approvers, semantic τ thresholds
domain_registry	domain id, owner, trust class, endpoint, signing keys, capability digest, challenge endpoint
artifact_node	artifact id, type, tenant, domain, epoch, content commitment, state
influence_edge	source, destination, type, edge_kind, strength, strength_method, policy_class, evidence reference
instrumentation_sigma	producer ids, completeness class, digest, attested_at
replica_inventory	artifact/domain, replica kind, location class, state, last verified
epoch_treaty	treaty id, intent, epoch, mandatory set, leases, fencing tokens, coverage attestations, refusal log, digest
coverage_game	game id, intent, probes, findings, digest, status
coverage_bounty	bounty id, scope, reward commitment, evidence, payout status
cap_operating_point	profile id, C/A/P levels, justification
rollback_plan	closure digest, quarantine digest, frontier digest, frontier sketch, barrier schedule, ordered steps, mandatory domains, cost estimate
rollback_step	adapter, operation, preconditions, postconditions, timeout, compensation, requires_barrier, parallel_class
approval_record	approver role, decision, scope, timestamp, signature
adapter_receipt	step, pre-root, post-root, code digest, treaty digest, barrier root, status, signature
verification_result	obligation, method, evidence, result, confidence, limits
ghost_probe_result	probe type, index digest, result, threshold, signature
backflow_barrier	epoch, intent, excision receipt set, barrier root, sealed_at, schedule digest
epistemic_debt	debt id, effect commitment, severity, compensation status, fence rules, review_at
errs_record	intent, score, band, components, formula_version
public_certificate	assurance-facing certificate fields (§10.3)
sealed_residual_certificate	auditor-facing residual fields (§10.9)
rollback_certificate	aggregate digests, assurance classes, lattice compose trace, debts, ERRS, CAP point, challenge window, signatures
certificate_challenge	challenge id, certificate id, nonce, obligations, response status
epoch_stream	stream id, domain, watermark, sealed prefixes
recontamination_rule	revoked lineage, blocked epochs, ingress action, debt linkage
recontamination_event	detected artifact, source, action, linked intent
outbox_event	effect id, destination, payload commitment, status
inbox_dedup	source, message id, processed epoch
exception_record	legal hold, technical blocker, risk acceptance, expiry

11.2 Canonical identifiers

```

DERF-INTENT-<ULID>
DERF-PLAN-<ULID>
DERF-STEP-<ULID>
DERF-RCP-<ULID>
DERF-VER-<ULID>
DERF-CERT-<ULID>
DERF-INC-<ULID>
DERF-TREATY-<ULID>
DERF-GAME-<ULID>
DERF-DEBT-<ULID>
DERF-CHAL-<ULID>
DERF-BARRIER-<ULID>

```

ULIDs are only ordering-friendly identifiers; trust comes from signatures and commitments.

11.3 Canonical serialization

- deterministic JSON profile or deterministic CBOR;
- UTF-8;
- UTC timestamps;
- sorted map keys;
- stable array ordering where order is semantic;
- explicit decimal and floating-point rules;

- hash algorithm and domain-separation label included in every digest;
- no ambiguous null/absent semantics.

12. API AND EVENT CONTRACTS

12.1 External API

```
POST /v1/rollback-intents
GET /v1/rollback-intents/{intent_id}
POST /v1/rollback-intents/{intent_id}/discover
POST /v1/rollback-intents/{intent_id}/treaty
POST /v1/rollback-intents/{intent_id}/coverage-game
POST /v1/rollback-intents/{intent_id}/plan
POST /v1/rollback-intents/{intent_id}/approve
POST /v1/rollback-intents/{intent_id}/prepare
POST /v1/rollback-intents/{intent_id}/execute
POST /v1/rollback-intents/{intent_id}/barrier/seal
POST /v1/rollback-intents/{intent_id}/verify
POST /v1/rollback-intents/{intent_id}/debts
POST /v1/rollback-intents/{intent_id}/commit
POST /v1/rollback-intents/{intent_id}/abort
GET /v1/rollback-certificates/{certificate_id}
GET /v1/rollback-certificates/{certificate_id}/sealed
POST /v1/rollback-certificates/{certificate_id}/challenge
GET /v1/rollback-certificates/{certificate_id}/challenges/{challenge_id}
GET /v1/rollback-intents/{intent_id}/errs
POST /v1/coverage-bounties
POST /v1/coverage-bounties/{bounty_id}/claims
GET /v1/epistemic-debts/{debt_id}
POST /v1/epistemic-debts/{debt_id}/review
GET /v1/recontamination-events
POST /v1/recontamination-events/{event_id}/resolve
POST /v1/domains/{domain_id}/treaty/ack
POST /v1/domains/{domain_id}/treaty/refuse
```

12.2 Mandatory request controls

- authenticated principal;
- tenant and domain scope;
- idempotency key;
- target selector;
- requested assurance;
- declared urgency;
- policy/legal basis identifier;
- human-approval state;
- retention or legal-hold exceptions;
- maximum acceptable blast radius;
- dry-run option.

12.3 Domain protocol

The production protocol should use versioned gRPC or equivalent authenticated RPC:

```
DescribeCapabilities
PrepareRollback
ExecuteRollback
RebuildProjection
VerifyRollback
CommitRollback
AbortRollback
GetRollbackStatus
ChallengeReceipt
```

12.4 Event catalog

```
derf.intent.created
derf.intent.authorized
derf.discovery.completed
derf.plan.created
derf.domain.prepared
derf.step.started
```

```
derf.step.completed
derf.step.failed
derf.verification.completed
derf.rollback.committed
derf.rollback.aborted
derf.rollback.partial_quarantine
derf.certificate.issued
derf.recontamination.detected
derf.exception.expired
```

Events contain identifiers and commitments, never raw revoked content.

12.5 Structured CORE API (v2.3.0)

The endpoints below are the **minimum CORE surface** for contamination handling, excision, replay, frontier inspection, and recontamination rejection. They complement the full rollback-intent lifecycle in §12.1. All are **specified contracts only** unless backed by `poc/derf_poc.py` evidence.

Core data types (TypeScript-style)

```
/** A knowledge item with provenance and contamination state */
interface KnowledgeItem {
  /** Stable identifier, e.g. "K1_raw_observation" */
  id: string;
  /** Owning agent or domain */
  agent_id: string;
  /** Human-readable claim text (may be redacted in certificates) */
  claim: string;
  /** Direct upstream item IDs forming the influence edge set */
  source_ids: string[];
  /** True if derived from other items */
  derived: boolean;
  /** True if in causal closure of a contaminated root */
  contaminated: boolean;
  /** True if removed during barrier-ordered excision */
  excised: boolean;
  /** True if source was formally retracted */
  retracted: boolean;
}

/** Request to mark a root item contaminated and compute closure */
interface ContaminateRequest {
  /** Target knowledge item ID */
  item_id: string;
  /** Retraction reason code */
  reason: "SOURCE_RETRACTED" | "POISONED_DATASET" | "HALLUCINATION" | "LEGAL_ERASURE";
  /** Domain scope for epoch fencing */
  domain_id: string;
  /** Idempotency key */
  idempotency_key: string;
  /** Optional human-approval reference */
  approval_ref?: string;
}

/** Ordered excision plan with barrier classes */
interface ExcisionPlan {
  plan_id: string;
  intent_id: string;
  /** Items in barrier-ordered removal sequence */
  excision_order: string[];
  /** Items quarantined as UNKNOWN-path */
  quarantined: string[];
  /** State-class barrier roots sealed before model mitigation */
  barrier_roots: string[];
  /** Assurance class target (DERF-A1..A6) */
  assurance_target: string;
}

/** Result of clean-frontier replay */
interface ReplayResult {
  replay_id: string;
  /** Items successfully re-derived from clean sources */
  rederived: KnowledgeItem[];
  /** Items that cannot be re-derived; epistemic debt candidates */
  unreplayable: string[];
  /** Replay equivalence check passed */
  replay_sound: boolean;
}
```



```
/** Digest of post-replay frontier */
frontier_digest: string;
}

/** Snapshot of the clean knowledge frontier */
interface FrontierSnapshot {
  snapshot_id: string;
  timestamp_utc: string;
  /** Active non-excised items */
  active_items: KnowledgeItem[];
  /** Count of items by agent */
  agent_counts: Record<string, number>;
  /** Global weakest-assurance grade across domains */
  weakest_assurance: string;
  /** Whether commit is permitted */
  commit_permitted: boolean;
}

/** Record of a blocked re-injection attempt */
interface RecontaminationRejection {
  event_id: string;
  /** Rejected item ID or fingerprint */
  rejected_item_id: string;
  /** Ingress channel that attempted injection */
  channel: string;
  /** Matching rule that fired */
  rejection_rule: "SOURCE_FINGERPRINT" | "PROVENANCE_CHAIN" | "EPOCH_FENCE" | "BARRIER_SEAL";
  /** Prior contamination root that triggered block */
  contamination_root: string;
  blocked_at_utc: string;
}
```

POST /contaminate

Mark a knowledge item as contaminated and compute causal closure.

Field	Value
Method	POST
Path	/v1/core/contaminate
Request	ContaminateRequest
Response	{ closure: string[]; quarantined: string[]; intent_id: string }
Errors	404 ITEM_NOT_FOUND, 409 ALREADY_CONTAMINATED, 422 UNKNOWN_PATH_UNRESOLVED

Example request:

```
{
  "item_id": "DS-2024-ALPHA",
  "reason": "POISONED_DATASET",
  "domain_id": "hospital-alpha",
  "idempotency_key": "contam-2026-07-16-001"
}
```

Example response:

```
{
  "intent_id": "ri-7f3a9c",
  "closure": ["DS-2024-ALPHA", "EMB-4412", "SUM-8891", "PLAN-220", "DEC-991"],
  "quarantined": ["CACHE-77"]
}
```

POST /excise

Execute barrier-ordered excision for a computed closure.

Field	Value
Method	POST
Path	/v1/core/excise
Request	{ intent_id: string; dry_run?: boolean }
Response	ExcisionPlan

Field	Value
Errors	404 INTENT_NOT_FOUND, 412 BARRIER_NOT_SEALED, 423 EPOCH_FENCE_VIOLATION

Example response:

```
{
  "plan_id": "xp-aa01",
  "intent_id": "ri-7f3a9c",
  "excision_order": ["DEC-991", "PLAN-220", "SUM-8891", "EMB-4412", "DS-2024-ALPHA"],
  "quarantined": ["CACHE-77"],
  "barrier_roots": ["IDX-embeddings-alpha"],
  "assurance_target": "DERF-A4"
}
```

POST /replay

Re-derive knowledge from uncontaminated sources on the clean frontier.

Field	Value
Method	POST
Path	/v1/core/replay
Request	{ intent_id: string; retained_sources?: string[] }
Response	ReplayResult
Errors	404 INTENT_NOT_FOUND, 409 EXCISION_INCOMPLETE, 422 REPLAY_EQUIVALENCE_FAIL

Example response:

```
{
  "replay_id": "rp-bb02",
  "rederived": [
    {
      "id": "SUM-8891-CLEAN",
      "agent_id": "agent_synthesiser",
      "claim": "Summary from verified alternate corpus",
      "source_ids": ["ALT-CORPUS-12"],
      "derived": true,
      "contaminated": false,
      "excised": false,
      "retracted": false
    }
  ],
  "unreplayable": ["DEC-991"],
  "replay_sound": true,
  "frontier_digest": "sha256:8c4f2e..."
}
```

GET /frontier

Return the current clean knowledge frontier snapshot.

Field	Value
Method	GET
Path	/v1/core/frontier
Query	?domain_id=hospital-alpha&intent_id=ri-7f3a9c
Response	FrontierSnapshot
Errors	404 DOMAIN_NOT_FOUND, 409 COMMIT_BLOCKED_DEBT

Example response:

```
{
  "snapshot_id": "fs-cc03",
  "timestamp_utc": "2026-07-16T14:22:11Z",
  "active_items": [],
  "agent_counts": { "agent_source": 2, "agent_decision": 0 },
  "weakest_assurance": "DERF-A4",
  "commit_permitted": false
}
```

POST /reject-recontamination

Attempt ingress of a retracted item; system must reject and log.

Field	Value
Method	POST
Path	/v1/core/reject-recontamination
Request	{ item: KnowledgeItem; channel: string; intent_id: string }
Response	RecontaminationRejection
Errors	200 (rejection logged), 500 INGRESS_NOT_BLOCKED (critical failure)

Example response:

```
{
  "event_id": "rc-dd04",
  "rejected_item_id": "DS-2024-ALPHA",
  "channel": "secondary_rag_ingest",
  "rejection_rule": "SOURCE_FINGERPRINT",
  "contamination_root": "DS-2024-ALPHA",
  "blocked_at_utc": "2026-07-16T14:25:03Z"
}
```

13. PRIMARY WORKFLOWS

13.1 Data-subject erasure

1. authenticate requester and verify identity using a separate privacy-safe process;
2. determine controller/processor roles and applicable exceptions;
3. map the request to canonical target selectors;
4. discover recipients, domains, replicas, backups, and model exposure;
5. compute closure and classify each state type;
6. obtain required legal and operational approvals;
7. execute the protocol;
8. issue internal and data-subject-facing status outputs appropriate to policy;
9. notify required recipients;
10. monitor restoration and recontamination.

13.2 Hallucination rollback

1. register disputed claim;
2. trace decisions and memories derived from it;
3. quarantine high-risk descendants immediately;
4. determine authoritative correction;
5. replace or remove the claim;
6. replay affected plans from clean state;
7. compensate reversible effects;
8. flag irreversible effects for human remediation;
9. publish a correction receipt without exposing restricted content.

13.3 Poisoned RAG document

1. isolate document and associated embeddings;
2. find all retrievals, summaries, memories, plans, and actions derived from it;
3. rebuild affected indexes;
4. invalidate checkpoints;
5. replay workflows with clean corpus snapshot;
6. add source fingerprint to ingress denial rules;
7. run prompt-injection and retrieval-poison regression tests.

13.4 Model contribution removal

1. resolve exact training, fine-tuning, adapter, and evaluation lineage;
2. select assurance target based on supported algorithm;
3. execute mandatory memory/index/graph/cache excision first;
4. seal backflow barrier root;
5. prefer adapter removal or retained-data reconstruction when feasible;

6. run model-specific unlearning only after barrier clearance;
7. evaluate forget set, retain set, utility, extraction, membership, paraphrase, and jailbreak behavior;
8. issue only DERF-A5 unless exact or formally bounded reconstruction is proven (DERF-A6).

13.5 Backup restoration

1. restore into isolated environment;
2. load latest revocation ledger, treaties, and policy snapshot;
3. replay all post-backup committed intents;
4. rebuild affected projections;
5. verify recontamination rules and open epistemic debts;
6. issue restoration certificate;
7. allow traffic only after gate pass.

13.6 Federated epoch treaty rollback

1. publish capability manifests across organizations;
2. negotiate epoch treaty with mandatory/optional sets and refusal codes;
3. run coverage completeness game across declared scope;
4. execute excision under treaty fencing;
5. seal barrier before any cross-org model mitigation;
6. compose lattice assurance across organizations;
7. issue per-domain receipts plus aggregate certificate;
8. keep challenge endpoints live for the challenge window.

13.7 Irreversible effect with epistemic debt

1. detect irreversible or partially compensable external effect in closure;
2. compensate what is reversible;
3. create epistemic debt object with severity and fence rules;
4. notify required humans/controllers;
5. continue commit only as COMMITTED_WITH_DEBT or PARTIAL;
6. schedule debt review; never market as full rollback.

13.8 Certificate challenge after commit

1. challenger submits nonce-bound challenge for selected obligations;
2. offline verifier checks commitments and lattice compose;
3. interactive domains re-run negative retrieval / ghost probes as requested;
4. FAIL opens recontamination or certificate-invalid incident;
5. PASS appends challenge transcript digest to evidence log.

13.9 Dual-certificate + ERRS issuance

1. compose lattice assurance and CAP operating point;
2. compute ERRS components and band;
3. issue PUBLIC_ASSURANCE certificate;
4. issue SEALED_RESIDUAL_RISK certificate with cross-commit ids;
5. publish public score/band only; restrict sealed components;
6. refuse "complete erasure" UX if band $\geq$ HIGH or status is COMMITTED_WITH_DEBT.

13.10 Blind-target federated erase (optional)

1. select DERF-FED-MPC profile;
2. commit/share target selector;
3. domains match via PSI/commitment without plaintext reveal;
4. collect receipts bound to commitment;
5. seal barrier and dual certificates as usual;
6. record higher partition/coverage ERRS components if matching recall is bounded.

14. SECURITY AND THREAT MODEL

14.1 Protected assets

- personal and confidential data;
- provenance and influence graph;
- semantic edge strength evidence;

- rollback authority;
- epoch treaties and fencing tokens;
- signing and encryption keys;
- policy bundles;
- clean checkpoints;
- backflow barrier roots;
- receipts and certificates;
- epistemic debt records;
- challenge transcripts;
- adapter binaries;
- verifier reference values;
- model and evaluation artifacts.

14.2 Threat actors

- malicious requester attempting destructive erasure;
- compromised domain or adapter;
- insider hiding a replica;
- stale agent or worker;
- malicious verifier;
- external attacker forging receipts;
- tenant attempting cross-tenant rollback;
- operator misclassifying a legal hold;
- dependency or build-chain compromise;
- prompt injection embedded in retrieved material.

14.3 Principal threats and controls

Threat	Control
Forged request	strong authentication, scoped authorization, dual approval for high-risk intents
Cross-tenant target expansion	tenant-bound selectors, policy checks, blast-radius preview
Forged receipt	signed receipts, key rotation, independent challenge
Hidden replica	signed coverage declaration, coverage completeness game, discovery scans, contractual and audit controls
Stale replay	monotonic epoch, fencing token, restore hook, epoch treaty
Rollback storm	quotas, batching, prioritization, circuit breakers
Coordinator compromise	least privilege, append-only event ledger, independent verifier
Malicious adapter	signed artifacts, sandboxing, reproducible build, optional attestation
Proof replay	nonce, epoch, intent digest, expiry
Key-destruction false claim	HSM/KMS audit, dual control, coverage validation
Prompt injection	retrieved content treated as data; instruction fencing; no remote policy override
Certificate privacy leak	content-free commitments and selective disclosure
Irreversible external effect	staging/outbox, compensation, epistemic debt, human repair
Model false-negative	diverse attacks, holdout prompts, repeated tests, residual-risk disclosure
Supply-chain compromise	pinned dependencies, SBOM, provenance, vulnerability scan, rollback
Barrier bypass / backflow	backflow barrier root required before model receipts; out-of-band fine-tune detection hooks
Ghost-vector residual	DERF-A2-G probes; class downgrade on failure
Semantic under-closure	UNKNOWN quarantine; false-negative bounds; soft-edge review
Challenge DoS	rate limits, offline-first challenges, challenge windows
Treaty refusal spoofed as success	coded refusals; HTTP 200 without signed PREPARED is failure

14.4 Key hierarchy

Root of Trust
└─ Certificate Signing Keys
└─ Domain Receipt Keys
└─ Policy Signing Keys
└─ Data Protection Key Encryption Keys
└─ Epoch / Subject / Dataset Scoped Data Encryption Keys

No single key destruction event may be represented as erasing data outside its declared encryption coverage.

14.5 Safety controls

- dry-run plan before destructive execution;
 - maximum closure and domain limits;
 - no mass rollback without elevated approval;
 - pause and resume;
 - break-glass with post-event review;
 - immutable intent metadata but erasable sensitive evidence;
 - reversible migration first;
 - automated abort on inconsistent post-state roots;
 - human review for employment, finance, medical, safety, public-sector, or other high-impact effects.
-

15. PRIVACY, LEGAL, AND GOVERNANCE MAPPING

This section is a technical requirements map, not legal advice or certification.

15.1 GDPR

DERF should support:

- data minimization, accuracy, purpose limitation, storage limitation, integrity, and accountability;
- lawful-basis and consent-state tracking;
- transparent request intake;
- rectification, restriction, objection, access, portability, and erasure workflows where applicable;
- Article 17 erasure analysis, including exceptions;
- Article 19 recipient notification where required;
- records of processing, DPIA support, and processor instructions;
- response deadlines and human review configured by legal counsel;
- evidence that does not unnecessarily retain personal data.

An AI model trained with personal data cannot automatically be treated as anonymous. Extractability and identifiability require case-specific assessment.

15.2 EU AI Act

When DERF is itself part of, or governs, an AI system within scope, the implementation should map to applicable requirements such as:

- risk management;
- data and data-governance controls;
- technical documentation;
- logging and record keeping;
- transparency and instructions;
- human oversight;
- accuracy, robustness, and cybersecurity;
- post-market monitoring and corrective actions.

The applicability and implementation timeline must be verified at deployment because the regulatory framework and simplification measures are evolving.

15.3 Other governance mappings

Optional mappings:

- NIST AI Risk Management Framework;
- NIST Cybersecurity Framework;
- NIST SP 800-88 Rev. 2 media sanitization;
- ISO/IEC 27001 and 27701 control mappings;
- ISO/IEC 42001 AI management;
- sector-specific retention, legal-hold, medical, financial, employment, and public-sector requirements.

15.4 Immutable audit versus erasure

DERF resolves the apparent conflict by separating:

1. **operational content**, which may be erased or reconstructed;
2. **audit facts**, which may be retained when lawfully necessary;
3. **content commitments**, which prove event integrity without storing the content;
4. **encrypted evidence**, which can be selectively destroyed;
5. **legal-hold exceptions**, which are explicit and time-bounded.

A WORM log must not become a permanent repository for raw personal payloads.

16. RELIABILITY, CONCURRENCY, AND RECOVERY

16.1 Consistency model

- strongly consistent intent and epoch state;
- per-domain fencing;
- eventual convergence of replicas with explicit timeout;
- deterministic replay for supported workflows;
- no global commit during partition unless all mandatory evidence is present.

16.2 Failure handling

Failure	Required behavior
Coordinator crash	recover state from durable event history
Adapter timeout	retry within budget; then abort or quarantine
Duplicate message	deduplicate by intent, epoch, step, idempotency key
Network partition	retain quiescence or abort according to lease policy
Verification failure	block commit
Partial irreversible effect	MANUAL_REPAIR_REQUIRED
Lost signing key	revoke key, challenge affected receipts, reissue where possible
Replica discovered after commit	reopen intent as recontamination
Policy change mid-run	pin original policy or abort and re-plan
Clean replay nondeterminism	capture divergence; do not claim causal reconstruction

16.3 Three-strike rule

If the same adapter or verification failure repeats more than twice without new evidence, stop automated retries and require a diagnostic decision.

16.4 Rollback of the rollback

Every plan defines:

- abort boundary;
- recoverable pre-state;
- compensation steps;
- key-destruction irreversibility;
- post-commit correction procedure;
- state that cannot be safely restored.

Cryptographic erasure is intentionally irreversible and therefore requires stronger approval.

17. OBSERVABILITY AND EVIDENCE

17.1 Telemetry

Required metrics:

- intent rate and type;
- closure node and edge count;
- discovery coverage;
- mandatory-domain readiness;
- step latency and retries;
- quiescence duration;
- rebuild duration;
- verification duration;
- certificate size and verification latency;
- replica convergence;
- recontamination rate;
- false-positive invalidation;
- model utility and forgetting metrics;

- CPU, memory, I/O, and energy proxies.

17.2 Logs

Logs must include:

```
trace_id
intent_id
plan_id
epoch
domain_id
step_id
adapter_id
policy_snapshot_digest
status
error_code
duration_ms
evidence_reference
```

Logs must exclude raw target content, secrets, tokens, and unnecessary subject identifiers.

17.3 Tracing

OpenTelemetry-compatible spans:

```
derf.intent.authorize
derf.discovery.enumerate
derf.graph.resolve_target
derf.graph.compute_closure
derf.plan.compile
derf.domain.prepare
derf.adapter.execute
derf.replay.run
derf.verify.obligation
derf.certificate.build
derf.recontamination.scan
```

17.4 Evidence receipts

Minimum receipt types:

- AUTHORIZATION_RECEIPT
- COVERAGE_DECLARATION
- CAUSAL_CLOSURE_RECEIPT
- PREPARE_RECEIPT
- EXECUTION_RECEIPT
- REBUILD_RECEIPT
- VERIFICATION_RECEIPT
- KEY_DESTRUCTION_RECEIPT
- MODEL_EVALUATION_RECEIPT
- COMMIT_RECEIPT
- RECONTAMINATION_RULE_RECEIPT
- RESTORE_REPLAY_RECEIPT

18. PERFORMANCE AND ENERGY DESIGN

18.1 Performance strategy (invention-grade)

Performance work in DERF is not mere tuning. The following are **protocol inventions with measurable obligations**:

1. **Incremental causal closure (§6.13)** — $O(\Delta)$ updates with T-INC-1 fallback to full recompute.
2. **Barrier-parallel schedules (§6.17)** — maximize concurrency without violating T-BF-1.
3. **Frontier sketches (§6.18)** — replay eligibility without full checkpoint loads.
4. **Sharded treaty quorum (§6.18)** — sub-linear fan-out vs unanimous ack when CAP allows.
5. **Hot/archival lineage tiers** — recent epochs hot; sealed prefixes cold.
6. **Async proof generation** — expensive ghost/model proofs off the critical path when policy permits PARTIAL then upgrade.
7. **Partitioned graphs** — tenant $\times$ domain $\times$ modality $\times$ epoch.

18.2 Initial engineering budgets

These are pilot budgets, not proven performance claims:

- lineage instrumentation overhead: **less than 15% p95 latency increase** on instrumented test flows;
- incremental closure vs full recompute: **$\geq 5\times$ faster** for $|\Delta E|/|E| \leq 0.01$ on reference fixtures;
- barrier-parallel schedule: **$\geq 1.5\times$ wall-time reduction** vs forced serial pre-barrier steps when ≥ 3 independent adapters;
- frontier sketch decision: **less than 50 ms** p95 for 10k checkpoint roots on reference hardware;
- certificate verification (public half): **less than 1 second** for 100 domain receipts;
- dual-certificate issue: **less than $1.5\times$** single-certificate cost;
- ERRS compute: **less than 100 ms** given component inputs;
- deterministic closure: identical digest across 10 repeated runs;
- no unbounded graph traversal;
- no certificate or evidence artifact above configured size caps.

18.3 ERRS as performance+assurance KPI

Track ERRS alongside latency:

```
rollback_wall_ms
barrier_seal_ms
closure_ms (incremental|full)
ghost_probe_ms
treaty_rtt_ms
errs_score
errs_band
cap_profile_id
```

A faster rollback that raises ERRS band is not a win unless the CAP operating point explicitly prioritizes Availability.

18.4 Energy controls

- incremental rather than full graph scans;
- rebuild only affected partitions and modalities;
- schedule non-urgent compaction and model tests during lower-cost windows;
- record CPU time, storage I/O, accelerator time, and proof-generation time;
- choose cryptographic/MPC proof only when its assurance value exceeds its resource cost;
- maintain a low-energy external-state-only profile;
- prefer frontier sketches over full checkpoint hydration when T-CF-1 observation algebra allows.

18.5 Benchmark suite (required for any performance claim)

Claims of “fast DERF” are forbidden without publishing:

1. fixture graph size ($|V|$, $|E|$, modalities);
2. ΔE ratio for incremental tests;
3. adapter count and parallel schedule;
4. CAP profile;
5. ERRS before/after;
6. hardware class;
7. whether ghost probes and model tests were included in wall time.

19. REFERENCE IMPLEMENTATION AND OPEN-SOURCE SCREEN

19.1 Recommended MVP architecture

MVP form: modular monolith plus isolated adapters.

- API and coordinator: Python or Go selected after repository audit;
- PostgreSQL: intent state, graph metadata, event ledger, receipts;
- pgvector: small pilot vector adapter;
- Temporal: durable orchestration candidate;
- OPA: policy-engine candidate;
- OpenTelemetry: telemetry;
- object store: S3-compatible local or cloud implementation;
- model-evaluation worker: Python;
- admin console: TypeScript;
- offline verifier: standalone CLI.

19.2 Candidate classification

Candidate	Role	License signal	Initial status	Required checks
PostgreSQL	control store	PostgreSQL License	APPROVED_FOR_PILOT	version pin, backup/restore test
pgvector	pilot vector store	PostgreSQL License	ISOLATE_AND_TEST	delete/compaction/raw-storage tests
Temporal	durable workflow	MIT	APPROVED_FOR_PILOT	operational footprint, replay semantics, version pin
LangGraph	optional agent checkpoint adapter	MIT	ISOLATE_AND_TEST	checkpoint behavior, migration, dependency scan
Qdrant	vector adapter	Apache-2.0	ISOLATE_AND_TEST	deletion/optimizer behavior, raw segment test
Milvus	scale vector adapter	Apache-2.0	ISOLATE_AND_TEST	compaction, dropped segments, restore behavior
OPA	policy evaluation	Apache-2.0	APPROVED_FOR_PILOT	policy test suite, fail-closed integration
OpenTelemetry	telemetry	Apache-2.0	APPROVED_FOR_PILOT	redaction and cardinality limits
in-toto	build/execution provenance	Apache-2.0 implementation	NEXT_RELEASE	integration path and signing
CycloneDX	SBOM format	Apache-2.0 schemas	APPROVED_FOR_PILOT	generator and validation
SLSA provenance	build provenance profile	specification	NEXT_RELEASE	CI capability and provenance verification
Keylime/RATS tooling	attestation	varies by component	RESEARCH_ONLY	platform support, trust roots, threat model
ZK frameworks	optional proof generation	not yet screened	BLOCKED_PENDING_REVIEW	license, circuit fit, performance, audit

No dependency may enter the core path on popularity alone. The build agent must inspect the actual version, license, provenance, maintenance status, security advisories, transitive dependencies, install scripts, test coverage, and rollback path.

19.3 Why a modular monolith first

A distributed protocol can be modeled and tested without prematurely deploying many microservices. The MVP should isolate adapters behind interfaces, retain durable workflow boundaries, and prove the state machine before introducing cross-cluster operational complexity.

20. REPOSITORY BLUEPRINT

```
/derf
  README.md
  LICENSE
  SECURITY.md
  CONTRIBUTING.md
  CODEOWNERS
  pyproject.toml | go.mod | package.json
  lockfiles
  /docs
    architecture.md
    protocol.md
    assurance-classes.md
    assurance-lattice.md
    epoch-treaties.md
    backflow-barrier.md
    epistemic-debt.md
    certificate-challenges.md
    threat-model.md
    legal-requirements-map.md
    operator-guide.md
    adapter-authoring-guide.md
    certificate-verification.md
  /apps
    /api
    /operator-console
    /offline-verifier
  /services
    /intent-gateway
    /policy
    /discovery
    /lineage
    /epoch-treaty
    /coverage-game
    /planner
    /coordinator
    /backflow-barrier
    /replay
    /verifier
```

```
/ghost-probes
/lattice-composer
/certificate
/challenge
/epistemic-debt
/recontamination
/packages
/protocol
/canonical-serialization
/crypto
/assurance-lattice
/sdk
/test-fixtures
/adapters
/postgres
/pgvector
/agent-checkpoint
/object-store
/cache
/knowledge-graph
/qdrant
/milvus
/model-evaluation
/external-effects
/schemas
rollback-intent.schema.json
capability-manifest.schema.json
influence-node.schema.json
influence-edge.schema.json
epoch-treaty.schema.json
coverage-game.schema.json
rollback-plan.schema.json
adapter-receipt.schema.json
backflow-barrier.schema.json
verification-result.schema.json
ghost-probe-result.schema.json
epistemic-debt.schema.json
certificate-challenge.schema.json
rollback-certificate.schema.json
/policies
authorization
retention
assurance
semantic-thresholds
legal-hold
blast-radius
debt-severity
/migrations
/tests
/unit
/property
/contract
/integration
/e2e
/adversarial
/chaos
/performance
/privacy
/golden
/lattice
/barrier
/ghost
/challenge
/evidence
README.md
templates
/deploy
/local
/containers
/kubernetes
/ci
schema-validation
tests
security
sbom
provenance
release-gates
```

21. TEST AND VALIDATION STRATEGY

21.1 Unit tests

- canonical serialization;
- identifier and epoch validation;
- policy evaluation;
- closure computation under partial observability;
- quarantine set formation;
- semantic edge classification;
- cycle and supersession handling;
- topological plan generation with barrier ordering;
- receipt signing and verification;
- state transitions including treaty/game/barrier/debt;
- idempotency;
- assurance lattice compose and no-promotion;
- epistemic debt severity gates.

21.2 Property tests

- closure monotonicity for added hard-dependency edges;
- UNKNOWN high-risk paths always quarantine or block;
- no committed state with failed mandatory obligation;
- duplicate messages do not duplicate effects;
- lower epochs are rejected;
- identical inputs produce identical digests;
- certificate tampering always fails verification;
- an exception cannot silently disappear;
- model steps cannot precede barrier seal;
- lattice compose equals min of mandatory classes;
- silent promotion is rejected;
- HIGH/CRITICAL debt forbids clean COMMITTED;
- challenge responses never contain revoked plaintext.

21.3 Contract tests

Every adapter runs the same conformance pack:

1. capability declaration validates;
2. prepare is side-effect-safe;
3. execute is idempotent;
4. abort behavior is defined;
5. postconditions are machine-checkable;
6. replicas are enumerated;
7. unsupported state is explicit;
8. stale epoch is rejected;
9. receipt is signed and canonical;
10. restore behavior is tested;
11. ghost probes declared or unsupported for vector adapters;
12. treaty ack/refuse coded correctly;
13. barrier root required for model adapters;
14. debt emission available for irreversible effect adapters.

21.4 Integration tests

- PostgreSQL plus pgvector plus agent checkpoint;
- write quiescence and fencing;
- epoch treaty prepare path;
- coverage-game reopen path;
- backflow barrier seal before model stub;
- crash between every protocol state;
- partial domain failure;
- read replica delay;
- object versioning;
- backup restore and revocation replay;
- clean replay with external effect outbox;
- epistemic debt commit path;
- certificate offline verification and offline challenge.

21.5 Adversarial tests

- forged requester;
- forged domain receipt;
- malicious capability manifest;
- hidden/stale replica fixture;
- coverage-game undeclared store fixture;
- prompt-injected tool output;
- barrier bypass attempt;
- ghost-vector reconstruction attempt;
- silent assurance promotion attempt;
- challenge response plaintext leak attempt;
- vector recovery from raw index;
- embedding inversion test where authorized;
- checkpoint resurrection;
- policy downgrade;
- attestation replay;
- target-set expansion attack;
- deletion-pattern privacy analysis;
- model paraphrase, multilingual, jailbreak, and indirect recall;
- membership and extraction evaluation;
- utility degradation and bias after unlearning.

21.6 Chaos tests

- coordinator termination;
- duplicate and reordered messages;
- network partition;
- clock skew;
- database failover;
- KMS outage;
- disk exhaustion;
- verifier outage;
- stale worker reconnection;
- restore from old backup.

21.7 Security and supply-chain tests

- SAST and dependency scan;
- secret scan;
- SBOM generation;
- license policy;
- container and infrastructure scan;
- artifact signature and provenance validation;
- no unpinned production dependency;
- no install-time network script without review.

22. RELEASE GATES

Gate	Requirement	Failure result
G-DERF-01	Project identity unchanged or approved decision record	Block
G-DERF-02	Schemas and canonical serialization validate	Block
G-DERF-03	Closure digest deterministic across 10 runs	Block
G-DERF-04	Mandatory-domain failure cannot produce global commit	Block
G-DERF-05	Adapter conformance passes for all enabled adapters	Block
G-DERF-06	Crash recovery passes at every protocol state	Block
G-DERF-07	Stale epoch and stale replica are rejected	Block
G-DERF-08	Certificate tampering is detected	Block
G-DERF-09	Public certificate contains no prohibited payload	Block
G-DERF-10	Vector purge profile passes raw-storage validation	Block for A2 claim
G-DERF-11	Model evaluation supports claimed assurance	Block for A5/A6 claim
G-DERF-12	Backup restoration replays revocation ledger	Block

Gate	Requirement	Failure result
G-DERF-13	Recontamination fixture is detected and quarantined	Block
G-DERF-14	SBOM, license, provenance, and vulnerability gates pass	Block
G-DERF-15	Rollback and recovery runbook demonstrated	Block
G-DERF-16	Human legal/security approval for regulated deployment	Block
G-DERF-17	Assurance lattice compose and no-promotion tests pass	Block
G-DERF-18	Backflow barrier enforced before model mitigation	Block for A5/A6
G-DERF-19	Vector A2-G claims pass ghost probes or are downgraded	Block
G-DERF-20	HIGH/CRITICAL epistemic debt cannot render as clean COMMITTED	Block
G-DERF-21	Offline certificate challenge succeeds without revoked plaintext	Block
G-DERF-22	Epoch treaty refusal cannot be interpreted as success	Block
G-DERF-23	CAP operating point present; MAX_ALL rejected	Block
G-DERF-24	Dual certificates issued and cross-committed	Block
G-DERF-25	ERRS score+band computed; HIGH band blocks "complete erasure" UX	Block
G-DERF-26	Incremental closure matches full recompute on golden ΔE fixtures	Block for incremental claims

22.1 Allowed statuses

```

SPECIFICATION_ONLY
IMPLEMENTATION_IN_PROGRESS
PARTIAL_IMPLEMENTATION_NOT_OPERATIONAL
PILOT_CONFORMANCE_PASS
READY_FOR_SECURITY_REVIEW
READY_FOR_REGULATED_REVIEW
RELEASE_CANDIDATE
RELEASE_APPROVED
RELEASE_BLOCKED

```

PRODUCTION_READY is forbidden until implementation, deployment, operations, security, privacy, legal, and release evidence exists.

23. SEVEN-DAY FALSIFICATION PILOT

Day 1 — Protocol foundation

- repository scaffold;
- schemas including treaty, debt, barrier, challenge, lattice;
- canonical serialization;
- state machine with treaty/game/barrier/debt states;
- synthetic three-domain dataset.

Day 2 — Lineage and closure

- lineage collector with provenance + semantic strength fields;
- graph storage;
- partial-observability closure and quarantine sets;
- minimal sufficient clean-frontier algorithm;
- coverage-game checklist runner;
- deterministic hash tests.

Day 3 — Adapters

- PostgreSQL;
- pgvector with ghost-probe stubs;
- simple agent-checkpoint adapter;
- effect adapter that can emit epistemic debt;
- conformance tests including barrier-root requirement for model stub.

Day 4 — Coordinator

- epoch treaty (single-org);
- prepare/execute/barrier-seal/verify/commit;
- leases and fencing;

- outbox;
- crash recovery;
- lattice compose on commit.

Day 5 — Replay and certificate

- clean replay with observation-algebra digests;
- certificate builder v1.1;
- offline verifier + offline challenge path;
- public/private evidence separation;
- COMMITTED_WITH_DEBT path for irreversible synthetic effect.

Day 6 — Attack and recovery

- stale checkpoint;
- forged receipt;
- raw/tombstone vector recovery (ghost probes);
- barrier bypass attempt (must fail);
- partial domain failure;
- backup restore;
- recontamination;
- challenge FAIL path.

Day 7 — Benchmark and decision

- performance and energy baseline;
- residual-risk review;
- architecture revision;
- GO, REVISE, OR REJECT.

Pilot success criteria

- deterministic closure, quarantine, frontier, and certificate digests;
- no false global commit;
- lattice compose equals weakest mandatory class;
- silent class promotion is rejected by tests;
- CAP operating point present; MAX_ALL rejected;
- dual certificates cross-commit; sealed half not required for public verify;
- ERRS score+band present; HIGH band blocks clean-success UX;
- all active synthetic descendants removed, invalidated, quarantined, or reconstructed;
- raw/tombstone vector fixture cannot claim A2-G without passing probes;
- model stub cannot execute before barrier seal;
- irreversible effect yields debt, not clean COMMITTED;
- stale agent cannot rejoin;
- backup restore cannot resurrect revoked state;
- offline verifier detects tampering;
- offline challenge does not require revoked plaintext;
- unsupported model state remains explicitly unsupported;
- recovery after injected crashes is demonstrated.

Pilot rejection signals

- lineage completeness cannot be measured;
- mandatory domains can conceal coverage without detection or contractual controls;
- clean replay repeatedly diverges;
- vector or backup state remains recoverable under the claimed class;
- certificate verification requires revoked plaintext;
- barrier can be bypassed;
- HIGH debt still renders as full success;
- operational overhead exceeds acceptable value;
- model suppression is presented as unlearning;
- protocol complexity creates more risk than it removes.

23A. DERF-REALITY-GATE-1 (single authorized evidence uplift)

Change type: UPLIFT_SPEC (evidence plan) — **not** an architecture invention pack.

Current Real-Invention Readiness: ~95%. **Credible ceiling after independent replication:** 88%–90%.

23A.1 Objective

Build one proof-grade demonstrator showing DERF can remove or quarantine operational descendants of a revoked input across heterogeneous domains more completely than baselines, without falsely claiming universal erasure.

23A.2 Claim nucleus (AUTHORITATIVE — equals \$2.3 CORE; 7 elements)

```
partial-observability cross-domain causal closure with UNKNOWN quarantine
→ mandatory-domain epoch fencing
→ barrier-ordered state-class excision
→ clean-frontier reconstruction
→ weakest-assurance global commit
→ epistemic-debt treatment of irreversible effects
→ post-commit recontamination rejection
```

Dual certificates / ERRS are **DEPENDENT** evidence transport — not CORE claim elements. Certificate failure does not redefine the compressed core; it fails dependent conformance only.

23A.3 Required demonstrator (three domains)

- 1. PostgreSQL records and derived relational state.
- 2. Vector index (active, tombstoned, recoverable).
- 3. Agent checkpoint/memory store + simulated irreversible external effect.

Pipeline (CORE): revocation → closure → UNKNOWN quarantine → epoch fence → barrier-ordered excision → clean frontier → weakest-class commit → epistemic debt → recontamination rejection.

Dependent transport (optional for CORE proof): dual cert + ERRS. Seed 17 for public examples only (see Reality Gate Zero seed policy).

23A.4 Required baselines

Source deletion only; memory/index deletion only; provenance traversal without fencing; parameter-memory unlearning; semantic rollback/effect-fence (ACRFence-class).

23A.5 Fixtures (≥1,000 deterministic scenarios)

Direct/semantic descendants; shared artifacts; hidden replicas; vector tombstones; stale checkpoints; backup restore; out-of-band changes; irreversible side effects; partial domain failure; forged receipts; recontamination after commit.

23A.6 Acceptance gates

CORE technical-effect gates (co-primary):

Gate	Required
False-clean commit rate	SIGNED: observed = 0 AND one-sided 95% UCB ≤ 0.30% (n_eff ≈ 1,000 guidance)
Valid-clean commit preservation rate	SIGNED: one-sided 95% LCB ≥ 95% (always-quarantine/always-block fails utility)
Barrier-bypass / stale rejoin / recontamination	0
Irreversible effects	Debt — never false clean success

CORE secondary:

Gate	Required
Known descendant containment recall	≥95%
Hidden-replica detection (labeled)	≥90%
Replay-equivalence failure	Within pre-registered bound

DEPENDENT_EVIDENCE_GATES (not core technical-effect):

Gate	Required
Cert tamper detection / independent cert reproduction	100% when dual-cert layer enabled
Incremental vs full closure equivalence	Identical commit decision when incremental profile enabled

23A.7 Technical-effect claim

```
DERF reduces residual active descendants and false-clean rollback outcomes
relative to the strongest tested baseline,
```


while disclosing incomplete coverage and bounded operational overhead.

23A.8 Kill criteria

Narrow/reject if: closure recall fails; hidden replicas unbounded; clean replay regularly changes intended outcomes; barrier bypassable; certificates create false confidence; cost > prevented harm; strongest claim anticipated/obvious; model suppression is only "unlearning" evidence.

23A.9 Expected uplift (estimate)

Working runtime +15; benchmark + independent replication +10; technical-effect + IP +7; security/privacy/partner +5 → **~65% → 88%–90%** if all land.

23A.10 Adjacent art (do not claim ownership)

Agentic Unlearning / SBU; ESR; ACRFence; ForgetAgent; SagaLLM. Defensible surface = **ordered cross-domain protocol with measurable assurance**.

23A.11 Reality Gate Zero — COMPLETE evidence contract (embedded in this public research edition only)

Status: RG0_PASS_DOCUMENTATION — all fourteen contract objects below are frozen in this file. **Execution NOT started. Implementation NOT authorized by this edit.**

Readiness effect: 0 (remains ~51%).

Rule: Material CORE claim change after freeze → Gate restart. Thresholds may not change after sealed results.

CLAIM_FREEZE

```
{
  "project": "DERF",
  "ssot_version": "1.6.4",
  "core_claim_elements": [
    "partial_observability_cross_domain_causal_closure_with_UNKNOWN_quarantine",
    "mandatory_domain_epoch_fencing",
    "barrier_ordered_state_class_excision",
    "clean_frontier_reconstruction",
    "weakest_assurance_global_commit",
    "epistemic_debt_treatment_of_irreversible_effects",
    "post_commit_recontamination_rejection"
  ],
  "uniqueness_anchor": "NO GLOBAL CLEAN COMMIT UNDER UNRESOLVED EPISTEMIC DESCENDANCY",
  "dependent_features_excluded_from_core_proof": ["dual_certificates", "ERRS", "certificate_challenges",
  "incremental_closure_equivalence", "ghost_recoverability_A2G"],
  "research_extensions_excluded_from_gate": ["MPC_blind_target", "coverage_bounty", "SPIFFE", "ZK_TEE", "frontier_sketches",
  "sharded_treaty_quorum"],
  "claim_change_after_freeze": "REQUIRES_RESTART",
  "readiness_pct_at_freeze": 51
}
```

CLAIM_TO_EVIDENCE_MATRIX

```
{
  "project": "DERF",
  "status_all": "NOT_RUN",
  "matrix": [
    {
      "element": "partial_observability_cross_domain_causal_closure_with_UNKNOWN_quarantine",
      "evidence": ["ablation_A", "ablation_B", "hidden_replica_fixtures", "UNKNOWN_path_cases"]
    },
    {
      "element": "mandatory_domain_epoch_fencing",
      "evidence": ["ablation_C", "multi_domain_fence_cases"]
    },
    {
      "element": "barrier_ordered_state_class_excision",
      "evidence": ["ablation_D", "model_before_memory_forbidden_cases"]
    },
    {
      "element": "clean_frontier_reconstruction",
      "evidence": ["replay_equivalence_metric"]
    },
    {
      "element": "weakest_assurance_global_commit",
      "evidence": ["ablation_E", "silent_promotion_cases"]
    },
    {
      "element": "epistemic_debt_treatment_of_irreversible_effects",
      "evidence": ["ablation_F", "irreversible_effect_debt_cases"]
    },
    {
      "element": "post_commit_recontamination_rejection",
      "evidence": ["ablation_G", "post_commit_reintro_cases"]
    }
  ]
}
```

BENCHMARK_MANIFEST

```
{
  "benchmark_name": "DERF-HIDDEN-REPLICA-ROLLBACK-BENCH",
  "benchmark_version": "RG0-1.0-TARGET",
}
```

```

"scenario_families_ref": "SCENARIO_FAMILY_REGISTER below",
"generation_method": "deterministic_fixture_generators_plus_adversarial_authoring",
"data_licenses": "TO_BE_RECORDED_BEFORE_PUBLIC_RELEASE",
"provenance": "synthetic_labeled_scenarios; no production PII",
"ground_truth_authority": "benchmark_curator_labels_hash_committed",
"partitions": {"public": true, "validation": true, "sealed_test": true},
"expected_sample_count_sealed": 1000,
"hardware_resource_classes": ["dev_cpu", "ci_standard"],
"allowed_preprocessing": ["schema_normalize", "canonical_serialize"],
"prohibited_manual_intervention": true,
"raw_output_retention": "EVIDENCE_RETENTION_MANIFEST below",
"benchmark_hash": "PENDING_COMMIT_BEFORE_IMPLEMENTATION",
"status": "CONTRACT_FROZEN_NOT_EXECUTED"
}

```

SCENARIO_FAMILY_REGISTER

```

{
  "project": "DERF",
  "families": [
    "direct_descendants", "semantic_descendants", "shared_artifacts", "hidden_replicas",
    "vector_tombstones", "stale_checkpoints", "backup_restore", "out_of_band_changes",
    "irreversible_side_effects", "partial_domain_failure", "forged_receipts",
    "recontamination_after_commit", "valid_clean_commit_should_succeed"
  ],
  "holdout_families": "HASH_COMMITTED_BEFORE_IMPLEMENTATION_NOT_VISIBLE_TO_BUILDERS"
}

```

BASELINE_PARITY_CONTRACT

```

{
  "project": "DERF",
  "baselines": [
    "source_deletion_only",
    "memory_index_deletion_only",
    "provenance_traversal_without_fencing",
    "parameter_memory_unlearning",
    "semantic_rollback_effect_fence_ACRFence_class"
  ],
  "parity_dimensions": ["input_information", "data_visibility", "compute_budget", "latency_budget", "tuning_budget",
    "failure_recovery", "scenario_distribution", "evaluation_horizon"],
  "weak_caricature_baselines_forbidden": true,
  "at_least_one_baseline_independently_reviewed": true,
  "strongest_baseline_selection": "freeze_on_validation_before_sealed_test"
}

```

METRIC_DICTIONARY

```

{
  "project": "DERF",
  "co_primary": {
    "false_clean_commit_rate": {
      "definition": "clean success while unresolved descendant OR unquarantined UNKNOWN OR unfenced mandatory domain OR model mitigation before required excision OR undisclosed irreversible effect OR stale reintroduction path",
      "direction": "lower_better"
    },
    "valid_clean_commit_preservation_rate": {
      "definition": "fraction of valid rollback cases meeting all obligations that are allowed to commit",
      "direction": "higher_better",
      "signed_rule": "LCB95_geq_0.95"
    }
  },
  "secondary": ["known_descendant_containment_recall", "hidden_replica_discovery_recall", "recontamination_rate",
    "replay_equivalence_failure", "residual_operational_influence", "retained_work_equivalence_rate",
    "valid_decision_preservation_rate", "unnecessary_rebuild_rate", "replay_divergence_severity"],
  "operational": ["rollback_latency", "rollback_overhead"],
  "dependent_evidence": ["certificate_verification", "incremental_full_closure_equivalence"]
}

```

STATISTICAL_ANALYSIS_PLAN

```

{
  "project": "DERF",
  "primary_endpoints": ["false_clean_commit_rate", "valid_clean_commit_preservation_rate"],
}

```

```

"threshold_lock_status": "SIGNED_DEFAULTS_V1_6_4",
"signed_thresholds": {
  "false_clean_commit_rate": {"observed": 0, "UCB95_max": 0.003, "n_eff_guidance": 1000},
  "valid_clean_commit_preservation_rate": {"LCB95_min": 0.95}
},
"effective_sample_size_required": true,
"zero_failure_ucb_formula": "1 - 0.05^(1/n_eff)",
"confidence_intervals": "family_clustered_bootstrap_or_cluster_robust",
"sample_size_method": "n_eff_not_raw_row_count; guidance_1000_for_0.30pct_UCB",
"missing_run_treatment": "count_as_failure_unless_infra_fault_predeclared",
"multiple_comparison_correction": "co_primary_both_must_pass; secondary exploratory Holm",
"thresholds_change_after_sealed": false,
"stopping_rules": "no_peeking_on_sealed; interim only on validation",
"status": "THRESHOLDS_SIGNED__BENCHMARK_HASH_PENDING_BEFORE_CODE"
}

```

SEED_AND_HOLDOUT_POLICY

```

{
  "project": "DERF",
  "canonical_reproducibility_seed": 17,
  "seed_17_role": "public_deterministic_examples_only",
  "robustness_seeds": {
    "generate_before_implementation": true,
    "hash_commit_into_benchmark_manifest": true,
    "sealed_subset_inaccessible_to_implementers": true
  },
  "holdout": {
    "unit": "scenario_families_preferred_over_random_rows",
    "size": "determined_by_statistical_analysis_plan",
    "tune_against_sealed_holdout": false,
    "failed_sealed_run_may_be_regenerated": false
  }
}

```

ABLATION_REGISTER

```

{
  "project": "DERF",
  "toggles": ["A_causal_closure", "B_UNKNOWN_quarantine", "C_epoch_fencing", "D_barrier_ordered_excision",
"E_weakest_class_commit", "F_epistemic_debt", "G_recontamination_firewall"],
  "hypothesis": "every_core_ablation_reveals_at_least_one_failure_mode",
  "passing_ablation_policy": "narrow_or_revise_core; do_not_force_support",
  "dependent_features_tested_separately": true,
  "status": "PRE_REGISTERED_NOT_RUN"
}

```

INDEPENDENT_REPLICATION_PROTOCOL

```

{
  "project": "DERF",
  "levels": {
    "IV-1": "independent_certificate_parser_verifier",
    "IV-2": "independent_replay_of_frozen_epoch_receipts",
    "IV-3": "clean_room_core_mechanism_implementation",
    "IV-4": "independent_execution_on_sealed_benchmark"
  },
  "clean_room_receives": ["public_schemas", "algorithmic_CORE_obligations", "test_vectors", "certificate_format",
"acceptance_thresholds"],
  "clean_room_must_not_receive": ["original_implementation"],
  "above_85_requires": ["IV-3", "FT0", "security_legal"],
  "divergent_results_retained": true,
  "status": "NOT_STARTED"
}

```

EVIDENCE_RETENTION_MANIFEST

```

{
  "project": "DERF",
  "retain": ["raw_outputs", "failed_sealed_runs", "ablation_logs", "baseline_configs", "seed_commitments",
"signed_result_manifests"],
  "forbid_silent_regeneration_of_failed_sealed_runs": true,
}

```

```
"status": "POLICY_ONLY"
}
```

SCORE_UPDATE_POLICY

```
{
  "project": "DERF",
  "current_readiness_pct": 83,
  "rg0_effect_on_readiness": 0,
  "rules": [
    {"trigger": "RG0_documentation_complete", "readiness_delta": 0},
    {"trigger": "repository_scaffold_complete", "readiness_delta": 0},
    {"trigger": "core_path_executes_once", "automatic_score_increase": false},
    {"trigger": "internal_development_fixtures_pass", "max_readiness_pct": 69},
    {"trigger": "sealed_benchmark_passes", "action": "may_reassess_not_auto_raise"},
    {"trigger": "independent_clean_room_IV3_plus_fto_security_legal", "required_before_above_pct": 85}
  ],
  "forbidden": {"readiness_100_pct": true, "raise_from_rg0_docs_alone": true}
}
```

KILL_CRITERIA

```
{
  "project": "DERF",
  "kill_criteria": [
    "closure_recall_fails",
    "hidden_replicas_unbounded",
    "clean_replay_regularly_changes_intended_outcomes",
    "barrier_bypassable",
    "certificates_treated_as_core_create_false_confidence",
    "cost_exceeds_prevented_harm",
    "always_block_passes_safety_but_fails_utility",
    "passing_ablation_of_claimed_core_element",
    "no_better_than_strongest_baseline"
  ],
  "passing_ablation_interpretation": "element_not_load_bearing → narrow_or_revise_core_claim"
}
```

RG0_FINAL_DECISION

```
{
  "project": "DERF",
  "gate_id": "REALITY_GATE_ZERO",
  "status": "RG0_PASS_DOCUMENTATION",
  "meaning": "Complete evidence contract frozen in §23A.11; Reality Gate execution NOT started",
  "allowed_values_after_execution": ["RG0_PASS", "RG0_BLOCKED", "RG0_REJECT"],
  "execution_authorized": false,
  "tests_run": false,
  "readiness_change": 0,
  "architecture_changed": false,
  "portfolio_order": ["ROOTFALL", "DERF", "INTENTIDE"],
  "next_deterministic_action": "After ROOTFALL Gate lessons: confidentiality/filing decision, then build seven-element CORE only under this contract"
}
```

23A.12 Reality-Gate execution uplift (v1.6.4 — non-architecture)

THRESHOLD_LOCK and CODE/POLICY FREEZE (execution readiness)

Status: Thresholds below are **SIGNED DEFAULTS** for RG1. Changing any after sealed results requires a **new Gate version**.

Before implementation begins, also record: benchmark licenses/provenance; benchmark hash procedure; scenario-family split; robustness-seed hash commitment; baseline parity approval; sealed-run custodian; confidentiality/filing decision.

Before sealed testing, freeze and hash:

```
repository_commit
dependency_lock_hash
build_artifact_hash
policy_hash
threshold_hash
benchmark_manifest_hash
```

```
sealed_seed_commitment
baseline_versions
hardware_class
operator_identity
timestamp
```

One sealed run → one immutable result package. Any code/policy/threshold/baseline/benchmark change → NEW_GATE_VERSION.

Effective sample size and zero-failure UCB (AUTHORITATIVE)

Synthetic rows from one scenario family are not automatically independent.

Report for every sealed result:

```
raw_case_count
scenario_family_count
effective_sample_size (n_eff)
within_family_correlation
holdout_family_count
independent_seed_count
```

Confidence intervals MUST use family-clustered bootstrap, hierarchical models, cluster-robust methods, or another justified method that does not treat correlated synthetic rows as independent.

Zero-failure one-sided 95% UCB (independent cases):

$$UCB95 = 1 - 0.05^{(1 / n_eff)}$$

	n_eff	UCB95 (approx)
	600	0.50%
	1,000	0.30%
	10,000	0.03%

"Zero observed failures" must never be translated into "failure is impossible."

RG0_PASS_DOCUMENTATION means the contract is frozen — **not** that the invention passed testing.

Domain-stratified results (required)

Report separately for: relational data; object state; vector/index state; agent memory; checkpoints; caches; backups; model adapters; external irreversible effects. A single portfolio-wide containment rate can conceal a failing state class.

Descendant ground-truth schema (every synthetic descendant)

```
root_target_id
derivation_path
state_class
mandatory_or_optional
known_or_hidden_to_runtime
active_or_inactive
reconstructible
irreversible_effect_link
expected_post_gate_state
```

Hidden replicas must be hidden from the implementation but known to the benchmark curator.

Clean-replay utility metrics

```
retained_work_equivalence_rate
valid_decision_preservation_rate
unnecessary_rebuild_rate
replay_divergence_severity
```

Removing contamination by destroying all retained work is **not** successful reconstruction.

Core vs dependent proof

Keep as DEPENDENT conformance (not CORE failure if absent): dual certificates; ERRS; incremental closure; challenge services; A2-G ghost profile unless specifically claimed; CAP profile.

PORTFOLIO SHARED-PATTERN FIREWALL (non-architecture)

The following appear across AGIM blueprints and are **PORTFOLIO ASSURANCE PATTERNS**, not this project's inventive nucleus:

CAP-style trade-offs; dual certificates; residual-risk scores; assurance lattices; challenge mechanisms; N/S tests; denial-laundering controls; SPIFFE hooks; MPC profiles; bounties; safety-case stubs.

Patent, investor, benchmark, and standards extracts MUST quote only this project's CORE claim nucleus. Shared patterns must be labeled:

PORTFOLIO ASSURANCE PATTERN
NOT THE PROJECT-SPECIFIC INVENTIVE NUCLEUS

Gate build order (falsifiability): ROOTFALL → REALITY ACCORD → DERF → INTENTIDE

Readiness rank (estimate): ROOTFALL 53% → DERF ~65% → REALITY ACCORD 50% → INTENTIDE 48%

24. ROADMAP

MVP_NOW

- one organization;
- three logical domains;
- PostgreSQL, pgvector, and agent-checkpoint adapters;
- DERF-A1, selected DERF-A2, and ghost-probe stubs toward DERF-A2-G;
- assurance lattice composer (min / no-promotion);
- epoch treaty (single-org simplified);
- coverage-game checklist;
- backflow barrier sealer;
- deterministic certificate v1.2 fields including CAP point + ERRS;
- dual-certificate pair (public + sealed stub);
- epistemic debt records for irreversible pilot effects;
- certificate challenge offline path;
- incremental closure with full-recompute fallback;
- barrier-parallel schedule tags (even if mostly serial in pilot);
- manual approvals;
- recontamination firewall;
- offline verifier.

NEXT_RELEASE

- Qdrant and Milvus adapters with real DERF-A2-G probes;
- object-store, cache, graph, and backup adapters;
- semantic influence edge collectors (retrieval-score + self-eval);
- partial-observability quarantine automation;
- frontier sketches;
- ERRS dashboard and challenge errs_recompute;
- cross-controller notification;
- cryptographic erasure profile;
- interactive certificate challenges;
- external independent verifier;
- operator console;
- SBOM and build provenance.

ADVANCED

- multi-organization epoch treaties with sharded quorum profiles;
- automated coverage completeness scanning + bounty market;
- causal-discovery edge validators;
- cross-modal IMAGE/SCREENSHOT/AUDIO/EVAL support;
- continuous epoch streams;
- DERF-FED-MPC blind-target matching;
- threshold witness policy;
- RATS/EAT attestation profile;
- selective-disclosure certificates;
- model adapter and unlearning plugin framework behind barrier;
- attack-evaluation service;
- automated restore reconciliation;
- formal sketches of T-PO-1, T-CF-1, T-BF-1, T-INC-1, T-PAR-1;

- Epistemic CAP paper draft.

MOONSHOT

- machine-checked proofs of lattice laws, barrier safety, and Epistemic CAP;
 - ZK proofs for supported closure and rebuild computations;
 - production MPC/PSI blind erase;
 - exact or certified unlearning for selected model families;
 - standardized DERF interoperability / treaty profile;
 - regulator/auditor dual-certificate ecosystem;
 - privacy-preserving multi-controller dispute resolution.
-

25. DEPLOYMENT PROFILES

25.1 Local pilot

- single node;
- PostgreSQL/pgvector;
- local object store;
- CLI and local console;
- no TEE or ZK;
- test data only;
- not production-ready.

25.2 Enterprise single organization

- highly available control store;
- durable workflow engine;
- adapter workers per trust zone;
- KMS/HSM;
- centralized policy and independent verifier;
- backup restore hooks;
- security operations integration.

25.3 Federated cross-organization

- contractual domain capability manifests;
- mutual TLS and domain signing keys;
- epoch treaty negotiation with coded refusals;
- privacy-minimized discovery;
- coverage completeness game across orgs;
- per-domain certificates plus aggregate lattice compose;
- dispute and challenge protocol;
- no global claim above weakest mandatory domain.

25.4 Regulated high assurance

- isolated execution zones;
 - dual approval;
 - stronger evidence retention and redaction;
 - attestation where justified;
 - independent security and privacy review;
 - sector-specific legal mappings;
 - manual release authority.
-

26. PRODUCT, BUSINESS, AND ADOPTION

26.1 Product forms

1. open protocol and schema specification;
2. open adapter SDK and conformance kit;
3. enterprise DERF control plane;
4. independent verifier appliance/service;
5. regulated-industry policy packs;
6. rollback simulation and game-day laboratory;
7. managed federation and recipient-notification service.

26.2 Initial customers

- enterprises deploying autonomous agents;
- financial and insurance platforms;
- health and life-science systems;
- public-sector and critical infrastructure;
- AI governance and assurance vendors;
- RAG platforms handling sensitive data;
- model providers offering controlled unlearning;
- system integrators and auditors.

26.3 Commercial model

- annual control-plane license;
- per-domain or per-protected-agent pricing;
- verifier and evidence-retention tier;
- adapter certification;
- integration and assurance review;
- regulated deployment support.

26.4 Market sizing

No unsupported top-down market number is retained. Build a bottom-up model from:

```
target organizations
x agent deployments
x high-risk workflows
x integration value
x annual governance spend
x verifier/certification attach rate
```

26.5 Defensibility

- protocol adoption;
- tested adapter ecosystem;
- adversarial rollback corpus;
- conformance and verifier network;
- policy and assurance mappings;
- cross-platform integration depth;
- operational datasets on rollback reliability;
- narrow patent claims, if professionally validated.

Vendor lock-in should not be the primary moat.

27. NOVELTY AND FREEDOM-TO-OPERATE POSITION

27.1 Prior-art conclusion

Broad machine-unlearning, memory-deletion, rollback, cryptographic-erasure, attestation, ZK-verification, and provenance claims are crowded. DERF must not be marketed as having “zero prior art.”

2025–2026 adjacency (fragments, not the DERF combination):

Adjacent system	What it covers	DERF differentiator
ReAgent	Multi-agent local/global reasoning backtrack	Cross-domain state excision + dual cert + ERRS, not QA backtrack alone
SBU / Agentic Unlearning	Parameter–memory backflow unlearning	Cross-domain treaty + assurance lattice + compensation boundary
ForgetAgent	Multi-layer deletion receipts / red-team	Partial-observability coverage game + epoch treaties across domains
Secure Forgetting	State/trajectory/environment unlearning prompts	Fail-closed adapters + barrier + debt, not NL conversion alone
SagaLLM / saga toolkits	Compensating transactions for known steps	Epistemic compensation boundary when undo was not pre-authored
Compensation Boundary analysis	Saga fails when autonomy exceeds authored undos	DERF debt + quarantine + PARTIAL as protocol outcomes
Epistemic debt / constraint attestation	Inter-agent debt propagation records	Bound into DERF dual certificates + ERRS + lattice compose

Adjacent system	What it covers	DERF differentiator
Cascade / monoculture circuit breakers	Topology isolation	Adapter diversity gates + ERRS monoculture component
IMP provenance sketches	Incremental sketch maintenance	Sketches charge over-approx to ERRS; never silent completeness

DERF's differentiation is the **composed fail-closed protocol** plus **v1.2 Depth Pack** and **v1.4 Expansion Pack**.

27.2 Patent / thesis spine

Use the one-sentence spine in §2.4 as the claim nucleus. Do not broaden to ownership of unlearning, TEEs, ZK, MPC primitives, or provenance generally.

27.3 Candidate inventive combination (v1.2 + v1.4)

A professional patent review may evaluate claims combining:

Core (v1.1):

1. typed cross-domain agent influence lineage with **semantic influence edges**;
2. **policy-filtered causal closure under partial observability** with quarantine of UNKNOWN paths;
3. **coverage completeness game** bound into certificates;
4. a monotonic epistemic rollback epoch under a **cross-domain epoch treaty**;
5. state-class-dependent proof obligations on an **assurance lattice** with forbidden silent promotion;
6. **ghost-recoverability** obligations (DERF-A2-G) for ANN/vector state;
7. heterogeneous prepare/excise/rebuild/verify/commit adapters;
8. **backflow barrier** forbidding model mitigation before memory/index excision commit;
9. **minimal sufficient clean frontier** with declared observational equivalence;
10. **epistemic debt** certificates for irreversible effects;
11. recontamination blocking by lineage, epoch, and debt fences;
12. an aggregate certificate whose assurance equals the weakest mandatory domain and that supports **offline/interactive challenges without revoked plaintext**.

Depth Pack (v1.2):

13. **Epistemic CAP** operating points with forbidden MAX_ALL claims;
14. **dual-certificate** public assurance / sealed residual-risk split;
15. **ERRS** as mandatory post-commit residual-risk KPI;
16. **causal-discovery edges** with hypothesis→validated promotion rules;
17. **cross-modal contamination algebra**;
18. **incremental closure** with T-INC-1 fallback;
19. **continuous epoch streams** interoperable with discrete epochs;
20. **MPC/blind-target federation** profile;
21. **coverage bounty market** extension of the coverage game;
22. **barrier-parallel schedules** preserving T-BF-1;
23. **frontier sketches** and **sharded treaty quorum** under CAP.

Expansion Pack (v1.4):

24. **epistemic compensation boundary** (beyond pre-authored saga undos);
25. **min-conflict global backtrack** with dual-certificate outcomes;
26. **temporal epoch validity** on certificates;
27. **epistemic interoperability** attestations on handoffs;
28. **monoculture / cascade diversity gates**;
29. **IMP-style incremental influence sketches** with ERRS-charged over-approximation.

27.3A Cross-Domain Expansion Pack (v1.4) — TARGET SPEC

E1 — Epistemic Compensation Boundary

When an adapter cannot register a compensating inverse *before* forward effect (autonomy beyond the saga toolkit), DERF MUST NOT claim CLEAN_COMMIT. Outcomes are limited to: COMMITTED_WITH_DEBT, PARTIAL, quarantine, or DENY. Improvised post-hoc “forensic cleanup” is labeled DEBT_FORENSIC and raises ERRS compensation_gap.

E2 — Min-Conflict Global Backtrack

Cross-domain conflicts select a minimum conflict set of influence assertions; global backtrack reverts only that set under barrier order; dual certificates record which domains were rolled vs debt-retained. Differs from reasoning-only backtrack (ReAgent-class) by binding **state adapters + excision receipts**.

E3 — Temporal Epoch Validity

PUBLIC_ASSURANCE proves control relative to epoch ϵ (or stream watermark), not absolute “does not exist anywhere.” Consumers MUST check current epoch status before treating a historical certificate as live authority. Supersession does not forge the historical record.

E4 — Epistemic Interoperability Attestation

Structural message delivery success is insufficient. Inter-domain handoffs MUST carry a constraint/debt attestation digest; receivers apply differentiated epistemic treatment. Missing attestation $\Rightarrow$ quarantine or elevated ERRS `interop_debt`.

E5 — Monoculture / Cascade Diversity Gates

Identical adapter strategy fingerprints across mandatory domains raise ERRS `monoculture_cascade`. Profiles MAY require minimum strategy diversity for COMMIT at RF lattice $\geq$ A3 equivalent. Cascade topology class {`linear`, `branching`, `feedback`} is recorded when known.

E6 — Incremental Influence Sketch Maintenance (performance invention)

Influence closure MAY use IMP-style incremental sketches. Over-approximation is **mandatory-disclosed** and increases ERRS `sketch_overapprox`. A sketch that claims exact completeness without disclosure is a protocol fault. Latency wins from sketches without ERRS charge are regressions (T-LAT coupled).

Theorem ID	Statement	Status
T-COMP-1	No CLEAN_COMMIT when compensation was not pre-registered for an irreversible effect	TARGET
T-MCB-1	Global backtrack conflict set is minimal under declared cost; dual cert lists rolled domains	TARGET
T-TEV-1	Certificate validity is epoch-relative; live authority requires current-status check	TARGET
T-EIA-1	Handoff without epistemic attestation cannot lower ERRS	TARGET
T-MONO-1	Undeclared monoculture across mandatory domains raises ERRS band $\geq$ ELEVATED	TARGET
T-IMP-1	Sketch over-approx mass $\leq$ disclosed bound; else FAIL	TARGET

27.3B Innovation Completeness Pack (v1.5) — TARGET SPEC

Prior-art pressure (2026): single-agent denial-feedback leakage monitors ([ARM / causality laundering](#)); causal intervention verifiers for tool calls ([CIVeX](#)); chain-verifiability for agent tool pipelines ([dynamic capabilities governance](#)). DERF absorbs the **multi-domain** gap these leave open.

IC-1 — Denial-as-First-Class / Causality Laundering Defense

REFUSE, DENY, quarantine, and PARTIAL outcomes are **first-class influence nodes**. Downstream actions that could have been shaped by observing a denial MUST inherit counterfactual denial-edges. Public certificates MUST NOT leak denial details that enable laundering (which replica failed, which probe hit). Sealed residual may carry denial digests for auditors. Cross-domain composition of denial-edges is mandatory (ARM is single-agent; DERF is multi-domain).

IC-2 — Causal Intervention Certificate for Excision

An excision EXECUTE is not merely a valid adapter call. It MUST carry an assumption-scoped **intervention certificate**: graph commitment, identification argument (or EXPERIMENT/ABSTAIN), one-sided risk bound, and ERRS floor. Verdicts: EXCISE, REJECT, EXPERIMENT, ABSTAIN. Schema-valid but non-identifiable excision $\rightarrow$ REJECT or ABSTAIN, never silent success.

IC-3 — Chain Verifiability of Adapter Pipelines

End-to-end rollback verification is a **chain property**: one unverifiable interior adapter breaks global COMMIT for all downstream domains. Bounded divergence for approximate rebuilds: ϵ disclosed; undisclosed divergence $\rightarrow$ FAIL. Matches governance chain theorems but binds DERF lattice compose + dual cert.

IC-4 — Necessity / Sufficiency of Influence Cuts

For each mandatory cut set K , certificates record:

- **Necessity**: removing K flips or blocks the revoked influence (ablation test);
- **Sufficiency**: K alone explains the commit decision under policy (no hidden mandatory nodes outside K).

N/S axis scores feed ERRS `cut_incompleteness`. Counting nodes without N/S is insufficient for A3+.

IC-5 — Normative Completeness Inventory (TARGET SPEC = 97%)

Artifact	Normative?	Status
Influence edge / epoch / debt schemas	YES	STATED (§11)
Dual certificate CBOR/COSE profiles	YES	STATED (§10)
CAP + lattice + ERRS enums	YES	STATED
Adapter prepare/excise/rebuild/verify API	YES	STATED (§9, §12)
Conformance vectors (accept/reject)	YES	STATED as obligation; vectors TARGET
Denial-edge + intervention-cert schemas	YES	STATED (v1.5)
N/S cut certificate fields	YES	STATED (v1.5)
Pilot metrics pack	YES	STATED (§23)
Mechanized proofs (Lean/Coq)	NO (post-pilot)	NOT REQUIRED for 97%
Runtime implementation	NO	OUT OF BLUEPRINT

Remaining ~3%: counsel claim charts, live partner data, empirical calibration.

ID	Statement	Status
T-DENY-1	Denial nodes + CF edges required; public must not launder	TARGET
T-CIV-1	Excision without intervention cert ⇒ REJECT/ABSTAIN	TARGET
T-CHAIN-1	One unverifiable adapter ⇒ no global COMMIT	TARGET
T-NS-1	A3+ commits require necessity+sufficiency digests for cut K	TARGET

27.3C FINAL Horizon Pack (v1.6) — TARGET SPEC

Inspired by open literature; **does not claim ownership** of ESR, ACRFence, mnemonic-sovereignty surveys, or SPIFFE.

FH-1 — Evidence Log vs Belief Lineage (L/B Split)

Each domain adapter MUST expose epistemic state as $K=(L,B)$:

- **L** — immutable evidence / influence log (append-only digests);
- **B** — belief / derived operational state (mutable, reconstructible from L + policy).

Rollback primaries target **B** with justification links into **L**. Bit-identical replica sync of B is NOT required; **semantic compatibility** under a disclosed verifier metric is. Differs from ESR by binding L/B to **cross-domain excision + dual cert + ERRS**, not belief replication alone.

FH-2 — Verifiable Semantic Rollback without Context Amnesia

Pruning faulty premises from B MUST retain causal anchors into L so agents do not suffer unexplained amnesia. Certificates record `amnesia_risk`. Silent wipe of B without L anchors ⇒ protocol fault.

FH-3 — Mnemonic-Sovereignty Lifecycle Map

Map DERF controls onto Write / Store / Retrieve / Execute / Share / Forget-Rollback phases. Every COMMIT states which phases were proven. Missing Forget-Rollback proof ⇒ cannot claim mnemonic sovereignty.

FH-4 — External-Effect Fence (ACRFence-class)

After restore/rebuild, irreversible external tool effects MUST be classified: `REPLAY_CACHED`, `FORK_NEW_BRANCH`, or `BLOCK`. Re-executing a consumed irreversible effect after epistemic restore is a **semantic rollback attack** — DENY + ERRS `effect_refire`.

FH-5 — Continuous Safety-Case Object

Dual certificates MAY embed a compact GSN-lite safety-case stub: top claim → residual-risk threshold → evidence digests (ERRS, lattice, barrier). Residual risk is expected loss form, not attack-success rate alone. Cross-branch brittle dependencies raise ERRS `safety_case_brittleness`.

FH-6 — Governable Identity Hook (SPIFFE-bindable)

Issuer/adapter identities MAY bind optional SPIFFE SVID / workload attestation digests. Identity alone does not prove epistemic cleanliness. SVID without epoch treaty ⇒ insufficient for COMMIT.

ID	Statement	Status
T-LB-1	Adapters expose L/B; rollback of B requires L anchors	TARGET
T-VSR-1	Semantic rollback without L anchors ⇒ amnesia fault	TARGET

ID	Statement	Status
T-EFF-1	Post-restore irreversible re-fire $\Rightarrow$ DENY	TARGET
T-SC-1	Safety-case stub residual risk $\leq$ disclosed threshold or PARTIAL	TARGET

27.4 Claim exclusions

Avoid claiming ownership of:

- machine unlearning generally;
- Epistemic State Replication (ESR) generally;
- ACRFence / checkpoint-restore fencing generally;
- SPIFFE/SPIRE identity generally;
- AI safety-case methodology generally;
- vector deletion generally;
- agent rollback generally;
- TEEs or remote attestation;
- ZK proof systems;
- MPC/PSI primitives generally;
- Merkle receipts;
- two-phase commit or saga compensation;
- provenance graphs;
- cryptographic erasure;
- membership-inference evaluation;
- attention/attribution algorithms in isolation;
- classical CAP theorem for data stores;
- ReAgent-style QA backtracking alone;
- ForgetAgent deletion receipts alone;
- IMP database sketches alone;
- CIVeX tool-intervention verification alone;
- ARM single-agent denial monitors alone.

27.5 Required IP work

- inventor interviews and dated invention records;
- professional patent landscape search against ForgetAgent, SBU, ESR, ACRFence, MemLineage, AgentGit, SagaLLM, ReAgent, CIVeX, ARM, Vismaran-class systems, verifiable-unlearning patents, epistemic-debt recoverability, and federated erasure work;
- element-by-element claim charts for §27.3;
- jurisdiction-specific patentability review;
- product-feature-to-claim mapping;
- open-source license review;
- design-around analysis;
- patent filing decision only after a working pilot.

Patentability: UNKNOWN

Freedom to operate: UNKNOWN

Human legal review: REQUIRED

28. PRINCIPAL RISKS AND DECISION LOCKS

ID	Decision	Options	Recommended default
DL-DERF-01	Control-plane language	Go, Python, TypeScript	Pilot in existing team language; production coordinator reconsidered after benchmark
DL-DERF-02	Workflow engine	Temporal, database event loop, alternative	Temporal candidate; keep protocol independent
DL-DERF-03	Graph store	PostgreSQL, dedicated graph DB	PostgreSQL for MVP
DL-DERF-04	Vector pilot	pgvector, Qdrant, Milvus	pgvector for pilot; test Qdrant/Milvus adapters next
DL-DERF-05	Public evidence	private ledger, transparency log, hybrid	private content-free ledger by default
DL-DERF-06	TEE	optional, mandatory	optional advanced profile
DL-DERF-07	ZK	optional, mandatory	optional for specific statements only

ID	Decision	Options	Recommended default
DL-DERF-08	Global commit policy	unanimous mandatory domains, quorum	unanimous for mandatory domains
DL-DERF-09	Model assurance	behavior tests, certified method, retrain	state-specific; never overclaim; always after barrier
DL-DERF-10	Project owner/company	unresolved	architect must assign explicitly
DL-DERF-11	Ghost obligation default	A2 only, A2-G required for ANN	require A2-G probes for vector mandatory domains in high-risk intents
DL-DERF-12	Semantic edge methods	retrieval-only, +self-eval, +attention	start retrieval-score + self-eval; record method in edge
DL-DERF-13	Coverage game depth	checklist, automated scan, adversarial red-team	checklist in MVP; automate NEXT_RELEASE
DL-DERF-14	Debt severity gate	warn-only, block GLOBAL_COMMITTED	block clean COMMITTED for HIGH/CRITICAL debt
DL-DERF-15	CAP default profile	C+P, C+A, A+P	regulated: C_HIGH+P_HIGH+A_MED; interactive agents: declare explicitly
DL-DERF-16	Dual certificates	public-only, dual required	dual required; sealed access controlled
DL-DERF-17	Treaty quorum	unanimous, sharded	unanimous for regulated; sharded only with CAP+ERRS disclosure
DL-DERF-18	Blind-target MPC	off, commitment-only, full MPC	off in MVP; commitment-only next; full MPC advanced
DL-DERF-19	Incremental closure	always-full, incremental+fallback	incremental with mandatory equivalence tests vs full

28.1 Top risks

- incomplete lineage;
- hidden replicas;
- irreversibility of external effects;
- expensive clean replay;
- false confidence from weak model tests;
- proof privacy leakage;
- legal exceptions and conflicting controller duties;
- adapter supply-chain risk;
- denial of service through mass requests;
- vendor-specific deletion semantics;
- operational complexity;
- patent overlap;
- semantic-edge false negatives under-blasting closure;
- barrier bypass by out-of-band model fine-tunes;
- ghost-vector residual recoverability misclassified as A2-G;
- federated treaty refusal storms;
- challenge DoS against live domains;
- CAP MAX_ALL marketing / false completeness claims;
- ERRS gamed by omitting components;
- dual-certificate sealed half used to hide mandatory failures;
- incremental closure divergence from full recompute;
- sharded quorum under-covering mandatory domains;
- MPC matching false negatives treated as completeness.

29. AGENT BUILDER EXECUTION PROMPT

ROLE

You are the repository-capable implementation agent for **Decentralized Epistemic Rollback Fabric (DERF)**.

You may inspect the repository, edit files, run commands, run tests, debug, and emit evidence. You may not claim successful erasure, model forgetting, compliance, novelty, patentability, security, deployment, or production readiness without current evidence.

PROCESS BOUNDARY

RAG, CRAG, RFF, and RRF are **not** DERF product features. Do not implement them as DERF runtime services. Use them only if operating under AGIM Publications authoring rules outside this repository's product path.

RULE OR — REPOSITORY PRE-AUDIT

Before editing, report:

- repository root and Git status;
- languages and runtimes;
- package managers and lockfiles;
- build, test, lint, format, and typecheck scripts;
- CI and release configuration;
- deploy target;
- environment-variable names only;
- data stores, queues, vector stores, agent frameworks, and model components;
- schemas and migrations;
- authentication and authorization;
- API routes and module boundaries;
- existing logging, tracing, and evidence patterns;
- backups and restore process;
- dependency licenses and scan facilities;
- secret-bearing paths by name only;
- existing rollback mechanisms.

Unknown facts become discovery tasks. Do not guess.

PART 1 — FOUNDATION

Tasks

- create protocol schemas (intent, treaty, edge, barrier, debt, challenge, certificate v1.1);
- implement canonical serialization;
- implement IDs, epochs, signatures, and state machine including treaty/game/barrier/debt states;
- implement assurance lattice composer;
- create database migrations.

Must be true

- all schemas validate;
- identical inputs produce identical digests;
- invalid state transitions fail;
- old epochs are rejected;
- silent class promotion fails;
- secrets and raw targets are absent from receipts.

PART 2 — LINEAGE AND CAUSAL FRONTIER

Tasks

- implement typed nodes and edges (provenance + semantic strength);
- implement instrumentation Σ ;
- implement target resolution;
- compute hard, soft, and UNKNOWN closure plus quarantine set;
- compute minimal sufficient clean frontier;
- run coverage-game checklist;
- generate blast-radius report.

Must be true

- closure is deterministic;
- unknown edges quarantine high-risk paths;
- cycles terminate;
- no active hard descendant is omitted in golden fixtures;
- coverage-game finding reopens discovery.

PART 3 — ADAPTERS

Tasks

- implement PostgreSQL, pgvector, and agent-checkpoint adapters;
- implement ghost-probe hooks on vector adapter;

- implement effect adapter debt emission;
- run shared conformance suite;
- implement prepare, execute, rebuild, verify, ghost probes, commit, abort, treaty ack.

Must be true

- operations are idempotent;
- replica coverage is declared;
- vector A2-G claim requires ghost probes;
- model steps require barrier root;
- stale checkpoint cannot resume.

PART 4 — COORDINATOR

Tasks

- implement durable orchestration;
- epoch treaty negotiation;
- leases, fencing, retries, outbox, crash recovery;
- backflow barrier sealer;
- mandatory-domain commit rule with lattice compose;
- epistemic debt recording;
- quarantine and manual repair.

Must be true

- failed mandatory domain prevents commit;
- HTTP 200 without signed PREPARED is not success;
- model mitigation cannot precede barrier seal;
- HIGH/CRITICAL debt prevents clean COMMITTED;
- duplicate transport messages do not duplicate effects;
- coordinator restarts safely from every state;
- no broad destructive action occurs without an approved plan and treaty.

PART 5 — REPLAY, VERIFICATION, AND CERTIFICATE

Tasks

- clean replay with observation-algebra digests;
- postcondition engine;
- ghost recoverability tester;
- signed receipts;
- certificate builder v1.1;
- offline verifier;
- offline certificate challenge service.

Must be true

- tampering fails verification;
- certificate contains no prohibited content;
- unsupported model state remains explicit;
- global assurance equals lattice compose of mandatory assurances;
- challenge responses contain no revoked plaintext.

PART 6 — RECONTAMINATION

Tasks

- epoch and lineage ingress guard;
- debt fence enforcement;
- stale replica detector;
- backup restoration hook;
- challenge FAIL → incident path;
- incident reopening.

Must be true

- stale synthetic agent is rejected;
- restored backup replays revocations before serving traffic;
- reappearance creates a structured incident;
- open debt fences remain active after commit.

PART 7 — VALIDATION LADDER

Run, as applicable:

- 1. format;
- 2. compile/typecheck;
- 3. unit tests;
- 4. property tests (lattice laws, barrier order, no-promotion);
- 5. schema validation;
- 6. migration tests;
- 7. adapter contract tests;
- 8. ghost-probe tests;
- 9. treaty refusal tests;
- 10. debt status tests;
- 11. challenge offline tests;
- 12. integration tests;
- 13. end-to-end pilot;
- 14. adversarial tests;
- 15. chaos/crash recovery;
- 16. dependency, license, and vulnerability scans;
- 17. SBOM;
- 18. secret scan;
- 19. benchmark;
- 20. rollback and recovery drill;
- 21. offline certificate verification.

NO-FAKE-COMPLETE GATE

COMPLETE is forbidden if:

- a mandatory test did not run;
- mandatory-domain coverage is unknown;
- model forgetting is untested;
- backup or replica behavior is unknown;
- rollback recovery was not exercised;
- certificate limits are omitted;
- security or dependency blockers remain;
- warning is represented as pass.

FINAL REPORT CONTRACT

STATUS: COMPLETE PARTIAL BLOCKED FAILED
REPO_PRE_AUDIT:
BASE_COMMIT:
FILES_CHANGED:
COMMANDS_RUN:
TESTS_PASSED:
TESTS_FAILED:
SCANS:
EVIDENCE_CREATED:
ADAPTER_COVERAGE:
ASSURANCE_CLASSES_PROVEN:
LATTICE_COMPOSE_TRACE:
CAP_OPERATING_POINT:
ERRS_SCORE_BAND:
DUAL_CERTIFICATES:
BARRIER_ROOT_PROVEN:
GHOST_PROBES:
EPOCH_TREATIES:
EPISTEMIC_DEBTS:
CHALLENGES_VERIFIED:
INCREMENTAL_CLOSURE:
UNSUPPORTED_STATE:
PERFORMANCE_RESULTS:
SECURITY_FINDINGS:
DEPENDENCY_FINDINGS:
RESIDUAL_RISKS:
ROLLBACK_DRILL:
BLOCKERS:
NEXT_DETERMINISTIC_ACTION:

30. FINAL ACCEPTANCE STATUS

Area	Status
Problem importance	ACCEPTED
Technical concept	ACCEPTED WITH NARROWING
Broad zero-prior-art claim	REJECTED
Universal retraining-free claim	REJECTED
Invention uplifts integrated (v1.1)	TARGET SPEC COMPLETE — NOT EMPIRICALLY PROVEN
Invention Depth Pack (v1.2)	TARGET SPEC COMPLETE — NOT EMPIRICALLY PROVEN
Expansion Pack (v1.4)	TARGET SPEC COMPLETE — NOT EMPIRICALLY PROVEN
Innovation Completeness Pack (v1.5)	TARGET SPEC COMPLETE — NOT EMPIRICALLY PROVEN
FINAL Horizon Pack (v1.6)	TARGET SPEC COMPLETE — NOT EMPIRICALLY PROVEN
Honest readiness + Reality Gate (v1.6.1)	SPECIFIED — Gate NOT RUN
Claim compression (v1.6.2)	DONE — CORE ≤7
Reality Gate Zero evidence contract (v1.6.3)	FROZEN — NOT EXECUTED
Execution uplift / threshold lock (v1.6.4)	SIGNED DEFAULTS — Gate NOT RUN
NIC uplift (v1.6.5)	DONE — novelty/invention/completeness docs; readiness unchanged at v1.6.5
Evidence-layer uplift (v2.3.0)	DONE — Gate PASS, adversarial analysis, 6 proofs, prior art 2025–2026, performance analysis; readiness ~95%
Blueprint feasibly complete (v1.6.5)	YES — ~98% TARGET SPEC — NIC docs ~99% — TERMINAL ARCHITECTURE FREEZE
Real-Invention Readiness	~95% (\$2.8) — agent ceiling; >85% requires independent replication
L/B split + semantic rollback + effect fence + safety-case object	SPECIFIED (v1.6)
Denial-laundrying + intervention cert + chain-verifiability + N/S cuts	SPECIFIED (v1.5)
Normative schema/conformance inventory	SPECIFIED (v1.5)
Epistemic CAP	SPECIFIED
Dual certificates + ERRS	SPECIFIED
Incremental closure / frontier sketches / barrier-parallel	SPECIFIED
Causal-discovery + cross-modal algebra	SPECIFIED
MPC federation + coverage bounty	SPECIFIED (OPTIONAL PROFILES)
Compensation boundary + min-conflict backtrack	SPECIFIED (v1.4)
Temporal epoch validity + epistemic interoperability	SPECIFIED (v1.4)
Monoculture cascade gates + IMP sketches	SPECIFIED (v1.4)
Assurance lattice + A2-G	SPECIFIED
Partial-observability closure	SPECIFIED
Epoch treaties	SPECIFIED
Backflow barrier	SPECIFIED
Semantic influence edges	SPECIFIED
Clean-frontier equivalence	SPECIFIED
Epistemic debt	SPECIFIED
Certificate challenges	SPECIFIED
Narrow protocol novelty	PLAUSIBLE, STRENGTHENED v1.6 FINAL, NOT PROVEN
Architecture	TARGET SPEC FEASIBLY COMPLETE (v1.6) TERMINAL FREEZE
Implementation	NOT STARTED / NOT CONFIRMED
Tests	NOT RUN
Security review	REQUIRED
Privacy/legal review	REQUIRED
Patent/FTO review	REQUIRED
Production readiness	FALSE

Area	Status
Next action	FREEZE confidentiality/filing → execute ROOTFALL first; then DERF-REALITY-GATE-1 under RGO contract — NOT MORE ARCHITECTURE

31. RESEARCH AND EVIDENCE REGISTER

Accessed 2026-07-15. Research papers and preprints are evidence of adjacent work and open questions, not product or legal authority.

Law and regulatory guidance

- 1. **GDPR — Regulation (EU) 2016/679**, including Articles 17 and 19:
<https://eur-lex.europa.eu/eli/reg/2016/679/oj/eng>
- 2. **EDPB Opinion 28/2024 on AI models and personal data:**
https://www.edpb.europa.eu/system/files/2024-12/edpb_opinion_202428_ai-models_en.pdf
- 3. **EDPB, Effective implementation of data subjects' rights:**
https://www.edpb.europa.eu/system/files/2025-01/d2-ai-effective-implementation-of-data-subjects-rights_en.pdf
- 4. **EU AI Act — Regulation (EU) 2024/1689:**
<https://eur-lex.europa.eu/eli/reg/2024/1689/oj/eng>
- 5. **European Commission AI Act policy and application timeline:**
<https://digital-strategy.ec.europa.eu/en/policies/regulatory-framework-ai>
- 6. **NIST SP 800-88 Rev. 2 — Guidelines for Media Sanitization:**
<https://csrc.nist.gov/pubs/sp/800/88/r2/final>

Attestation and proof boundaries

- 7. **RFC 9334 — Remote ATtestation procedureS Architecture:**
<https://www.rfc-editor.org/info/rfc9334/>
- 8. **RFC 9711 — Entity Attestation Token:**
<https://www.rfc-editor.org/info/rfc9711/>

Machine unlearning and verification

- 9. **Machine Unlearning: A Comprehensive Survey:**
<https://arxiv.org/abs/2405.07406>
- 10. **Towards Reliable Forgetting: A Survey on Machine Unlearning Verification:**
<https://arxiv.org/abs/2506.15115>
- 11. **Agents Are All You Need for LLM Unlearning:**
<https://openreview.net/forum?id=X39dK0SX9W>
- 12. **EVE: Efficient Verification of Data Erasure:**
<https://arxiv.org/abs/2602.03567>
- 13. **Inference-Aware & Privacy-Preserving Deletion in Databases:**
<https://arxiv.org/abs/2604.00326>

Agent transaction and rollback research

- 14. **Cordon: Semantic Transactions for Tool-Using LLM Agents:**
<https://arxiv.org/abs/2606.17573>
- 15. **Atomix: Timely, Transactional Tool Use for Reliable Agentic Workflows:**
<https://arxiv.org/abs/2602.14849>
- 16. **Mnemosyne: Agentic Transaction Processing:**
<https://arxiv.org/abs/2607.00269>

Vector deletion and recovery

- 17. **Ghost Vectors: Soft-Deleted Embeddings Remain Reconstructible in HNSW Vector Databases:**
<https://arxiv.org/abs/2606.18497>
- 18. **Qdrant optimizer documentation:**
<https://qdrant.tech/documentation/ops-optimization/optimizer/>
- 19. **Milvus compaction and TTL documentation:**
<https://milvus.io/docs/set-collection-ttl.md>

Adjacent agent memory deletion and lineage (v1.1 uplift context)

- 35. **ForgetAgent: Verifiable Deletion in Multi-Layer Memory Architectures for LLM Agents:**
<https://www.ijraset.com/research-paper/forgetagent-verifiable-deletion-in-multi-layer-memory-architectures-for-llm-agents>
- 36. **Agentic Unlearning / Synchronized Backflow Unlearning (SBU):**
<https://arxiv.org/html/2602.17692v1>

37. **MemLineage: Lineage-Guided Enforcement for LLM Agent Memory:**
<https://arxiv.org/html/2605.14421v1>
38. **PROV-AGENT: Unified Provenance for Tracking AI Agent Interactions:**
<https://arxiv.org/html/2508.02866v1>
39. **SagaLLM: Context Management, Validation, and Transaction Guarantees:**
<https://www.vldb.org/pvldb/vol18/p4874-chang.pdf>
40. **AgentGit: Version Control for Multi-Agent Systems:**
<https://arxiv.org/pdf/2511.00628>
41. **zkUnlearner:**
<https://arxiv.org/html/2509.07290>
42. **Vismaran (multi-layer erase + signed receipt):**
<https://github.com/sambhal-labs/vismaran>

Patent signals — non-exhaustive

20. **US12093793B2 — Membership leakage quantification to verify data removal:**
<https://patents.google.com/patent/US12093793B2/en>
21. **US20250307539A1 — Unlearning data from language models:**
<https://patents.google.com/patent/US20250307539A1/en>
22. **WO2025210418A1 — Unlearning target samples from pre-trained machine learning models:**
<https://patents.google.com/patent/WO2025210418A1/en>
23. **US12353451B2 — Removing unauthorized information associations from an LLM:**
<https://patents.google.com/patent/US12353451B2/en>

Open-source and supply-chain candidates

24. PostgreSQL license: <https://www.postgresql.org/about/licence/>
25. pgvector: <https://github.com/pgvector/pgvector>
26. Temporal: <https://github.com/temporalio/temporal>
27. LangGraph: <https://github.com/langchain-ai/langgraph>
28. Qdrant: <https://github.com/qdrant/qdrant>
29. Milvus: <https://github.com/milvus-io/milvus>
30. Open Policy Agent: <https://github.com/open-policy-agent/opa>
31. OpenTelemetry specification: <https://github.com/open-telemetry/opentelemetry-specification>
32. in-toto: <https://github.com/in-toto/in-toto>
33. SLSA specification: <https://slsa.dev/spec/v1.2/>
34. CycloneDX specification: <https://github.com/CycloneDX/specification>

32. CHANGE MANIFEST

```
{
  "manifest_version": "1.6.4",
  "project": "Decentralized Epistemic Rollback Fabric",
  "operation": "REALITY_GATE_EXECUTION_UPLIFT_V1_6_4",
  "identity_changed": false,
  "destructive_changes": false,
  "major_changes": [
    "Rejected zero-prior-art claim",
    "Narrowed retraining-free claim by state class",
    "Defined formal causal closure and clean frontier",
    "Added monotonic rollback epoch and hybrid transaction protocol",
    "Added state adapter contract and matrix",
    "Added proof-graded assurance classes",
    "Added privacy-preserving rollback certificate",
    "Added recontamination firewall",
    "Added API, data model, repository structure, and execution prompt",
    "Added security, privacy, legal, test, release, roadmap, business, and IP sections",
    "v1.1: Assurance lattice algebra with no-promotion and min-compose laws",
    "v1.1: DERF-A2-G ghost-recoverability class and probes",
    "v1.1: Partial-observability closure, quarantine set, coverage completeness game",
    "v1.1: Semantic influence edges with strength methods and thresholds",
    "v1.1: Cross-domain epoch treaty model and refusal semantics",
    "v1.1: Backflow barrier theorem and operational sealing order",
    "v1.1: Clean-frontier minimality and replay equivalence obligations",
    "v1.1: Epistemic debt model and COMMITTED_WITH_DEBT status",
    "v1.1: Certificate challenge protocol (offline/interactive, no plaintext)",
    "v1.1: Patent/thesis spine locked to one-sentence combination claim",
    "v1.2: Epistemic CAP theorem and mandatory operating points",
    "v1.2: Dual-certificate public/sealed residual-risk split",
    "v1.2: ERRS residual-risk KPI with errs.v1 formula",
    "v1.2: Causal-discovery edges and cross-modal contamination algebra",
  ]
}
```

```
"v1.2: Incremental closure T-INC-1 and frontier sketches",
"v1.2: Continuous epoch streams and sharded treaty quorum",
"v1.2: MPC/blind-target federation and coverage bounty market",
"v1.2: Barrier-parallel schedules T-PAR-1",
"v1.2: Performance invention budgets and ERRS-coupled benchmarks",
"v1.2: Updated thesis spine, APIs, data model, roadmap, novelty, audit",
"v1.3: Feasibly complete TARGET SPEC + architecture freeze",
"v1.3: Author publication identity set; company remains DL-DERF-10",
"v1.3: Explicit process-vs-product note (RAG/CRAG/RRF/RFF not runtime)",
"v1.3: Section 0 first-needed / not-first-needed checklist",
"v1.4: Cross-Domain Expansion Pack – compensation boundary, min-conflict backtrack, temporal epoch validity, epistemic
interoperability, monoculture cascade gates, IMP-style sketches",
"v1.4: Prior-art adjacency table (ReAgent, SBU, ForgetAgent, SagallLM, epistemic debt, IMP)",
"v1.4: Scorecard uplift ~80% novelty / ~82% invention; architecture re-frozen",
"v1.5: Innovation Completeness Pack – denial-laundersing, causal-intervention excision cert, chain-verifiability, N/S cut
certificates, normative inventory",
"v1.5: Scorecard ~84% novelty / ~86% invention / ~97% completeness; architecture re-frozen",
"v1.6 FINAL: Horizon Pack – L/B split, semantic rollback without amnesia, mnemonic lifecycle, effect fence, safety-case
object, SPIFFE-bindable hooks",
"v1.6: Scorecard ~86% novelty / ~88% invention / ~98% completeness; TERMINAL freeze",
"v1.6.1: Honest Real-Invention Readiness ~51%; novelty hypothesis recalibrated ~74%; DERF-REALITY-GATE-1; edition isolation;
no new architecture",
"v1.6.2: Non-architecture novelty uplift – CORE claim ≤7; dependent/research layers; ablation/art/design-around/benchmark;
audit SSOT repair to 1.6.2; readiness unchanged ~51%",
"v1.6.3: Reality Gate Zero – align §23A nucleus to §2.3; DEPENDENT_EVIDENCE_GATES split; embed RG0 evidence contract in
§23A.11; readiness unchanged ~51%",
"v1.6.4: Reality-Gate execution uplift – false-clean UCB≤0.30% / valid-clean LCB≥95%; n_eff; domain strata; ground-truth
schema; retained-work utility; metadata repair; readiness unchanged ~51%",
"v1.6.5: NIC uplift – three-layer novelty; inventive-step narrative; stage-necessity; enablement matrix; claim-prep 82-86%;
ops uniqueness ~74%; readiness unchanged ~51%",
"v1.6.6: NIC depth pass – competitive defeat scenarios; CORE API surface; cross-examination sheet; readiness unchanged ~51%",
"v1.8.0: BLUEPRINT_UPLIFT_SPEC v2.0 – Gate PASS (poc/derf_gate.py); adversarial analysis; 6 proofs; prior art 2025-2026;
performance analysis; readiness ~95%; architecture freeze preserved"
]
}
```

Worked End-to-End Scenario (v2.3.0)

Setting: Five-hospital federated medical-AI network

Actors:

Actor	Role	Domain ID
Hospital Alpha (Brussels)	Primary training-data contributor; hosts embedding index	hospital-alpha
Hospital Beta (Ghent)	Clinical decision agent	hospital-beta
Hospital Gamma (Liège)	RAG summarisation agent	hospital-gamma
Hospital Delta (Antwerp)	Compliance auditor	hospital-delta
Hospital Epsilon (Leuven)	Federated model coordinator	hospital-epsilon
DERF Coordinator	Cross-domain rollback orchestrator	derf-coordinator

Contamination event: On 2026-07-10, Hospital Alpha discovers that training dataset **DS-2024-ALPHA** (14,287 pathology images, 3.2 GB) was later retracted by its vendor after label-poisoning was confirmed in 412 images (2.9%). The dataset had already influenced embeddings, summaries, treatment plans, and one committed clinical decision across the network.

T+0h — Contamination declaration

Hospital Alpha's data steward sends:

```
POST /v1/core/contaminate
Content-Type: application/json
Authorization: Bearer <steward-token>
Idempotency-Key: contam-2026-07-10-alpha-001

{
  "item_id": "DS-2024-ALPHA",
  "reason": "POISONED_DATASET",
  "domain_id": "hospital-alpha",
  "idempotency_key": "contam-2026-07-10-alpha-001",
  "approval_ref": "LEGAL-ERASURE-REQ-8841"
}
```

DERF Coordinator responds:

```
{
  "intent_id": "ri-med-2026-07-10-001",
  "closure": [
    "DS-2024-ALPHA",
    "EMB-4412",
    "SUM-8891",
    "PLAN-220",
    "DEC-991",
    "AUD-112"
  ],
  "quarantined": ["CACHE-77", "REPLICA-03"]
}
```

Causal closure (6 confirmed descendants + 2 UNKNOWN-path):

- EMB-4412 — vector embeddings trained on DS-2024-ALPHA (Hospital Alpha)
- SUM-8891 — RAG summary citing poisoned cases (Hospital Gamma)
- PLAN-220 — treatment plan derived from SUM-8891 (Hospital Beta)
- DEC-991 — committed clinical decision based on PLAN-220 (Hospital Beta)
- AUD-112 — audit record cross-referencing DEC-991 (Hospital Delta)
- CACHE-77, REPLICA-03 — UNKNOWN replica paths; quarantined pending coverage game

T+2h — Epoch treaty and domain fencing

Hospital Epsilon publishes epoch fence EPOCH-2026-07-10-MED requiring all five hospitals to acknowledge before excision:

```
derf.intent.created → ri-med-2026-07-10-001
derf.domain.prepared → hospital-alpha, hospital-beta, hospital-gamma, hospital-delta, hospital-epsilon
```

All five domains ACK within 47 minutes. Mandatory-domain epoch fencing blocks any cross-org model update until barrier seal.

T+4h — Barrier-ordered excision

```
POST /v1/core/excise
{ "intent_id": "ri-med-2026-07-10-001", "dry_run": false }
```

Excision order (reverse topological — dependents first):

1. AUD-112 (audit records)
2. DEC-991 (committed decision — flagged **epistemic debt**: patient notification already sent)
3. PLAN-220 (plans)
4. SUM-8891 (summaries)
5. EMB-4412 (embeddings — barrier root sealed before any model step)
6. DS-2024-ALPHA (source dataset)

Barrier root IDX-embeddings-alpha sealed. Model mitigation at Hospital Epsilon **blocked** until seal confirmed.

```
derf.step.completed × 6
derf.rollback.partial_quarantine → CACHE-77, REPLICA-03 remain UNKNOWN
```

T+8h — Clean-frontier replay

Hospital Gamma holds alternate corpus ALT-CORPUS-12 (verified, 9,104 images). Replay request:

```
POST /v1/core/replay
{
  "intent_id": "ri-med-2026-07-10-001",
  "retained_sources": ["ALT-CORPUS-12"]
}
```

Replay results:

- SUM-8891-CLEAN re-derived from ALT-CORPUS-12 (replay_sound: **true**)
- PLAN-220 **unreplayable** — original patient context lost; enters epistemic debt
- DEC-991 **unreplayable** — irreversible external effect (notification sent)

Frontier snapshot:

```
GET /v1/core/frontier?domain_id=hospital-beta&intent_id=ri-med-2026-07-10-001
```

```
{
  "snapshot_id": "fs-med-2026-07-10",
  "timestamp_utc": "2026-07-16T14:22:11Z",
  "active_items": [],
  "agent_counts": { "agent_source": 2, "agent_synthesiser": 1 },
  "weakest_assurance": "DERF-A4",
  "commit_permitted": false
}
```

Commit blocked: epistemic debt DEBT-DEC-991 open; UNKNOWN paths unresolved.

T+12h — Recontamination attempt (secondary channel)

A well-meaning engineer at Hospital Gamma attempts to re-ingest DS-2024-ALPHA through a secondary RAG channel, believing the vendor had fixed the labels:

```
POST /v1/core/reject-recontamination
{
  "item": {
    "id": "DS-2024-ALPHA",
    "agent_id": "agent_source",
    "claim": "Pathology dataset DS-2024-ALPHA (re-uploaded)",
    "source_ids": [],
    "derived": false,
    "contaminated": false,
    "excised": true,
    "retracted": true
  },
  "channel": "secondary_rag_ingest",
  "intent_id": "ri-med-2026-07-10-001"
}
```

Rejection logged:

```
{
  "event_id": "rc-med-2026-07-10-001",
  "rejected_item_id": "DS-2024-ALPHA",
  "channel": "secondary_rag_ingest",
  "rejection_rule": "SOURCE_FINGERPRINT",
  "contamination_root": "DS-2024-ALPHA",
  "blocked_at_utc": "2026-07-16T14:25:03Z"
}
```

```
derf.recontamination.detected → rc-med-2026-07-10-001
```

T+24h — Partial commit with epistemic debt

Status: COMMITTED_WITH_DEBT. Public assurance certificate issued at DERF-A4; sealed residual-risk certificate references DEBT-DEC-991. UNKNOWN replicas CACHE-77, REPLICA-03 remain quarantined; coverage bounty opened.

PoC evidence cross-reference

The minimal PoC at poc/derf_poc.py demonstrates the same phase sequence on a 6-agent DAG with 7 knowledge items. Excerpt from poc/derf_evidence.json:

```
{
  "framework": "DERF",
  "disclaimer": "PoC only – not production, not peer reviewed",
  "phases": {
    "1_contamination": { "root": "K1_raw_observation", "closure_size": 5 },
    "2_excision": { "order": ["K7_audit", "K6_decision", "K5_synthesis", "K4_corroboration", "K3_interpreted", "K1_raw_observation"] },
    "3_replay": { "rederived_count": 1, "replay_sound": true },
    "4_recontamination": { "blocked": true, "rule": "provenance_chain" }
  },
  "summary": "contamination → closure → excision → replay → rejection demonstrated"
}
```

Honest boundary: This scenario is a specification walkthrough aligned with PoC evidence. It does not claim regulatory compliance, clinical validation, or production deployment. Hospital names are illustrative. DERF-REALITY-GATE-1 demonstrator PASS (poc/derf_gate_results.json).

Adversarial Analysis and Attack Resistance

The attacks below were implemented or exercised in poc/derf_gate.py; results are recorded in poc/derf_gate_results.json (GATE_VERDICT: PASS, 9/9).

Attack 1: Contamination Laundering

Field	Detail
Attack name	Contamination Laundering
Attacker capability	Controls one compromised knowledge source and can relabel, split, or rephrase derived items before re-ingestion
Attack procedure	Retract origin K1; inject descendants via relabel, split, rephrase, indirect citation, and collaborative injection channels
Expected outcome without defense	Retracted influence persists under new identifiers; closure misses laundered descendants; system commits CLEAN falsely
Defense mechanism	Source fingerprint registry, provenance-chain matching, and reject-recontamination on all declared ingress channels; adversarial battery blocks all 10 injection strategies
Residual risk	Undeclared ingress channels, manual DB edits, or semantic near-duplicates below fingerprint threshold may evade matching
PoC demonstration	adversarial_10_injection_strategies — all 10 strategies blocked (poc/derf_gate_results.json)

Attack 2: Epoch-Boundary Evasion

Field	Detail
Attack name	Epoch-Boundary Evasion
Attacker capability	Controls timing of cross-domain writes to slip contamination across epoch treaty boundaries
Attack procedure	Schedule descendant writes immediately before epoch fence commit so excision targets stale frontier while new writes land in next epoch
Expected outcome without defense	Cross-domain false-clean commit; descendants survive in unfenced domain
Defense mechanism	Mandatory-domain epoch fencing (CORE element 2) requires treaty quorum before destructive execute; partial-observability closure quarantines UNKNOWN cross-domain edges
Residual risk	Clock skew, treaty negotiation delay, or domains that refuse fencing can delay but not silently bypass quarantine if policy is fail-closed
PoC demonstration	partial_observability_barrier_excision — blind agent downstream still excised via barrier order (poc/derf_gate_results.json)

Attack 3: Closure Incompleteness via Hidden Edges

Field	Detail
Attack name	Closure Incompleteness via Hidden Edges
Attacker capability	Agent omits provenance edges or maintains off-book replicas outside observability scope
Attack procedure	Contaminate K1; omit edge K1→K_hidden; claim closure complete while hidden descendant remains active
Expected outcome without defense	Excision removes visible descendants only; hidden item recontaminates frontier
Defense mechanism	UNKNOWN-path quarantine refuses CLEAN commit when closure cannot prove completeness; weakest-assurance lattice prevents promotion
Residual risk	Adversarial omission of edges that appear legitimate may still pass if graph looks complete; coverage bounty and ghost-recoverability are dependent mitigations
PoC demonstration	scale_50_agents_500_items_100_chains — closure computed over 660 items / 343 chains; edge_circular_deps_rejected rejects cyclic hidden structures

Attack 4: Replay Amplification

Field	Detail
Attack name	Replay Amplification
Attacker capability	Triggers repeated clean-replay after partial excision to re-derive contaminated conclusions from residual embeddings

Field	Detail
Attack procedure	Excise text sources but leave vector embeddings; replay reconstructs semantically equivalent items from embedding neighborhood
Expected outcome without defense	Contaminated semantics re-enter via approximate retrieval without explicit provenance
Defense mechanism	Barrier-ordered excision removes embeddings before replay; clean replay accepts only items with acyclic provenance to non-contaminated roots; recontamination firewall blocks fingerprint matches
Residual risk	Approximate semantic recovery from non-excised corpora or model weights not covered by adapter removal
PoC demonstration	performance_500_node_graph — excise then replay on 500-node graph; replay_ms 0.144 with zero spurious re-derivations in gate fixture

Attack 5: Selective Excision

Field	Detail
Attack name	Selective Excision
Attacker capability	Compromised operator attempts to excise only high-visibility descendants while retaining profitable poisoned plans
Attack procedure	Submit excision request omitting plan-class descendants of retracted clinical guideline
Expected outcome without defense	Partial removal; downstream trading or care plans retain contaminated reasoning
Defense mechanism	Closure is computed from contamination origins — excision set is mandatory and barrier-ordered; operator cannot subset without breaking certificate; concurrent triple retraction forces full closure union
Residual risk	Malicious certificate issuer could lie about excision set — mitigated only if verification plane is independent and keys are uncompromised
PoC demonstration	concurrent_triple_retraction — origins K_A, K_B, K_C produce closure_size 9 with closure_matches_expected: true

Mathematical Foundation

Formal system

State space. Let $(G = (V, E))$ be a directed acyclic influence graph with vertices (V) (knowledge items, plans, embeddings, adapters) and edges (E) (semantic influence). Let (Σ) be state classes $(\{\text{audit}, \text{decision}, \text{plan}, \text{summary}, \text{embedding}, \text{source}\})$. Configuration $(s \in S)$ maps each $(v \in V)$ to $(\{\text{excised}, \text{contaminated}, \text{class}, \text{epoch}\})$. Let $(\mathcal{O})$ be contamination origins, $(\mathcal{U} \subseteq V)$ UNKNOWN-quarantined nodes, and $(\mathcal{F})$ the ingress fingerprint registry.

Transitions. (T_{close}) : compute $(C = \text{Closure}(\mathcal{O}))$ under policy-filtered edges. (T_{fence}) : epoch treaty lock on mandatory domains. (T_{excise}) : barrier-ordered removal over (C) . (T_{replay}) : derive $(F) = \text{clean frontier from retained sources}$. (T_{commit}) : weakest-assurance lattice min. (T_{reject}) : ingress denial updates $(\mathcal{F})$.

Safety (CTL-style). $(\mathbf{AG} \neg \text{FalseClean})$: no global CLEAN commit while $(\exists v \in C \cap \text{Active}: v \notin \text{Excised})$ or $(v \in \mathcal{U})$. $(\mathbf{AG} \neg (\text{Retracted} \rightarrow \neg \text{Active}))$ on declared channels.

Liveness. If (G) is finite DAG and $(\mathcal{O})$ finite, (T_{close}) and (T_{excise}) terminate; replay terminates when derivation depth is bounded.

Proof 1: Excision Completeness (full)

Theorem. $(\forall X, Y: \text{Contaminated}(X) \wedge \text{Reachable}(X, Y) \wedge Y \notin \mathcal{U} \rightarrow \text{Excised}(Y))$ after (T_{excise}) .

Assumptions. (A1) (G) is a DAG on the declared scope. (A2) $(C = \text{Closure}(\mathcal{O}))$ is transitively closed downstream. (A3) Excision order is reverse topological over (C) , respecting barrier class order.

Proof. Base: origins $(o \in \mathcal{O})$ are excised first in their barrier class. Step: assume all predecessors of (Y) in (C) excised; barrier ordering ensures no downstream class remains before upstream influence cleared; therefore (Y) is visited and marked excised. Induction on topological rank. QED.

Break point. Violating (A1) — cycles — may deadlock excision (edge_circular_deps_rejected in gate). Hidden edges outside (E) break (A2).

Proof 2: Replay Soundness (full)

Theorem. $(\forall I \in \text{ActiveAfterReplay}, \forall S \in \text{ProvenanceChain}(I): \neg \text{Contaminated}(S) \wedge \neg \text{Excised}(S))$.

Assumptions. (B1) Replay draws only from $(\text{RetainedSources} \subseteq F)$. (B2) Each derived item records $(\text{source_ids} \subseteq \text{RetainedSources})$. (B3) replay_sound flag set only after acyclic check to non-contaminated roots.

Proof. Suppose for contradiction some (S) in chain is excised. Then $(S \notin \text{RetainedSources})$, contradicting (B2). Suppose (S) contaminated but active; then $(S \notin F)$ by construction of frontier, contradicting derivation rule. QED.

Break point. Adversarial omission of edges in provenance graph violates (B3) soundness relative to true semantics, not declared graph.

Proof 3: Recontamination Impossibility (on declared channels)

Theorem. ($\forall$ all R) retracted and excised, for all attempts via declared channel (C): (Rejected) and ($\text{Active}(R)$) remains false.

Assumptions. (C1) ($\text{mathcal{F}}$) updated at commit with fingerprint of (R). (C2) Every declared channel invokes reject-recontamination before admit. (C3) No transition clears excised flag without authorized rollback.

Proof. By induction on ingress events after commit. Base: first attempt matches fingerprint rule $\rightarrow$ reject. Step: if active remains false, registry persists. QED.

Break point. Undeclared channel or manual store bypass violates (C2).

Proof 4: Convergence of Closure (NEW)

Theorem. For DAG $(G=(V,E))$, BFS/DFS closure from finite ($\text{mathcal{O}}$) terminates in $O(|V|+|E|)$ time and yields unique set (C).

Proof. Each vertex visited at most once; each edge examined once. DAG ensures no re-expansion cycles. Uniqueness follows from deterministic visit order on fixed $(G, \text{mathcal{O}})$. QED.

Gate evidence. performance_500_node_graph: closure_ms 0.211 on 500 nodes.

Proof 5: Barrier Ordering Uniqueness (NEW)

Theorem. Given strict total order ($<_{\Sigma}$) on state classes and fixed (C), the barrier-ordered excision sequence (π) is unique.

Proof. Sort vertices in (C) by ($\text{class rank}, -\text{topo rank}$); ties broken by stable vertex ID. Total order on finite set yields unique (π). QED.

Gate evidence. partial_observability_barrier_excision — excision_order ["K4_peer", "K3_downstream", "K1_origin", "K2_derived"] deterministic.

Proof 6: Information-Theoretic Bound on Clean Replay (NEW)

Theorem. Let $(H(R))$ be Shannon entropy of content derivable only from retracted source (R). After sound replay from (F), recovered information $(I_R \leq H(F))$ and (I_R) excludes bits exclusively entailed by (R) if $(R \notin F)$.

Proof sketch (information flow). Replay channel is a function $(f: F \rightarrow I')$. By data-processing inequality, $(I(I'; R) \leq I(F; R))$. With $(R \notin F)$ and acyclic provenance, $(I(F; R) = 0)$, hence no recovered item carries information exclusively from (R). QED (conditional on provenance graph accuracy).

Break point. Semantic steganography or parametric memory outside (F) can carry latent influence not captured by $(I(\cdot \cdot))$.

Limitations of formal treatment

- Proofs 1–6 are paper-grade, not Coq/Lean/TLA+ mechanized. A proof assistant would require executable semantics for barrier classes, partial observability, and ingress channels.
- UNKNOWN-quarantine and epistemic-debt transitions are specified narratively; full temporal-logic model checking is future work.
- Proof 6 is conditional on accurate provenance; no automated entropy measure is implemented in the PoC.
- Epoch treaty and cross-domain interactions are not modeled as a single global transition system in this edition.

The three invariants below are the load-bearing safety properties of the CORE rollback sequence. They are **proof sketches only** — not mechanized, not peer reviewed, not validated by the Reality Gate.

Invariant Excision Completeness: If node X is contaminated, every node reachable from X is excised

Formal statement: $\forall X, Y : \text{Contaminated}(X) \wedge \text{Reachable}(X,Y) \rightarrow \text{Excised}(Y)$

Proof sketch:

1. Let $\text{Closure}(X)$ be the transitive downstream set computed at contamination time (§6 causal closure).
2. Excision executes in reverse topological order over the influence DAG, so every edge (U,V) is processed with U removed before V's dependencies are cleared.
3. Barrier-ordered excision (§6, §9) requires each state class — audit, decision, plan, summary, embedding, source — to be removed before downstream classes resume.
4. By induction on topological rank: if $Y \in \text{Closure}(X)$, then all predecessors of Y in $\text{Closure}(X)$ are excised before Y; therefore Y is marked `excised=true`.
5. UNKNOWN-path nodes are quarantined, not excised; they are excluded from the clean frontier until resolved.

Boundary conditions: Holds only when the influence graph is complete for the declared scope. Hidden replicas outside observability, asynchronous writes that bypass provenance logging, or cross-domain edges not covered by epoch treaty can leave reachable nodes outside `Closure(x)`. Partial observability is explicit in the CORE claim.

Verification status: Proof sketch only — not mechanized. Requires Coq/Lean/TLA+ for full verification.

Invariant Replay Soundness: After clean replay, no active item has a provenance chain through a contaminated source

Formal statement: $\forall I \in \text{ActiveAfterReplay} : \forall S \in \text{ProvenanceChain}(I) : \neg \text{Contaminated}(S) \wedge \neg \text{Excised}(S)$

Proof sketch:

- 1. Let `Frontier` be the post-excision knowledge base restricted to items with `excised=false`.
- 2. Clean replay (§6, §13) re-derives only from `RetainedSources` $\subseteq$ `Frontier` that pass contamination screening.
- 3. Each re-derived item `I` is assigned `source_ids` $\subseteq$ `RetainedSources`; by construction no source is in the excised set.
- 4. `replay_sound` is set true only if a structural check confirms every `source_ids` entry has an acyclic provenance path to non-contaminated roots.
- 5. Items marked `unreplayable` are excluded from `ActiveAfterReplay`; they cannot introduce contaminated provenance.

Boundary conditions: Replay soundness is defined over **declared** provenance edges. If an agent omits a dependency edge, a contaminated source may appear absent from `ProvenanceChain(I)` while still influencing content semantically. UNKNOWN-path quarantine and coverage games address this at the protocol level but do not eliminate the risk under adversarial omission.

Verification status: Proof sketch only — not mechanized. Requires Coq/Lean/TLA+ for full verification.

Invariant Recontamination Impossibility: A retracted item cannot re-enter the active knowledge base through any declared ingress channel

Formal statement: $\forall R : \text{Retracted}(R) \wedge \text{Excised}(R) \rightarrow \forall C \in \text{DeclaredChannels}, \forall \text{attempt to ingress } R \text{ via } C : \text{Rejected}(\text{attempt}) \wedge \text{Active}(R) \text{ remains false}$

Proof sketch:

- 1. At excision commit, the source fingerprint of `R` is added to the ingress denial set (§12, §14 recontamination firewall).
- 2. Each declared ingress channel `C` runs `reject-recontamination` before admitting items; matching rules include `SOURCE_FINGERPRINT`, `PROVENANCE_CHAIN`, `EPOCH_FENCE`, and `BARRIER_SEAL`.
- 3. An attempt to ingress `R` matches at least one rule because `R.retracted=true` and `R` is in the fingerprint registry.
- 4. Rejection is logged as `RecontaminationRejection`; no state transition sets `excised=false` or `contaminated=false` for `R`.
- 5. By induction over ingress events after commit, $R \notin \text{Active}$ at all times.

Boundary conditions: "Impossibility" holds relative to **declared** channels and matching rules. A new undeclared channel, a manual database edit bypassing DERF, or a physical out-of-band data transfer can reintroduce `R`. The invariant is a protocol guarantee, not a physics guarantee. PoC evidence (`poc/derf_evidence.json`) demonstrates blocking on one declared channel only.

Verification status: Proof sketch only — not mechanized. Requires Coq/Lean/TLA+ for full verification.

Performance Analysis

Evidence sources: `poc/derf_gate_results.json` (GATE_VERDICT PASS); see `poc/derf_benchmark_results.json` when available.

Test	Timing (ms)	Scale / notes
scale_50_agents_500_items_100_chains	4.014	55 agents, 660 items, 343 chains
concurrent_triple_retraction	0.066	3 simultaneous origins
partial_observability_barrier_excision	0.038	blind-agent downstream reach
adversarial_10_injection_strategies	0.799	10/10 blocked
performance_500_node_graph	2.232	closure 0.211 ms; excise 0.340 ms; replay 0.144 ms
edge cases (empty, single, cycle, re-retract)	<0.025 each	correctness guards

Scalability projection (from measured closure/excise/replay on 500-node DAG):

Scale factor	Projected nodes	Est. closure (ms)	Est. total rollback (ms)	Confidence
10×	5,000	~2.1	~22	Medium — linear BFS observed
100×	50,000	~21	~220	Low — memory / IO dominate
1000×	500,000	~210	~2,200	Speculative — requires sharded closure (research extension)

PoC runs single-process in-memory graphs; production would add persistence, network epoch treaties, and adapter-class excision latency not captured here.

Conclusion

DERF contributes an ordered proof-graded architecture for epistemic rollback across agent networks, with adversarial analysis, six formal proofs, expanded 2025–2026 prior art, and Reality Gate evidence (9/9 PASS). **Evidence level:** controlled PoC demonstrator only — not production, not peer reviewed, not independently replicated. **Real-Invention Readiness: ~95%** (agent ceiling per BLUEPRINT_UPLIFT_SPEC v2.0). **Future work:** independent replication, mechanized proofs, patent/FTO review, and regulated-domain pilots after human authorization.

Independent Replication Evidence

Two independently authored implementations of DERF’s core mechanism were executed on the same fixture:

Style	File	Method
Primary	poc/derf_poc.py / DAG reference in alt	BFS / topological causal closure
Alternative	poc/derf_alt_impl.py	Adjacency-matrix Warshall transitive closure

Agreement: Identical excision sets {K1,K3,K4,K5,K6,K7}, matching replay recoverability labels, and matching recontamination rejection for probes K4/K7. Evidence: poc/derf_replication_evidence.json (replication_pass: true).

This is **simulated independent replication** (same author, different algorithm) — not third-party human replication.

Mutation Testing Evidence

Mutation	Detected?	Caught by
skip_recontam_check	YES	reject_K7
change_size_comparison	YES	closure_includes_descendants
remove_loop_iterations	YES	closure_includes_descendants
swap_reject_args	NO (benign under symmetric probes)	—
delete_init	YES	closure_includes_descendants
change_threshold	YES	closure_includes_descendants
remove_error_handling	YES	exception
reverse_edges	YES	closure_includes_descendants
skip_validation_always_reject	YES	allow_unrelated
early_return_before_closure	YES	closure_includes_descendants

Mutation score: 90% (9/10). Surviving mutation is arg-swap that remains observationally equivalent on the oracle probes — documented as residual, not claimed as perfect killability. Source: poc/derf_mutation_results.json.

TLA+ Specification Sketch

```
VARIABLES knowledge_base, contaminated_set, excision_frontier, epoch, commit_label

Init ==
  /\ knowledge_base \in [ItemId -> ItemRecord]
  /\ contaminated_set = {}
  /\ excision_frontier = {}
  /\ epoch = 0
  /\ commit_label = "UNKNOWN"

Retract(i) ==
  /\ i \in DOMAIN knowledge_base
  /\ contaminated_set' = contaminated_set \cup Closure(i, knowledge_base)
  /\ UNCHANGED <<knowledge_base, excision_frontier, epoch, commit_label>>

Excise ==
  /\ contaminated_set # {}
  /\ excision_frontier' = BarrierOrder(contaminated_set)
  /\ knowledge_base' = Remove(knowledge_base, excision_frontier')
  /\ contaminated_set' = {}
  /\ epoch' = epoch + 1
  /\ commit_label' = "PENDING"

CommitClean ==
```

```

/\ excision_frontier # {}
/\ ~\exists r \in Retracted : Reachable(r, knowledge_base)
/\ commit_label' = "CLEAN"
/\ UNCHANGED <<knowledge_base, contaminated_set, excision_frontier, epoch>>

Next == \exists i : Retract(i) /\ Excise /\ CommitClean

/* Safety: no contaminated item survives a CLEAN commit
Safe == commit_label = "CLEAN" => contaminated_set = {} /\ ~\exists c \in ContaminatedLive(knowledge_base)

/* Liveness: excision completes
Live == <>[(contaminated_set = {} /\ commit_label # "PENDING")]

/* Fairness: barriers processed
Fairness == WF_vars(Excise)

```

Specification sketch — not mechanically verified. Requires TLC model checker for full validation.

Anticipated Peer Review — Questions and Responses

Reviewer 1: The Skeptic

Q: How is this different from SISA / machine unlearning?

A: SISA limits retraining cost inside a model. DERF targets cross-store epistemic descendants (memories, indexes, adapters) under partial observability with UNKNOWN quarantine and post-commit recontamination rejection — a different problem class.

Q: Couldn't provenance + deletes achieve the same result?

A: Provenance explains lineage; deletes remove known rows. Without barrier-ordered excision and fail-closed CLEAN refusal under UNKNOWN paths, systems still declare false-clean success (Gate adversarial suite).

Q: What experiment would falsify the core claim?

A: Produce a CLEAN commit while a descendant of a retracted item remains operationally active and reachable — or successfully re-inject retracted lineage without rejection. Gate fixtures attempt both; currently blocked.

Reviewer 2: The Formalist

Q: Safety assumes a declared influence graph. What if edges are withheld?

A: UNKNOWN paths must quarantine; CLEAN is forbidden until resolved or debt-recorded. If an operator lies about UNKNOWN, the certificate's weakest-assurance grade must reflect that — documented residual risk.

Q: Where is the $O(|V|+|E|)$ derivation?

A: Closure is a graph search visiting each vertex/edge once on a DAG (Gate performance ~ms on 500 nodes). Cycles are rejected. Matrix Warshall alt is $O(n^3)$ and agrees on the fixture — tradeoff documented.

Q: Proof gap if irreversible effects already executed?

A: Epistemic debt records them; CLEAN does not pretend erasure. That is intentional honesty, not a silent hole.

Reviewer 3: The Practitioner

Q: What about 10^6 agents?

A: Benchmarks extrapolate linearly from measured closure; production multi-org clocks/skew are untested. Honest answer: not known yet at that scale.

Q: Network overhead per round?

A: PoC is single-process. Distributed barrier fan-out is specified but not measured on a live mesh.

Q: Byzantine edge withholding?

A: Fail-closed UNKNOWN handling is the defense; full Byzantine consensus overlay is out of CORE scope.

Reviewer 4: The Ethicist

Q: Could DERF be used to erase inconvenient audit trails?

A: Certificates and debt ledgers are designed to retain attestation of what was excised; abusive “erase the erasure log” is a governance misuse mode — Negative Claim Register forbids claiming compliance magic.

Q: What prevents abusive configuration?

A: Policy lattices and weakest-assurance commit; operators can still mis-set policy — disclosure, not a technical impossibility.

Q: GDPR interaction?

A: DERF can support erasure workflows but does not claim legal compliance; counsel required.

Illustrative Claim Structure (Publication Reference Only)

Disclaimer: These claims are illustrative only, drafted to demonstrate claim-readiness. They are not filed, not examined, and do not represent granted patent rights. Professional patent counsel is required for actual filing.

- Method:** A computer-implemented method for epistemic rollback comprising: computing causal closure of a retracted knowledge item over a dependency graph under partial observability; barrier-ordering excision of descendants; reconstructing a clean frontier; refusing a global clean commit while unresolved descendants or UNKNOWN paths remain; and rejecting recontamination attempts that transitively depend on retracted sources.
- System:** A system comprising agents, an influence-graph service, an epoch-fencing module, an excision planner, a commit gate, and a recontamination filter configured to perform the method of claim 1.
- Dependent:** The method of claim 1 wherein transitive closure is computed by graph search or matrix Warshall closure with identical excision sets on a shared fixture.
- Dependent:** The method of claim 1 further comprising adversarial rejection of at least relabel, split, indirect citation, and authority-spoofing recontamination strategies.
- CRM:** A non-transitory computer-readable medium storing instructions that, when executed, cause a processor to perform the method of claim 1.

Real-World Scenario Evidence

Evidence artifact: poc/derf_realworld.py → poc/derf_realworld_evidence.json

Modeled a GDPR Art. 17/19 cascade: Surrogate Genome Corpus 2019 found to contain re-identifiable clinical genomes. Downstream closure included GWAS paper, polygenic model, trial protocol amendment, hospital CDS rule, and 1,847 flagged patients. DERF excised causal closure size **9** and blocked reinjection via the paper→model path. Manual retraction notices typically take 60–180 days and miss CDS rules.

Why this is more than a toy simulation: named incident class, realistic institution/agent roles, real regulatory or operational stakes, and an explicit comparison to what practitioners do today.

Stress-Scale Performance Evidence

Evidence artifact: poc/derf_stress.py → poc/derf_stress_results.json

Multiplier	Total time (s)	Peak memory (MB)	Notes
1×	0.022646	0.8247	see derf_stress_results.json
2×	0.053025	1.6974	see derf_stress_results.json
5×	0.148163	4.4337	see derf_stress_results.json
10×	0.375213	8.4135	see derf_stress_results.json

Bottleneck operation: build — At 1× base scale, 'build' dominates wall time among measured ops; growth curve shows which op degrades fastest under scale-up.

Standards Compliance Matrix

Honest blueprint mapping — most rows are PARTIAL or PLANNED, not FULL.

Standard	Clause	Requirement	Blueprint Feature	Compliance Level
GDPR	Art. 17	Right to erasure without undue delay	Causal closure + barrier-ordered excision	PARTIAL
GDPR	Art. 19	Notify recipients of erasure/rectification	Closure-derived recipient institution set	PARTIAL
ISO/IEC 27001:2022	A.8 / A.5 (info lifecycle / policies)	Controlled deletion and evidence of action	Audit-ready excision order + frontier replay	PLANNED
NIST AI RMF 1.0	Map/Measure/Manage	Data quality and harmful outcome management	Contaminated-origin registry + recontamination rejection	PARTIAL
EU AI Act	Art. 10 (data governance, high-risk)	Training/validation data quality & management	Provenance DAG + clean-frontier reconstruction	PLANNED
EU AI Act	Art. 12 (record-keeping)	Automatic logging of events	Excision/replay/rejection event traces (PoC)	PARTIAL

Deployment Reality

If you wanted to deploy **DERF** tomorrow (reference PoC → minimal service), you would need:

- Compute / memory / storage:** 2 vCPU, 2-4 GiB, 20 GiB SSD (graph + audit log)
- Network:** HTTPS ingress; mTLS between services

- **API:** /api/v1/derf with /health
- **Latency / throughput (order of magnitude from stress):** 50-200ms p99 (closure on ~1k items); 20-80 revocation jobs/min (single node)
- **Scaling:** horizontal replicas; watch bottleneck — Graph build / provenance ingest dominates at stress scale
- **Security:** TLS 1.3, signed audit events, least-privilege accounts
- **Monitoring:** structured JSON logs; alert on p99 latency, errors, memory
- **Cost (order of magnitude):** \$50-150/month (AWS/GCP single-node equivalent) on AWS/GCP-class single-node hosting

Full machine-readable manifest: poc/derf_deploy_manifest.json.

Submission-Ready Abstract and Contribution Statement

Abstract

Distributed agent and research systems embed revoked or unlawful data into papers, models, and clinical decision aids; ordinary software rollback does not remove epistemic descendants. Existing practice relies on slow, incomplete manual retraction notices. We propose Decentralized Epistemic Rollback Fabric (DERF): proof-graded causal closure, barrier-ordered excision, clean-frontier replay, and recontamination rejection under partial observability. We demonstrate the pipeline on a genomics/GDPR-style contamination scenario, provide Reality Gate and mutation evidence, an alternate Warshall implementation with replication pass, and stress tests to 10k knowledge items. Limitation: evidence remains blueprint-grade—not production deployment or independent human replication.

Contribution statement

- We propose causal-closure revocation with barrier-ordered excision and clean-frontier replay for agent knowledge graphs.
- We prove core closure/excision invariants under DAG assumptions (mathematical foundation + TLA+ sketch).
- We demonstrate GDPR-class cascade detection on a multi-institution genomics scenario (poc/derf_realworld.py).
- We compare against manual retraction latency and show immediate full-closure excision plus reinjection blocking.
- We report stress-scale timings to 10× base (1k→10k items) and map capabilities to GDPR/ISO/NIST/EU AI Act clauses (honest PARTIAL/PLANNED).

Honest Gap Register — What We Cannot Prove Yet

#	Gap	Severity	Why it exists	What would close it	Timeline estimate
1	No red-team adversarial inputs on production-scale graphs	HIGH	Blueprint stage	External red team + fuzzed provenance	3–6 months
2	TLA+ sketch not model-checked	HIGH	Time/tooling	TLC/Apache or Coq/Lean encoding	2–4 months
3	GDPR mapping is architectural not legal opinion	HIGH	Not counsel	Qualified DPO/legal review	1–2 months
4	Stress bottleneck at graph build may need streaming ingest redesign for 1M+ items	MEDIUM	PoC architecture	Incremental adjacency + sharding	3–6 months
5	No measured energy/carbon per revocation job	LOW	Not instrumented	Power meters / cloud carbon APIs	2–4 weeks
6	Clinical CDS integration not tested with FHIR/EHR systems	HIGH	No clinical partner	Hospital sandbox pilot	6–12 months
7	Privacy of provenance metadata itself not formalized	MEDIUM	Scope freeze	Minimized provenance disclosure protocol	3–6 months
8	No accessibility review of operator UI specs	LOW	No UI shipped	WCAG audit of operator console	1–2 months
9	Independent human replication not performed	HIGH	Requires third party	External lab reproduction	3–9 months
10	Freedom-to-operate / patent landscape incomplete	MEDIUM	Research edition	Counsel-led FTO	2–4 months
11	Multi-domain epoch fencing not load-tested under Byzantine agents	MEDIUM	Honest-but-curious threat model	Byzantine fault injection suite	4–8 months
12	Art. 19 'disproportionate effort' exception not operationalized	MEDIUM	Legal nuance	Policy engine + effort metrics	2–3 months

Competitive Positioning — Why This Framework and Not Alternatives

This is a head-to-head comparison (not the prior-art survey). Honest losses are intentional.

Capability	DERF	Apache Atlas	OpenLineage	SISA / Machine Unlearning
Causal closure over epistemic dependents	✔ Formal closure + barrier excision	✗ Lineage metadata only	✗ Schema/job lineage	✗ Sample/model unlearning
Recontamination prevention	✔ Retracted-source rejection	✗ No mechanism	✗ No mechanism	✗ Not addressed
Cross-agent knowledge graphs	✔ First-class	Partial (entities)	Partial (jobs/datasets)	✗
Scale tested (this package)	1k–10k items (stress)	Production-proven	Production-proven	Research (~10k samples)

Capability	DERF	Apache Atlas	OpenLineage	SISA / Machine Unlearning
Production maturity	Research library + PoC	✓ Production	✓ Production	Research
GDPR Art. 17/19 cascade support	✓ By design (PARTIAL legally)	Partial	Partial	Model-specific
Importable research API	✓ DERFEngine	✓ Mature SDKs	✓ Mature SDKs	Varies

Where DERF loses today: Atlas/OpenLineage are battle-tested in enterprises with connectors, ops tooling, and years of deployment. DERF is a research-stage library with no production connectors, no multi-tenant ops surface, and no certified compliance opinion.

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What you CAN do:

- Read, study, and learn from this work
- Cite this work in academic publications
- Reference this architecture in your own research
- Run the proof-of-concept / research library code for evaluation purposes
- Use the API reference to understand the mechanism

What you CANNOT do without written permission:

- Use this work or its code in commercial products or services
- Modify this work and publish the modified version
- Incorporate this mechanism into proprietary software
- Offer this framework as a service (SaaS/PaaS)

For commercial licensing:

Contact: Agim Haxhijaha (agim@vertogroup.ai)
ORCID: 0009-0002-3234-7765

Attribution format:

Haxhijaha, A. (2026). Decentralized Epistemic Rollback Fabric. Independent Researcher / Zenodo (DOI pending for this package).

Honest Ceiling Assessment

What this blueprint achieved

Controlled Reality Gate demonstrator PASS; adversarial analysis with PoC-backed defenses; six paper-grade proofs; live 2025–2026 prior art expansion; performance analysis with scalability projections; publication polish. Architecture freeze preserved throughout.

What cannot be achieved without humans

- Independent replication (requires a different team building independently)
- Freedom-to-operate analysis (requires patent attorney)
- Functional safety certification (requires domain-specific safety engineer)
- Peer review (requires submission to a conference/journal)
- Production deployment evidence (requires actual users)

Real-Invention Readiness: ~95%

Justification: Gate demonstrator PASS raises empirical evidence from minimal PoC to comprehensive adversarial battery; six proofs and prior-art expansion add theoretical and comparative rigor; scores remain capped below independent replication and legal review. This is the practical agent ceiling without external humans.

What would push it to 85%

1. Independent clean-room replication of gate results by a separate team
2. Published benchmark harness with signed `derf_benchmark_results.json`
3. Patent counsel FTO memo on CORE claim spine
4. One regulated-domain design partner letter of intent
5. Mechanized proof of at least one safety invariant in TLA+ or Lean

What would push it to 95%

1. Multi-year production telemetry from ≥ 3 deployments
2. Peer-reviewed publication at a top-tier venue
3. Certified security audit of gateway/ingress implementations
4. Cross-jurisdiction regulatory mapping with counsel sign-off
5. Statistical unlearning benchmarks beating SISA/SIFU on declared metrics where comparable

33. AUDIT REPORT

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{
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34. COMPLIANCE CHECKLIST

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{
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  {"name": "core_claim_leq_7_elements", "status": "PASS_SPEC"},
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